

DESCRIPTION	EXAMPLES OF WHERE THIS TECHNOLOGY IS USED
Mainframe  A mainframe is a powerful computer that has a high processing power and performance. It is big in size and can be used by several people at the same time.	It is used primarily by large organisations for critical tasks, such as: • Bulk data processing and storing of vast amounts of data, for example, census • It is also used in banks, company head offices and IT organisations
Server  A server is a computer that has really powerful processors, large hard drives and plenty of memory power.	Servers are computers that provide (or serve) information to other computers (for example, web pages). A server can be used in different environments, such as: • Offices and schools where data or a printer needs to be shared amongst many people
Workstation  A workstation is a type of computer with a high-resolution graphics screen and usually has a mass storage device, such as a disk drive.	It is used to perform special tasks, such as: • Precision drawings • Computer-aided design (CAD), 3D graphics and software development • It can also be used in different fields, such as engineering or medicine research
Desktop  A desktop computer is not designed to be carried around as it is made up of separate components. These computers are designed to be set up in a permanent location on a desk or table.	Desktops are used in a variety of environments, such as: • Schools: It helps school learners to gain knowledge by doing research for projects or assignments. • Entertainment: You can watch videos, listen to music, or stream videos. • Communication: Users can use desktops to make phone calls, send emails, and use instant messaging just by having a proper internet connection.

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DESCRIPTION	EXAMPLES OF WHERE THIS TECHNOLOGY IS USED
Laptop  Laptops are also known as <i>notebooks</i> and are portable PCs that combine the display, keyboard, processor, memory, hard drive and cursor-positioning device, such as a touch- or trackpad, all in one package that is operated by a battery.	Laptops are best used for the office or at home. It can be used for basic administration tasks and daily office use. It can also be used for personal use, such as browsing the internet, listening to music, or watching videos.
Tablet  Tablets are smaller than normal laptops and are ultra portable. Although tablets are generally cheaper than brand-new laptops, their processors and other components are less powerful than regular laptops. Tablets have virtual keyboards, or make use of a digital pen instead of physical keyboards.	Tablets generally have one user. With this device, you can access information and data from anywhere in the world at any time. Tablets can be used in a variety of environments: • Home: Can be used for domestic use, such as listening to music, browsing the internet, listening to music and streaming videos. • Work: Can be used as a laptop replacement; however, you will need one that has at least a nine-inch screen. • School: Can be used to access study material, as well as browsing the internet, listening to music, or doing research for projects.
Smartphone  These handheld-sized computers use a flash memory instead of a hard drive for storage. It has virtual keyboards and uses touch-screen technology. Smartphones are lightweight and have a good battery life, depending on the make. You can access data and information as long as you have a good internet connection.	Smartphones are used in a variety of ways: • To browse the internet • To listen to music • To take photos • To navigate • To make phone calls and send and receive messages.



TYPES OF COMPUTERS



<https://www.youtube.com/watch?v=7qOF5fTDI8&list=PL0aNAktW5HLRZAyE8XkucMuGZpKKCLi88&index=3>

DEDICATED (EMBEDDED) DEVICES

Embedded devices are devices designed for a single purpose; whether that purpose is to wake you up in the morning, control the temperature of the air conditioning or help navigate. As such, these devices are usually excellent at their specific tasks, but they are also unable to do anything else. Embedded devices are designed to perform specific dedicated computing tasks. Examples of these devices are:

- ATM machines
- DVD players
- Anti-lock braking systems
- Digital watches
- MP3 players
- Drones
- Airbag control systems

1.5 Economic benefits of computers

Over the years, it became impossible to imagine a world without computers or computer-controlled devices, such as ATMs, cameras and music players.

Computers have several economic benefits, such as:

- **Saving paper:** Companies and individuals can save paper by working electronically on computers. People can now send letters (emails), magazines and read books electronically instead of having to print out hard copies. This also saves money and time in the long run. Estimates have shown that there can be a 10 to 30% decrease in paper usage if the correct technology is used.
- **Saving labour:** Businesses can save money and labour (people) by using computer-controlled devices to perform repetitive functions that require several employees to perform.
- **Communication speed and costs:** Telephone and postage costs can be decreased drastically using computers and the internet. It also allows information to be sent and received faster than in the past. For example, the use of email allows instant communication with staff, clients and other individuals at any place or time. The use of smartphones will help employers to keep in touch with their employees regularly, which ultimately saves time and phone calls.
- **Efficiency:** Instead of being swamped with paperwork or sorting out piles of paperwork, files can be stored on computers. Retrieval of old documents is much easier when it is stored electronically. Also, computers can do the work of more than one person in less time than normal employees.
- **Accuracy:** Computers produce more accurate data or results than human beings. This, however, depends on the quality of the input data. If the information entered is incorrect, the results produced will most likely be incorrect. Computers can, however, detect certain human errors and might highlight these errors.
- **Reliability:** Computers that are maintained properly will be reliable in the long run. It is safe to say that we can rely on computers to perform the tasks that they have been programmed to do.



Activity 1.4

Read the following scenario and answer the questions that follow.

SAVING PAPER

The largest bank in the USA, Bank of America, greatly reduced its paper consumption by making use of electronic reports and forms, emails, double-sided copying and by using lighter-weight papers. As a result, it saved tonnes of paper, decreasing the consumption by 32%. (This is over a billion sheets of paper!)

- Give two examples of how computers benefit the economy by saving paper. Motivate your answer.
- List three ways of how computers have benefited the economy in South Africa. Motivate your answer.



Case Study

Thembisile and her smartphone

To understand how dependent we are becoming on smartphones, read the following case study about a day in Thembisile's life.

This is an example of a smartphone user, Thembisile, and how she uses her phone throughout the day.

Every morning at 07:00, Thembisile is woken up by the alarm on her smartphone. This alarm clock application has a function that plays soft music for five minutes before she has to wake up. This actually prepares her brain to wake up. After that, the tone becomes louder, which she cannot ignore. If she hits the snooze button, her alarm is set to snooze five times for five minutes at a time.

After Thembisile has woken up, she checks her emails, WhatsApp messages, other texts and her social media updates, such as Facebook and Twitter, which were set to not send any messages to her between 21:00 and 07:00.

She then has breakfast, which she logs into her *meal plan* application. This indicates the number of calories she consumed. She also adds toothpaste and soap to her *shopping list* application.

Her calendar application shows that she is supposed to be at the gym in 30 minutes. She gets into her car and mounts her phone on the hands-free device attached to the windscreen of her car. She then switches her phone to GPS navigation, which works out the best route to her three calendar appointments, i.e. gym, work and her 13:00 meeting.

At the gym, she uses her *gym application* and inputs her weight, height and energy level into the application. This gives her an exercise routine she should focus on for the session.

She realises she is running a little late for work, so she sends a quick message to her manager using an instant message service (IMS), such as WhatsApp. Her normal keyboard is replaced by **SwiftKey**, making it much quicker for her to type.

At work, Thembisile uses a *note-taking* application to make notes of the things she is currently working on, as well as the *scheduling* application, which books and shares new appointments. She then uses **CamCard** to scan the new business card she just received so that the details are automatically saved in her contact list.

Before going to bed, Thembisile quickly looks at the day's important news events on her *news* application and plays some of the several games on her phone, such as Pokémon, Candy Crush and Angry Birds.

At 22:00, her *sleep* application tells her it is time to sleep.

As you can see, Thembisile, and probably her entire family, are very active smartphone users. Information is transferred, stored and conveyed in their everyday lives. This is the direction that the world of technology is moving towards. It also shows us how dependent we are becoming on computers, or computer-operated devices.

Answer the following questions about the case study that you have just read:

1. Give a brief description of a smartphone.
2. Identify four applications that Thembisile uses during the day.
3. From the case study, identify five things for which Thembisile uses her phone. How does this improve her day-to-day life?
4. How does Thembisile's employer benefit from using a smartphone?
5. How does the SwiftKey keyboard differ from the normal keyboard on a smartphone?

REVISION ACTIVITY

1. Study the picture of a computer below and answer the questions that follow:



- Identify and name two input devices. (2)
 - Is device A an input device or an output device? (1)
 - What is the function of device B? (2)
2. Match the user in Column A with the type of computer in Column B.

COLUMN A	COLUMN B
1. A weather scientist working for the South African Weather Services	A. Tablet
2. A sales representative who drives around visiting customers each day	B. Supercomputer
3. A school teacher preparing work at home	C. Server
	D. Desktop
	E. Laptop

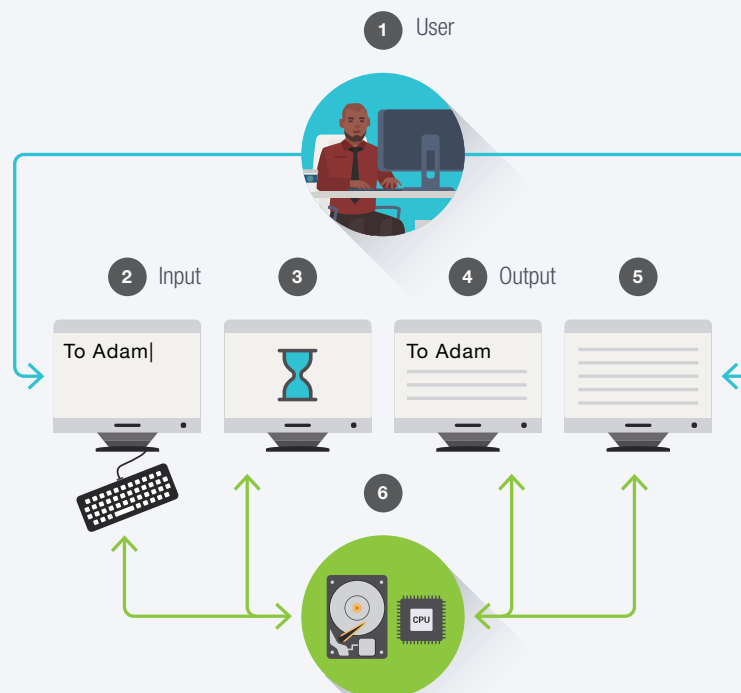
3. Your school has bought a new computer for the administration office. List two economic benefits of having a computer in the office. Explain how the school benefits in each case. (6)
4. a. Define what is meant by an embedded device. (2)
- b. Give an example of an embedded device that will be found in a motor car. (1)

... continued

REVISION ACTIVITY

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5. Look at the diagram below and answer the questions that follow:



- What stage of the information-processing cycle is taking place at 3? (1)
 - Does stage 6 represent storage or communication? Give a reason for your answer. (2)
6. Your class teacher ticks off each morning who is present at school. At the end of the term, she determines how many times each learner in the class was absent. This number is included in the learner's term report. Using this scenario, explain the difference between data and information. (4)

TOTAL: [24]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU...	YES	NO
1.	Understand ICT systems and how they are used in everyday life.		
2.	Distinguish between data and information.		
3.	Give examples of data and information used.		
4.	Identify the main concepts of a computer.		
5.	Understand the difference between hardware and software.		
6.	Describe the information-processing cycle with reference to how it uses hardware devices in each stage.		
7.	Describe the different types of computers and their uses.		
8.	Understand the economic reasons for using computers.		

USING A COMPUTER

CHAPTER OVERVIEW

Unit 2.1 Operating a computer

Unit 2.2 Posture and ergonomics

At the end of this chapter, you should be able to:

- Switch a computer on.
- Use a mouse correctly.
- Identify and explain the different icons when using a mouse.
- Understand the different features of the desktop.
- Use your desktop.
- Explain the importance of the correct posture when working with a computer.

INTRODUCTION

As a modern-day learner, you have probably used, seen, or read about computers. We use computers in our everyday lives, for example, in banks, schools, shops, hospitals, and maybe the people in your home use computers on a daily basis. Being able to use a computer is an important skill to learn; not only for school or university purposes, but also for the workplace.

In this chapter, we will introduce you to the practical side of using computers.

2.1 Operating a computer

Something to know

Note that login procedures may differ with different systems.

Something to know

A computer uses your username and password to verify your identity. This prevents unauthorised access to your computer and the information on the computer. It also protects you from anyone trying to invade your privacy, or trying to delete or copy your information.

Something to know

Remember to make sure that the mouse is on a flat surface with the buttons pointing forward.

Operating a computer may not be as difficult as you think. In this unit, we will focus on:

- Switching a computer on.
- Using the mouse correctly.
- Using your desktop.
- Applying basic file operations, such as opening, saving and printing a file.

SWITCHING THE COMPUTER ON



Guided Activity 2.1

To start up your computer, you need to do the following:

1. Make sure that the computer has been plugged into a power socket.
2. To switch the computer on, press the *Power* button. On some computers, this button lights up after the computer is switched on.
3. The computer will now **boot**. This is when the computer loads the operating system and checks that all the components are in good order.
4. After the computer has booted, a **log on** screen might appear. Logging on allows registered users access to the computer. If you are the only user of the computer, it will go straight to the desktop screen once it is turned on.
5. If you share the computer with other users, you might have to log in your details. You can do this by typing in your **username** and **password**, and then pressing the *Enter* key. After the computer accepts the login details, the desktop display will appear on the screen.

USING THE MOUSE

The mouse is one of the main hardware devices used to control the computer and interact with the GUI interface. Most mice are designed to have a scroller and two buttons – i.e. a left button and a right button, as seen in Figure 2.1.

When holding the mouse, rest your hand over it and place your index finger on the left button of the mouse. You can rest your thumb and pinkie (little finger) on the sides of the mouse. An example of how to do this is shown in Figure 2.2.



Figure 2.1: Mouse



Figure 2.2: How to hold a mouse

By **default**, a mouse is set for a right-handed person with the primary button on the left. You can change the function of the mouse buttons from the default left click to a right click.

However, most left-handed people either use the mouse in their right hand, or they place the mouse on the left-hand side of the keyboard and use their middle finger for the left button.

To explain how the mouse works, we will be using the left button of the mouse. The following table shows the basic things you need to know about a mouse.

Table 2.1: *Basics of the mouse*

ACTION	IMAGE
When moving the mouse, you will see that you are moving a pointer or cursor on the screen.	
Depending on what you are doing, the shape of the cursor will change.	
With the hand icon, you can open a link on the internet.	
This icon shows that you must wait while the computer is doing a task.	

A mouse can do a variety of tasks, such as:

- **Selecting an item:** To select something, move the cursor over the item, click once with the left button and then you can let go.
- **Opening an item:** To open something like a file, double click the left button. This is usually how it works, but sometimes a single click will also open a program. Through practice, you will learn when to use which one.
- **Moving items:** To move an item, you must drag and drop. You do this by pressing down on the left button above the item, as you drag the mouse along. Once you have placed the item where you want it, you can let go of the left button.
- **Scrolling up and down a page:** This is done by rolling the scroll- or mouse wheel up or down.

USING THE KEYBOARD

A keyboard is used for entering information, such as letters, words, numbers and symbols into a computer. Pressing any key on the keyboard is an input that the computer uses as data.

The basic keyboard, as seen in Figure 2.3, is called the QWERTY keyboard.

A keyboard is used for entering information, such as letters, words, numbers and symbols. Pressing any key on the keyboard is an input that the computer uses as data.



Figure 2.3: *QWERTY keyboard*

USING THE DESKTOP

The desktop screen is the first thing you see when you switch the computer on. It has a logical display surface and contains user-interface objects, such as windows, menus and buttons. Its main features are the start button, **taskbar** and icons. For you to use the desktop, each computer has an operating system (OS). An operating system is the system software that allows a user to communicate with a computer in the following ways:

- It manages a computer's hardware, such as the input and output devices, network devices, as well as storage devices.
- It provides services to facilitate the operation and management of additional software application programs.
- It assigns the necessary memory.
- It provides the **graphical user interface (GUI)**, pronounced as “gooey”, which allows a user to interact with and control software applications, or hardware devices using **graphical interface elements**, such as **windows**, **icons** and **menus**. This means that a program has graphical controls that a user can select using a mouse or keyboard.

The operating system has the following user interfaces:

- **Desktop:** This is the main screen of the GUI and consists of a system of icons on a screen.
- **Window:** Used to display the contents of an application with which the user can view and interact. It shows the information of each file, application, or folder in a separate window on the screen.
- **Dialogue boxes/message windows:** These are small or basic windows that are usually opened by a program to indicate to the user that more information is required.
- **Toolbars:** These are a set of icons at the top of a window, which can be clicked to perform certain functions.
- **List box:** This allows the user to select one or more items from a list. The user must click the box next to the item to select it.
- **Checkbox:** This is a small square box on the screen, which allows the user to click it to indicate an answer, or to allow a setting.
- **Windows Explorer:** Windows Explorer is a GUI component available in Microsoft Windows that enables users to access, edit, copy, delete and manage data, files and other content stored on a computer, or mobile device.

Figure 2.4 is an example of an operating system interface.

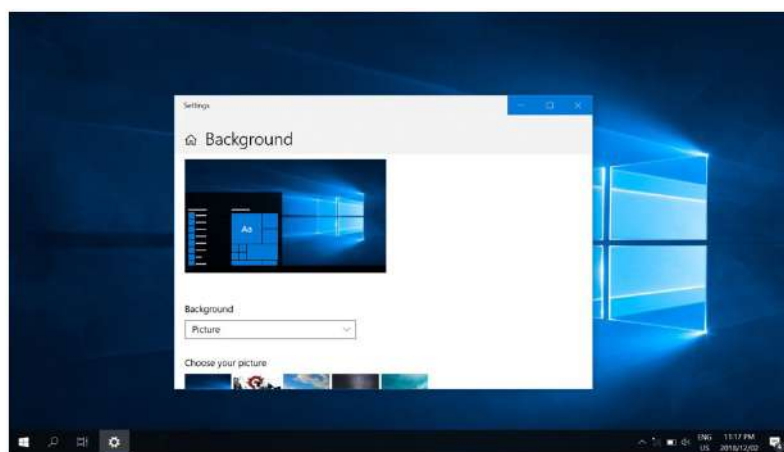


Figure 2.4: Example of an operating system interface

Something to know

Whenever you are working on, or interacting with a computer, you are called the *user*. And the way in which the computer interacts with you, is called an *interface*.

Something to know

With the new versions of Microsoft Windows, *Windows Explorer* is the default file manager, while *Finder* is the default file manager on Apple computers.

Figure 2.5 shows examples of some of the interfaces of an operating system:

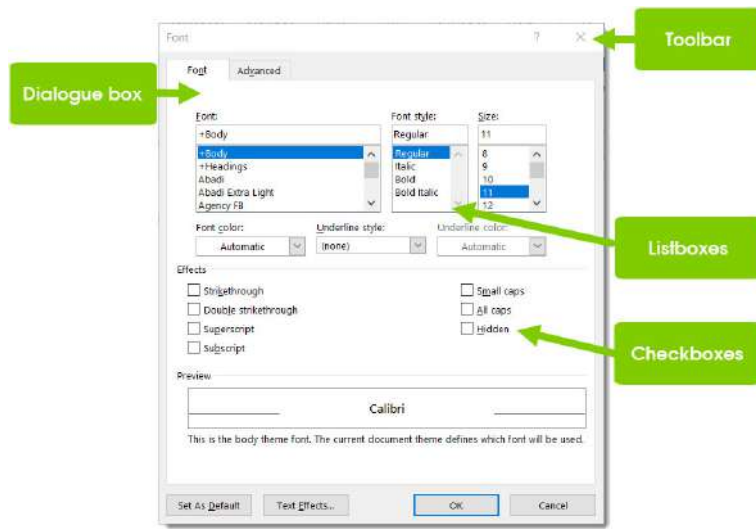


Figure 2.5: Different interfaces of an operating system

As most of you will start your computing experience with an operating system called Microsoft Windows, we will use Windows throughout the course.

With the desktop, you can use **shortcuts** to find folders, files, or programs quickly. You can identify a shortcut by checking if there is a curving arrow on the bottom left-hand side of the icon, as seen in Figure 2.6.



Figure 2.6: Example of a shortcut

Let's look in more detail at the desktop and icons that you can find on a computer:

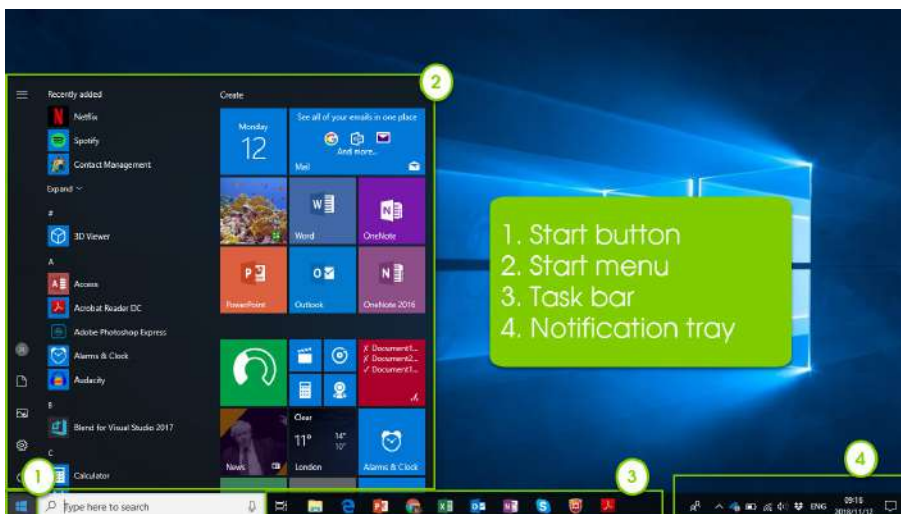


Figure 2.7: Desktop components



Something to know

An icon is a picture that shows a file, folder, or a program. Each icon is different for each type of file or program.

THE START BUTTON

The *Start* button can be found at the bottom left-hand corner of the desktop. It allows you to open programs, files, or folders on the computer. The *Start* menu appears when you click on the *Start* button. From there, you can select the application, folder, or file you want to open by clicking on it.

THE TASKBAR

The taskbar can be found at the bottom of the screen, as seen in Figure 2.7. It shows a button for each open folder, file, or application. When you click on any of the buttons on the taskbar, it will open the file, folder, or application, and therefore, becomes an active window.

The taskbar also has the *Start* button on the left-hand side and the **notification tray** on the right-hand side. You can perform certain functions, such as change the time, volume, or connect to the **Wi-Fi** by selecting the icons in the notification tray.



Activity 2.1

- The following steps are performed when starting up, using and shutting down a computer. The steps are in the incorrect order, however. Arrange the steps in the correct order.
 - Enter the password
 - Switch on the monitor
 - Session begins
 - Switch on the power button
 - Click on Log out
 - Enter the username
 - Computer session ends
 - Switch off the monitor and power
 - Shut down computer
- State whether the following are True or False. If false, give the correct answer.
 - A password should always be secret.
 - The keyboard is the most important hardware device used to control the computer.
 - The hour-glass icon shows that you must wait while the computer is doing a task.
- Match the terms in Column A with the correct description in Column B:

COLUMN A	COLUMN B
1. Desktop	A. This is the main screen of the GUI and consists of a system of icons on a screen
2. Shortcut	B. This is when the computer loads the operating system and checks that all the components are in good order
3. Boot	C. The computer identifies the user with this name
4. Username	D. You can use this to find files, programs, or files quickly
5. Icon	E. Allows you to access the time and volume icons on the computer
	F. This is a picture that shows a file, folder, or a program
	G. It allows you to open programs, files, or folders on the computer

... continued



Activity 2.1

... continued

4. Logging off and then switching off the computer when you are done using it, is actually quite important. Answer the following questions regarding this statement:
 - a. Give two reasons why switching off the computer after use is important.
 - b. Does logging off switch off the computer? Motivate your answer.
 - c. Briefly explain why it takes time for a computer to start up and shut down.

THIS PC, FILE MANAGER, MY DOCUMENTS AND RECYCLE BIN

After you click on the *This PC*, *My Documents*, or the *Recycle Bin* icon, a folder will open in a window on the desktop.

- **This PC:** When you open the *This PC* folder, it will give you information about the different storage disks, external hard drives, or hard disks on the computer.

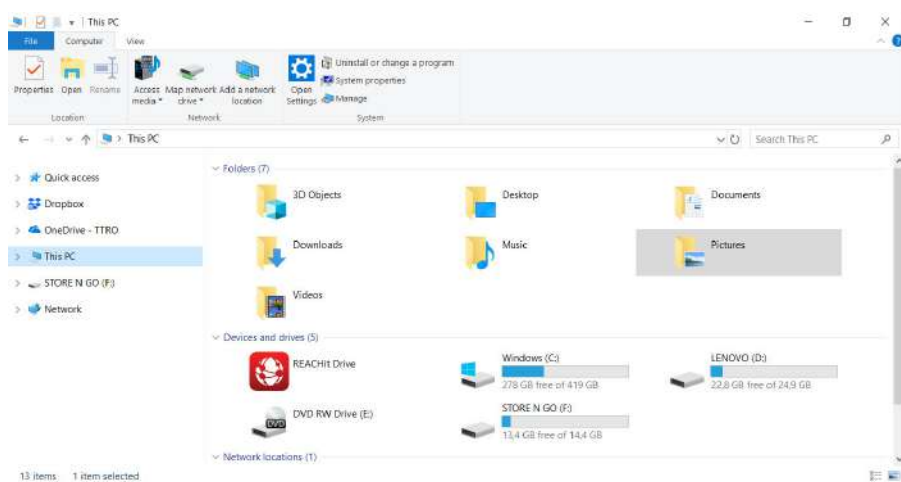


Figure 2.8: *This PC* folder

- **File manager:** A file manager is an application that helps manage the file system on a computer.
- **My Documents:** When you open this folder by double clicking on the icon, you can find your documents, music, pictures and other files. These files are stored in the *My Documents* folder on the computer's hard disk.
- **Recycle Bin:** When you delete items, such as folders or files from the computer, they are stored temporarily in the *Recycle Bin*. If you deleted a file by mistake, it can be restored to its original folder from the *Recycle Bin*. The files or folders in the *Recycle Bin* can also be permanently deleted by deleting them.

ACCESSING APPLICATIONS

You can access the different applications on the computer by using the *Start* menu. Click on the *Start* button and then scroll through the alphabetical list. If you want to save time, you can limit the scrolling by selecting any letter from the alphabetical list. From there, you can choose the letter that the application begins with.

If you are still having difficulty finding the application that you are looking for, type in the name of the application in the search box on the taskbar.



Something to know

In older versions of Windows, *This PC* is known as *Computer*.

GETTING TO KNOW WINDOWS

Zama needs to use Microsoft Word for a small project. Let's look at an example of how Zama uses Microsoft Word.



Guided Activity 2.2

1. To open Word, she clicks on the *Start* button and then moves the mouse pointer onto the Word icon and clicks on it, as seen in Figure 2.9.



Figure 2.9: Opening the Word application from the Start menu

2. Word will open in a separate window and the Word icon will appear on the taskbar.

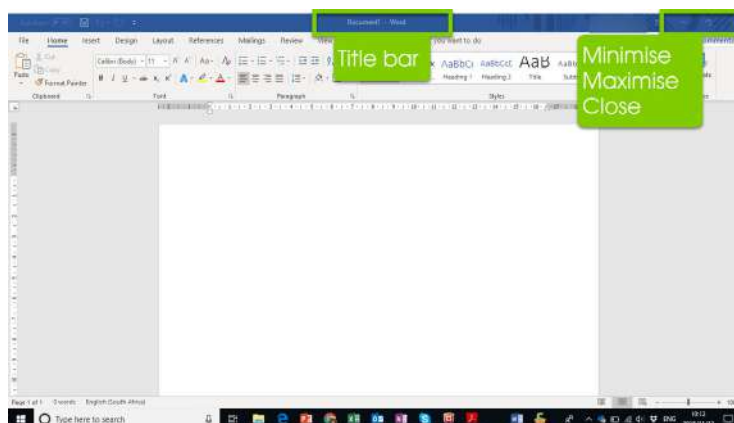


Figure 2.10: Word window

SOME THINGS TO NOTE ABOUT WINDOWS

The following section will look at the different features of windows, using the Word window as an example.

The title bar is found at the top of the window and the name of the open file can be seen on the title bar. Since the document is not named yet, the title bar will be *Document1-Word*.



Figure 2.11: The title bar

To minimise the taskbar, click on the *Minimise* button on the top-right corner of the window, as seen below.



Figure 2.12: The *Minimise* button

To enlarge the window or make it bigger, click on the *Maximise* button, as shown below. This can also be done by dragging the window from the title bar to the top of the screen.



Figure 2.13: The *Maximise* button

To close a window, click on the *Close* button (which is the x in the top-right corner of the window), as shown below.



Figure 2.14: The *Close* button

FILE MANAGEMENT

OPENING A DOCUMENT AND SAVING DATA

Now we must open the document and save the data we need on Word. Let us see how she does this.



Guided Activity 2.3

Do the following with the help of your teacher.

1. Open Word.
2. Type two short lines of the letter “a”, just so that you have typed something. (To begin a new line, press *Enter* on the keyboard.)
3. The text will always appear on the left of the cursor when you type.
4. Now that you have typed the two lines, you can save the file. To save the file, select *FILE*, then *SAVE* from the menu bar. When the *Save As* dialogue box appears, save the file with the file name ‘Test’ in the *File name* field in the dialogue box.
5. Click the *Save* button, so that the file can be saved and then close the dialogue box.
6. In the Word document, the name of the file will now be ‘Test’. Close the window by clicking on the *Close* button on the title bar.

You can follow this example in the video on the right.

When you are done using the computer, you can switch it off by first clicking on the *Start* button, then on the *Power* button where you choose *Shut down* to shut down the computer.



Video

Create a video showing the steps on how to open and save data in a Word document. These steps are explained on the left (Steps 1 to 6).



Something to know

Remember, shutting down a computer might differ from computer to computer!

OPENING A FILE

In the previous example, you saved a document with the file name 'Test'. Now let's learn how to open the saved file.



Guided Activity 2.4

Let's do the following:

1. Open Word. To open the required file, select *File* then *Open* from the menu bar.
2. When the *Open* dialogue box opens, select *Test.docx* from the list box.
3. Click the *Open* button, as highlighted in green in Figure 2.15, which then opens the document in Word.

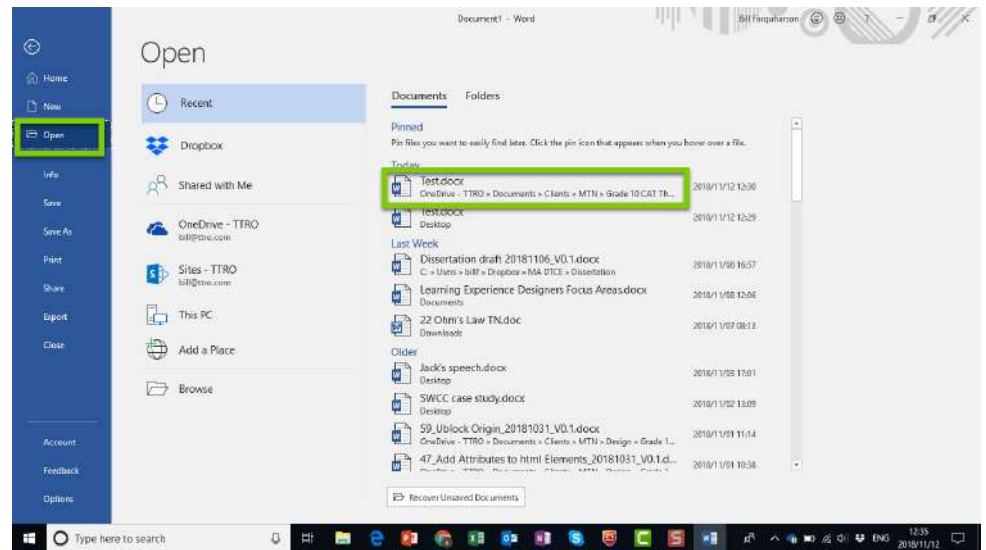


Figure 2.15: Open dialogue box

PRINTING A DOCUMENT

You will learn how to print documents when you study word processing. Refer to Chapter 1: Introduction to Word Processing in the Practical Book for more information.



Activity 2.2

Do the following at your computer:

1. Switch on the computer.
2. Find Word using the search box in the taskbar.
3. Use the mouse to open the Word application.
4. Type two to three sentences about why CAT, as a subject, is important.
5. Save the file and then close it.
6. Open the file that you just saved in (5).
7. Print Preview the document.
8. Close the application.

2.2 Posture and ergonomics

As computing takes over our lives more and more, typing is one of the most important and necessary skills that you will ever learn.

Touch typing is all about making sure that each finger has its own place on the keyboard. Because of that, you do not have to look down on the keyboard while typing, which makes your typing speed much faster.

Let's look at the following benefits of being able to touch type:

- **Speed:** This is the most obvious advantage of touch typing. On average, you can easily reach typing speeds of 75 to 80 words per minute. Comparing that to the average of ten words per minute of someone who does not touch type, you can see that it is much faster.
- **Time:** If you increase your typing time, you obviously decrease the time it will take you to complete a document. A person who types with two fingers at ten words per minute, will take 17 minutes to type a 170-word document. Typing at 60 words per minute will do the same document in four minutes.
- **Decrease fatigue:** Typing is very tiring when it is done over long periods of time. Learning to touch type reduces this exhaustion, as you do not have to focus on two things at a time. It also prevents you from looking up and down the whole time, placing less strain on your eyes, neck and spine.
- **Health:** Touch typing is better for your health, because you do not hunch over and look at the keys while using your fingers. This helps to reduce the strain on your joints.
- **Accuracy:** The better you can touch type, the less typing errors you will make, because you are looking at what you type, and not having to look down and find the right keys.

SOCIAL IMPLICATIONS: CORRECT TYPING POSTURE

Posture is defined as the position in which a person holds his or her body when sitting or walking. Sitting in front of a computer for long periods of time, even with a good posture, can lead to tiredness or eye discomfort.

Having the correct posture also helps you to type correctly and more efficiently.

A tip for having a good posture when using the computer, is to position yourself so that your belly button is in line with the letter “H” on the keyboard. You can see an example of this in Figure 2.16.

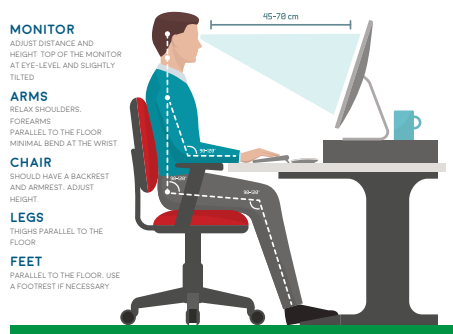


Figure 2.16: *Correct body posture*

The following are elements for maintaining a good posture:

- **Body:** Your back should be straight, upright and relaxed, but comfortable. For your back to be supported, it should be aligned with the chair's backrest. Do not lean too far forward as this will cause discomfort later.
- **Feet:** For your feet to feel comfortable, make sure that they are flat on the floor, or on a foot rest.
- **Knees and hips:** Push your hips as far back as they can go in the chair. As seen in Figure 2.16, your knees and hips should form a 90° angle when sitting on the chair. Also, make sure your knees do not touch the chair.
- **Arms, wrists, fingers and hands:** Your arms should be placed horizontally in a 90° angle, your wrists should be in a neutral position and your fingers should be placed on the home row.
- **Eyes:** Your eyes should focus on the text that you are typing on the screen. To position your screen, make sure it is slightly lower than your eye level.

Having the wrong posture when sitting in front of a computer, can cause various health problems, such as:

- **Posture-related injuries:** Posture-related injuries include back and neck pain, headaches, and shoulder and arm pain. Poor posture can affect your spine alignment, causing problems with blood vessels, and ultimately leading to blood clots and varicose veins. When your body is slumped over for long periods of time, you are more likely to clench your jaw and tighten your facial muscles. This can lead to jaw pain and headaches.
- **Computer-related overuse injuries:** The muscles and tendons in the elbows, wrists and hands can become painful with repetitive movement and bad posture. Symptoms include pain, swelling, stiffness of the joints, weakness and numbness.
- **Eye strain:** Eye strain is caused by focusing the eyes at the same distant point for long periods of time, working with the screen too close to the eyes and looking at the illuminated computer screen for too long.
- **Laptop-related injuries:** Laptops were never meant to be used for long periods of time. The fact that the screen and the keyboard are very close together, it can cause eye strain, as well as a bad posture. Even carrying the laptop around can put strain on your muscles and joints.

ERGONOMICS

Ergonomics is the study of the way in which humans work with the goal of increasing productivity, comfort and safety.

When speaking of computer ergonomics, the focus is mainly on the way in which people use computers, and how this can be improved to make people more comfortable and reduce discomfort and injuries.

Key ergonomic guidelines for safe computer usage include:

- Sit up straight with your back perpendicular to the ground.
- Your forearms should be at the same height as your mouse and keyboard.
- Your feet should be placed firmly on the ground or on a foot rest.

- The back and height of your chair, as well as the height of your armrests should be adjusted to support your body in this position.
- The monitor should be positioned at eye level and roughly 50 cm away from you. You may need to place something under the monitor to increase its height.
- The monitor should be tilted upwards to reduce glare.
- You should stand up and take regular breaks.

PREVENTION

Prevention is always better than cure. Many people suffer from posture-related injuries. Health problems related to poor posture and ergonomics can be prevented in different ways.

To prevent a repetitive strain injury, you should:

- Make sure your forearms are level with your mouse and keyboard.
- Make sure the mouse is positioned close to the keyboard.
- Take regular breaks and let your arms relax when you are not typing.

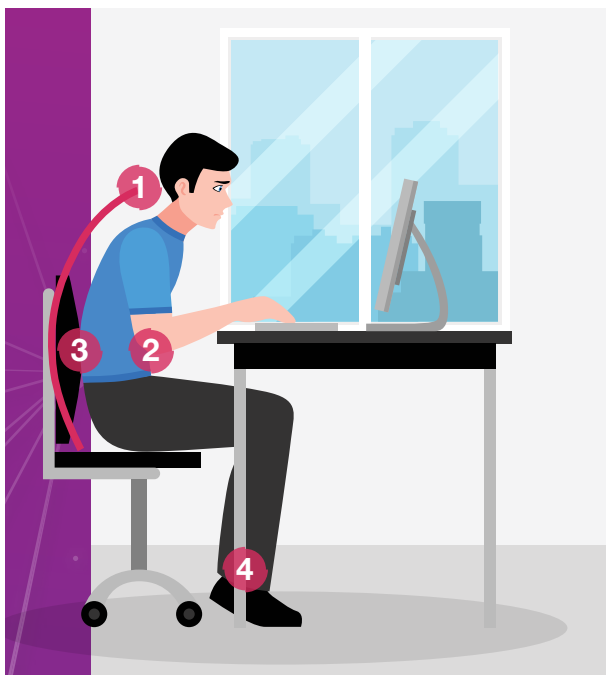
To avoid eye strain:

- Make sure that there is no glare on the monitor.
- Do not sit too close to the monitor.
- Make sure that the monitor is placed at eye level (or slightly lower).
- Take regular breaks to focus on far-away objects.



Activity 2.3

Use the image below to answer the questions that follow.



1. Is this person using good posture?
2. Motivate your answer by using the numbers in the diagram to guide you.
3. Explain why a good chair is important to assist you in maintaining a good posture.

REVISION ACTIVITY

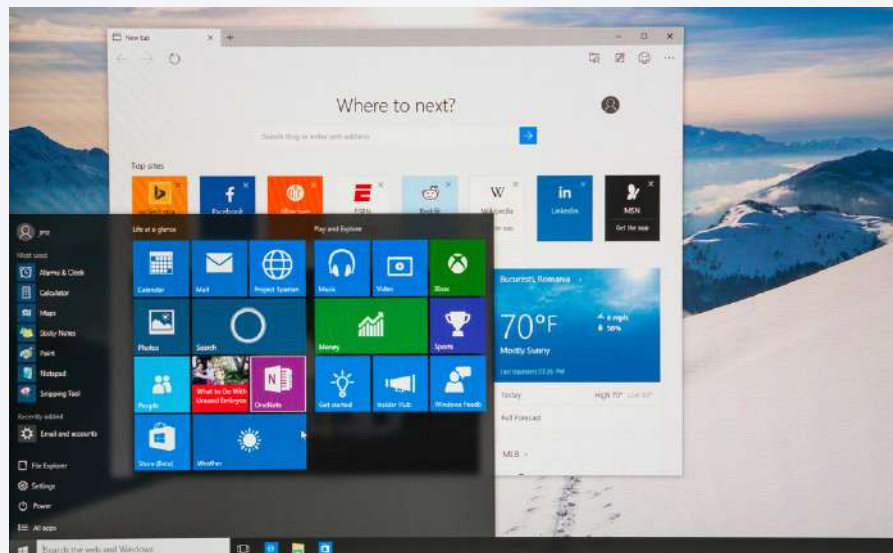
- When you are working on a computer that is used by more than one person, you will often be asked to enter a user name and password soon after the computer has been switched on. Give two reasons why you should never share your password with anybody else. (2)
- Match the mouse action in Column B with the desired outcome in Column A. Write down only the number from Column A and the matching letter from Column B.

COLUMN A	COLUMN B
1. Scroll up and down a web page	A. Click the left mouse button once
2. Select an icon on the desktop	B. Double-click the left mouse button
3. Open an application using a desktop icon	C. Click and hold the left mouse button while moving the mouse
4. Move an icon from one side of the desktop to another	D. Roll the scroll button of the mouse backwards and forwards

- Answer the following questions about a GUI. (4)
 - What does the acronym GUI stand for? (1)
 - What is the main screen of the Windows 10 GUI called? (1)
 - What is the function of an icon? (2)
- What does the arrow in the bottom left corner of the icon below indicate? (1)



- Study the image of a Windows 10 desktop below



- Label the parts of the desktop labelled A, B, C and D. (4)

... continued

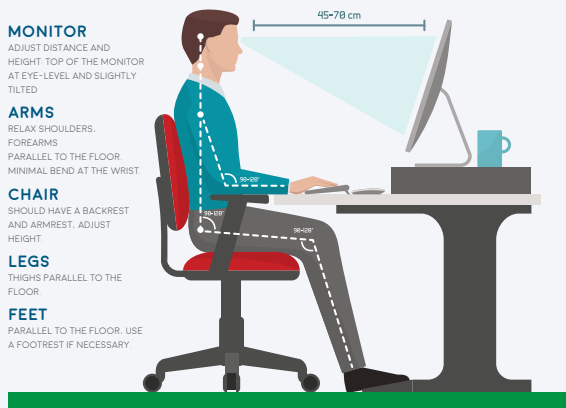
REVISION ACTIVITY

... continued

6. Study the icon below and answer the questions that follow:



- 6.1 What application can be accessed by double-clicking the above icon? (1)
- 6.2 Explain to someone who is NOT familiar with Windows 10 how you would use the application. (3)
7. Explain the difference between *saving* and *closing* a file. (2)
8. Give four tips to avoid eye strain when using a computer monitor. (4)
9. Use the image below to guide learners about having good posture when using a computer.



- 9.1 Mention at least four points.

(4)

TOTAL: [29]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Learn how to switch on a computer.		
2.	Learn how to use a mouse correctly.		
3.	Identify the different icons and what it means when using a mouse.		
4.	Use your desktop.		
5.	Learn about the operating system and the different components of the OS.		
6.	Learn about the GUI interfaces.		
7.	Understand the importance of using the correct posture when using the computer.		
8.	Understand the importance of ergonomics.		

FILE MANAGEMENT

CHAPTER OVERVIEW

Unit 3.1 Basic concepts of file management

Unit 3.2 File types

At the end of this chapter, you should be able to:

- Understand the basic concepts used in file management.
- Organise files and folders on the computer.
- Identify the different file types.



Take note

Before you can start with this chapter, you must have worked through the chapter on how to use a computer.

INTRODUCTION

Having an effective computer filing system can make a huge difference in your everyday life. Firstly, it helps to save time, but most importantly, it helps you to be more organised.

In this chapter, we look at what file organisation is, how we can achieve file organisation and the different types of files you can find on your computer. You will also learn about the importance of file management, and how you can organise, delete, restore, move, view and sort files and folders. Furthermore, you will learn about the basic accessories that Windows comes with, as well as how to install a printer.

3.1 Basic concepts of file management

File management on a computer is similar to filing documents in a filing cabinet. A filing cabinet is used to store paper files in cardboard folders. In the same way, we can store files and folders on a computer.

Figure 3.1 shows the *This PC* folder with the sub-folder *Music*. This folder contains sub-folders that are organised according to the name of the artist or group. This is to give you an idea of how file management looks on a computer. In Windows, folders are used to organise files.

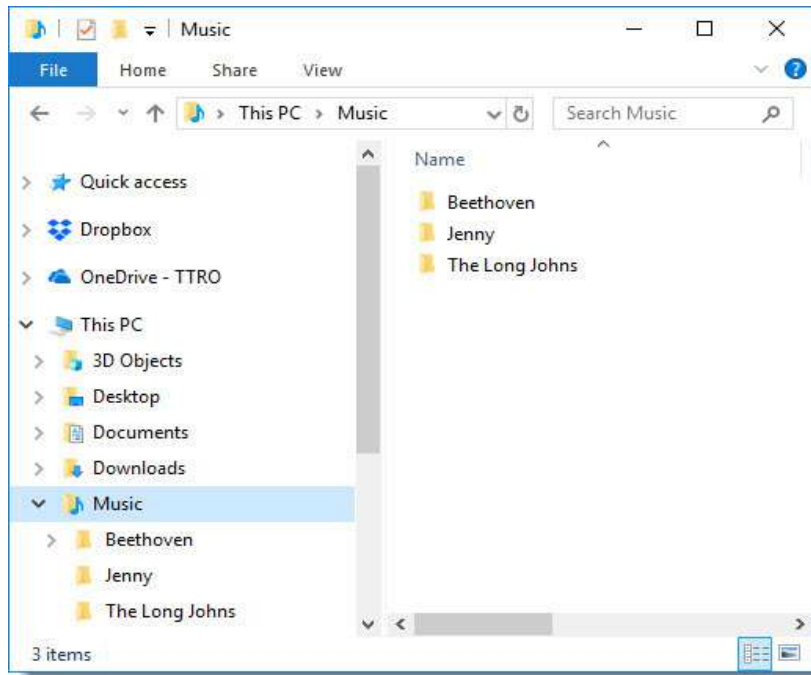


Figure 3.1: Folders used for file management

Folders are stored on a drive in a computer. Data is stored on and retrieved from a **disk drive**. Each drive will indicate the type of storage medium, for example, whether it is a hard disk, USB drive, CD, DVD, or so on. Figure 3.2 shows an example of the different drives that can be found on a computer.

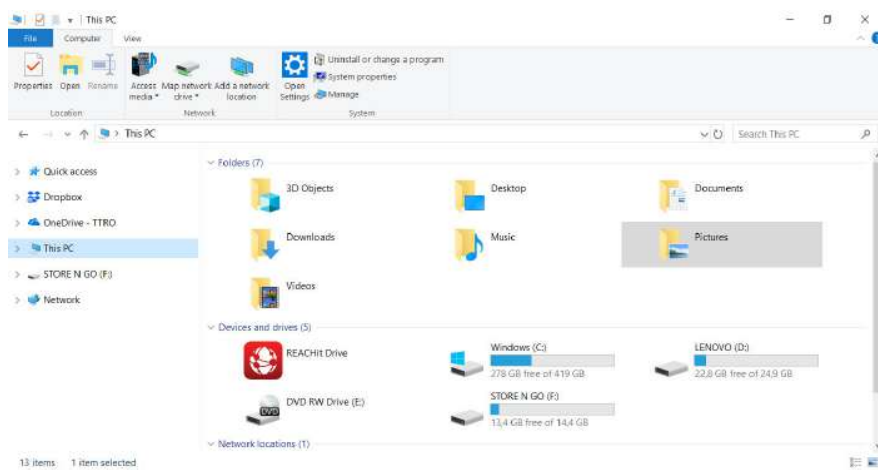


Figure 3.2: Different storage media on a computer

The capital letters and colon indicate what type of disk it is. For example, from Figure 3.2, you can see the following:

- The C: Which is also known as the C-drive, usually refers to the **hard disk drive (HDD)**.
- The D: Which is also known as the D-drive, usually refers to a CD or DVD drive.
- Other letters are also used to indicate storage media, for example, in this case **F:**, which refers to a removable disk (USB). Storage media, such as DVDs, USBs and hard drives, do not save data to the disks on the computer.

FOLDERS

A Windows operating system organises its drives, folders and files in a hierarchical folder structure. Files are stored on a computer inside folders. Folders are used to organise a computer's files so that it can be found easier. A folder will contain sub-folders and then files.

In the following example, we will explain a folder by comparing it to a tree. The structure is the shape of a pyramid, where each row of items is linked to the items beneath it. Because of this pyramidal structure, this hierarchical structure is also known as an “inverted tree”.

Figure 3.3 shows an example of an inverted tree.

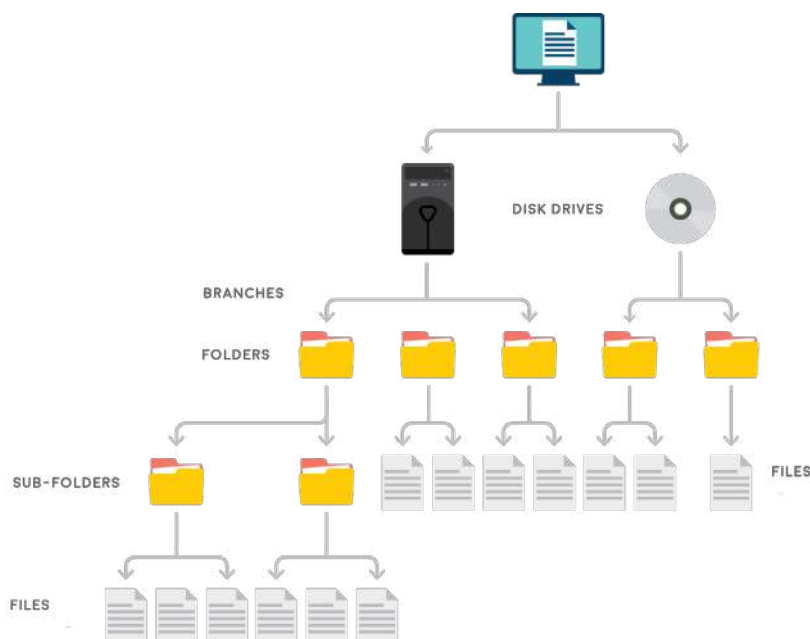


Figure 3.3: Hierarchical tree

The different drives, such as the different hard drives, CD/DVD drive, USB drives, as well as network drives, are found in the row below the root. Folders linked to the specific drives are then shown in the next row.

Any sub-folders and files found in the folder above are shown in the next row. This pattern continues until the final row only contains files.

The highest level of the tree structure of a drive is, therefore, called the **root directory** and is stored on the hard-disk drive of a computer. The documents then form a sub-folder of the root directory C-drive, which is written as C:\.

FILE PATH

A **file path** shows the location of where a file, web page, or other item is located on the computer. The file path shows you to which storage device the file is saved (i.e. the drive letter), in which folder and sub-folders the file is saved, the name of the file and finally, the type of file (given by the **file extension**).

For example, Jane has a music folder where she keeps her songs. Figure 3.4 shows an example of this file path.

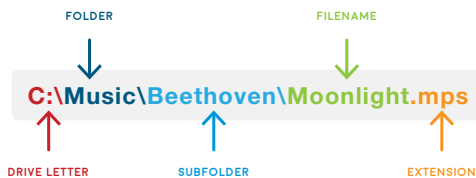


Figure 3.4: Example of a file path

As Figure 3.4 shows, each file's unique file path starts with the storage device drive letter. After the drive letter, you will see the name of the file's folders and sub-folders. A back slash (\) is used to separate the drive letter, and the different folders and files. Finally, the file path ends with the name and extension of the file.

Windows file names have two parts; the first part is the file name and then a period followed by an extension. A file extension or file name extension is found at the end of a file name and shows what type of file it is in Microsoft Windows. The extension is a three- or four-letter abbreviation that identifies the type of file. For example, in the file name, Moonlight.mp3, the extension is mp3.

Later in this chapter, you will learn more about the different file extensions.

FINDING A FILE PATH

To find the file path of a folder, click on the address bar in File explorer.



Guided Activity 3.1

You can do this as follows:

1. Open the *Computer* window from the *Start* menu.
2. Browse to the folder for which you want the file path.
3. Click on the address bar to see the folder's address.

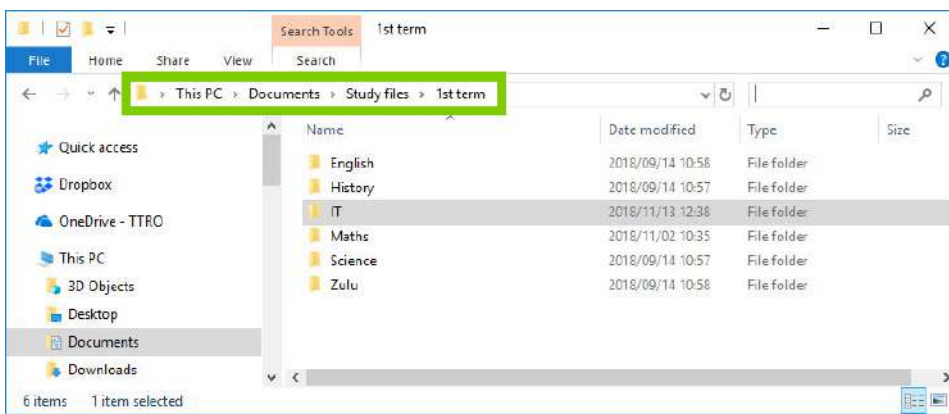


Figure 3.5: Address bar



Something to know

When using file paths in Windows, you must always use back slashes (and NOT forward slashes). Windows uses back slashes for paths, although everything else seems to use forward slashes.



FILE MANAGEMENT



https://www.youtube.com/watch?v=k-EID5_2D9U



Guided Activity 3.2

If you want to know what a file's unique file path is, you can find it using the *Properties* window.

1. Open the computer window from the *Start* menu.
2. Look for the file you want the file path for.
3. Right click on the file and select the *Properties* option.
4. Click on the *Security* tab.
5. The file path is given at the top of the security tab as the *Object* name.

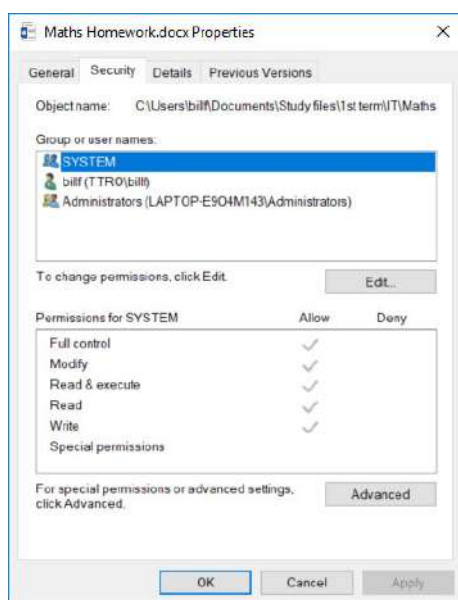


Figure 3.6: Reading a file path from the properties window



Activity 3.1

1. Define the following terms:
 - a. File path
 - b. Disk drive
 - c. Folders
2. Answer the following questions based on the file path below:

C:\Documents\School\Maths.docx

- a. What is shown in the file path above?
 - b. To what does "C:\" refer?
 - c. A file name consists of two elements. What are they in this diagram?
3. Using the figure below, provide the file path to get to the file: *Communication.pdf*.

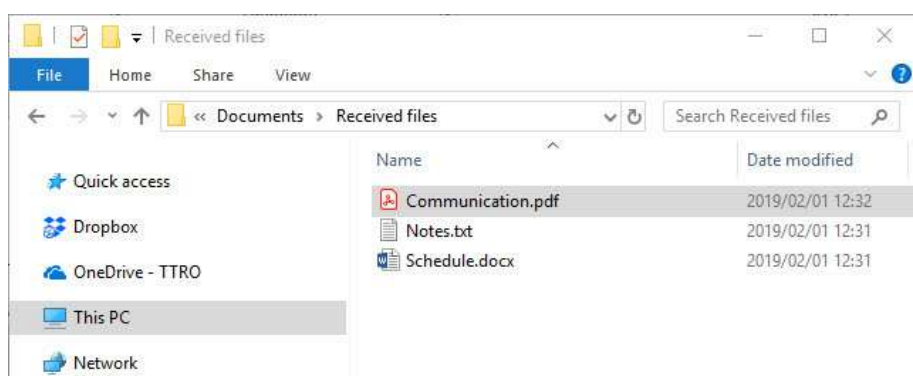


Figure 3.7: Determining the file path for the file, *Communication.pdf*

ORGANISING FILES AND FOLDERS

Organising files on your computer is an important task, as it can ultimately save you lots of time and effort! For you to find files and folders on the computer, it must be organised properly. It is, therefore, important for you to place files in folders and give them the correct names.

In Microsoft Windows, this can be done in different ways. This section will look at the easiest ways in which files can be copied or moved between folders, renamed and deleted. You will also learn about how to search for files on a computer.

Before you can start organising your computer, you first need to see how the computer is currently organised.

USING A FILE MANAGER (THIS PC)

A file manager is a built-in application that is used to organise files and folders. To open the file manager, double click on the *This PC* icon. The *This PC* folder will open up in a new window. From here, you can start working with the files and folders.



Guided Activity 3.3

Practise using the file manager on your computer by following the steps below:

1. Click on the *Start* menu button in the bottom-left corner of the screen.
2. Once the *Start* menu opens, type in the word “explorer”. You should see a *File explorer* option appear in the *Start* menu.
3. Click on the *File explorer* option, which will open the file explorer on the computer.
4. Once opened, click on the *This PC* option in the left panel. Here, it will show you the main folders. From here, you can explore the computer's organisational system by looking through the folder structure.

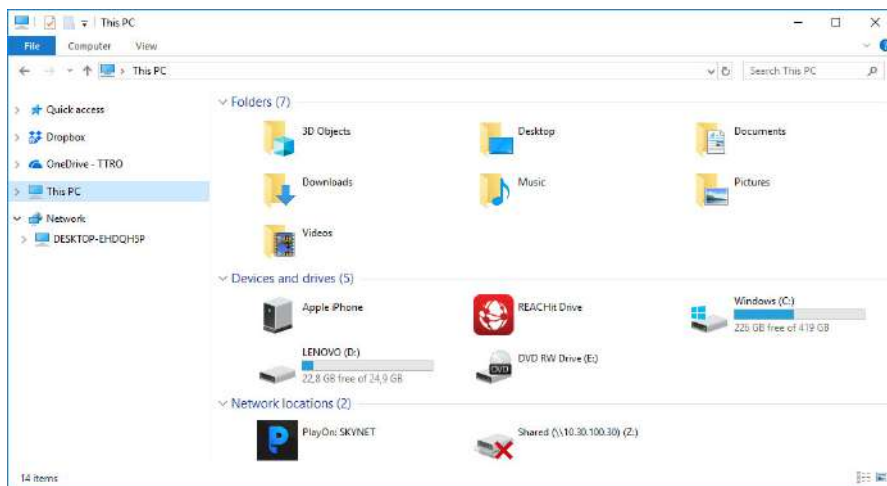


Figure 3.8: Checking how the computer is currently organised

In the *This PC* screen, the storage devices are indicated by grey and black icons, and include the drive letter in their name. The figure shows that there are four storage devices – i.e. Computer (C:), Data (D:), USB Drive (F:) and Recovery (E:). By double-clicking on any of these storage devices, it will open, and you can browse the files and folders stored in it. If you want to open any files or folders, you should double-click on them.

After you have examined the different files and folders on the computer, you can start organising them.

As you have learned, files are stored on a computer inside directories (called folders in Microsoft Windows). These folders are used to help organise a computer's files so that files can be found more easily. Each folder can contain any number of files or sub-folders. For example, a music listener might create a *Music* folder with all her music files in it.

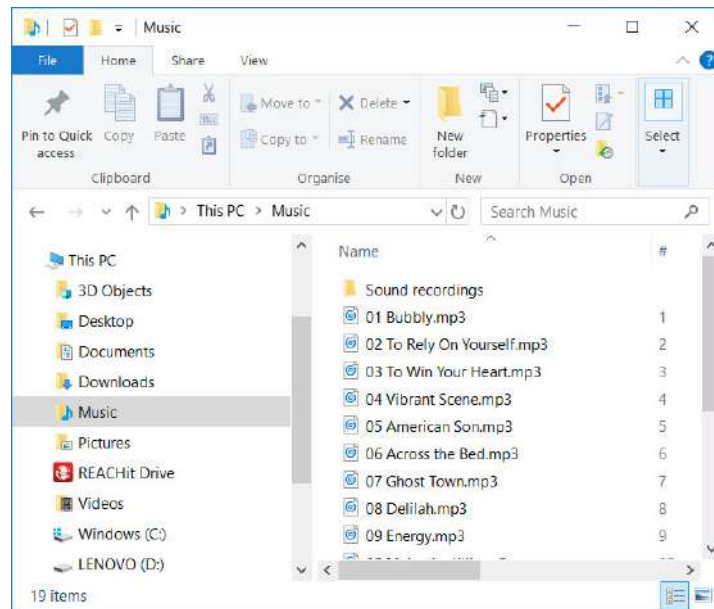


Figure 3.9: A folder containing a number of music files

A more organised music listener might have a *Music* folder as the first folder, and then sub-folders for each artist. It might even have a sub-folder for each style of music and then have folders inside of that for each artist. This would make it much easier to find songs by a specific artist.

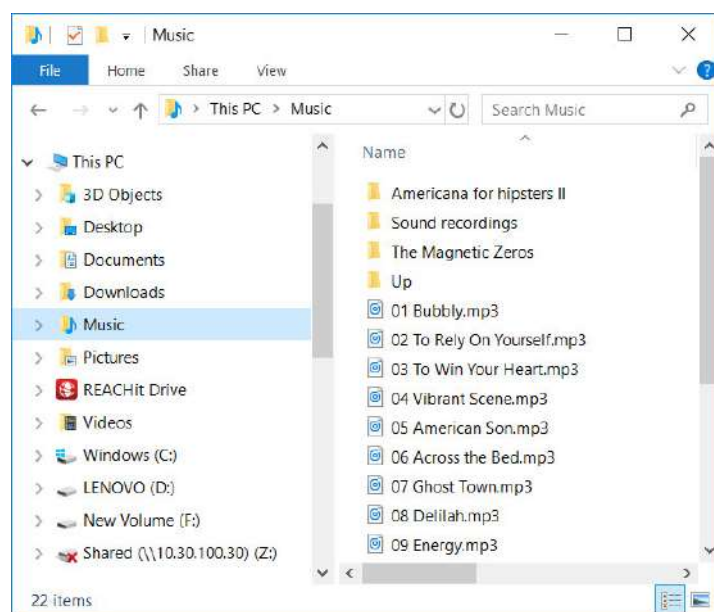


Figure 3.10: Folders can contain both files and folders



HOW TO ORGANISE
FILES AND FOLDERS



<https://www.youtube.com/watch?v=HDMwiJxZlrw>

A similar strategy can be used on all storage devices to make it easier to find important files and folders.

CREATING AND NAMING FOLDERS AND FILES

An important part of having an effective file structure is creating folders and sub-folders to organise your files.



Guided Activity 3.4

To create a new folder:

1. Using the *File explorer*, go to the location where you want to create a folder.
2. Open the *Action* menu by either right clicking, or by using the *ALT-F* shortcut key.
3. Hold the mouse cursor over the *New* option and select the *Folder* option.
4. Enter a name for the new folder and press *Enter* on the keyboard.

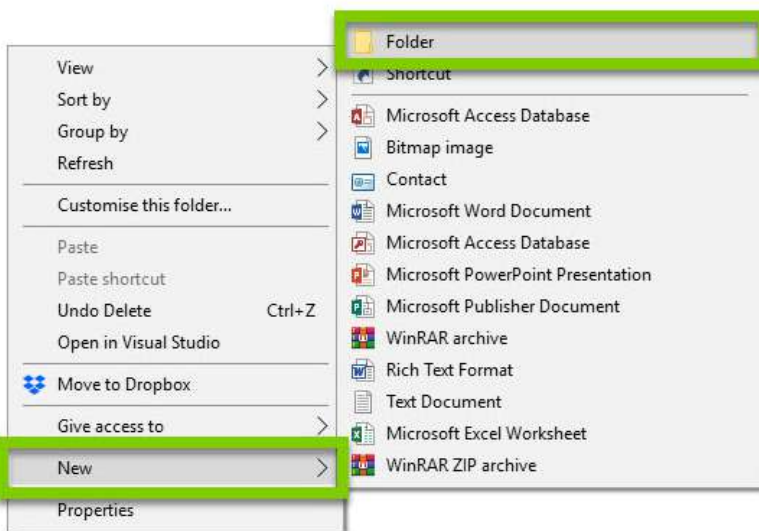


Figure 3.11: Creating a new folder

The same procedure can be used to create new files. However, instead of selecting the *Folder* option, the file type you want to create should be selected.

To create a new file:

1. Using the *File explorer*, go to the folder where you want to create a file, for example, *Documents*.
2. Right click on an empty section of the folder.
3. Click on *New*, select the type of file that you want to create.
4. Enter a new name for the file.
5. Press *Enter*.

FILE NAMING, CONVENTIONS AND PROPERTIES

FILE NAMING AND CONVENTIONS

A file name is used to identify a file on the computer. This is the best method of organising files and folders on the computer. Naming files properly helps you to navigate and locate files and folders easily.



MANAGING FOLDERS AND FILES

An interesting way to organise folders and files is shown in the following YouTube video. This is just to give you an idea of how you can do it. However, in the end, you must use a system that works for you.



<https://www.youtube.com/watch?v=uSnpAnhV2Jo>

A file-naming convention (FNC) helps keep the computer clutter-free and allows you to do the following:

- Know the contents of a file before you open it
- Navigate through long lists of files for the one that you are looking for
- Store different versions of the same document by naming them according to date

Best practices for naming files:

- Keep file names short, but meaningful.
- Include useful information, such as the name of the project or document on which you are working.
- Include the version number (e.g. v1 or -v1) and the date if there is more than one more version. Dates should always appear as yyyy/mm/dd to organise the files in order.

Things to avoid when naming files:

- Symbol characters, such as “\ / < > | “ ? [] ; = + & \$ α β”
- Abbreviations that are not easy to understand
- Simple words, such as “draft”, “current” or “document”.

FILE PROPERTIES

The *File properties* window shows you a range of information about a file, such as its file type, size, author, title and when it was changed last. The *Properties* window will also give you information on how you can manipulate the file.



Guided Activity 3.5

To see the properties of a file, right click on the file name and then click on *Properties*.

1. The *Properties* dialogue box will give you information about the file, as you can see in the figure below.

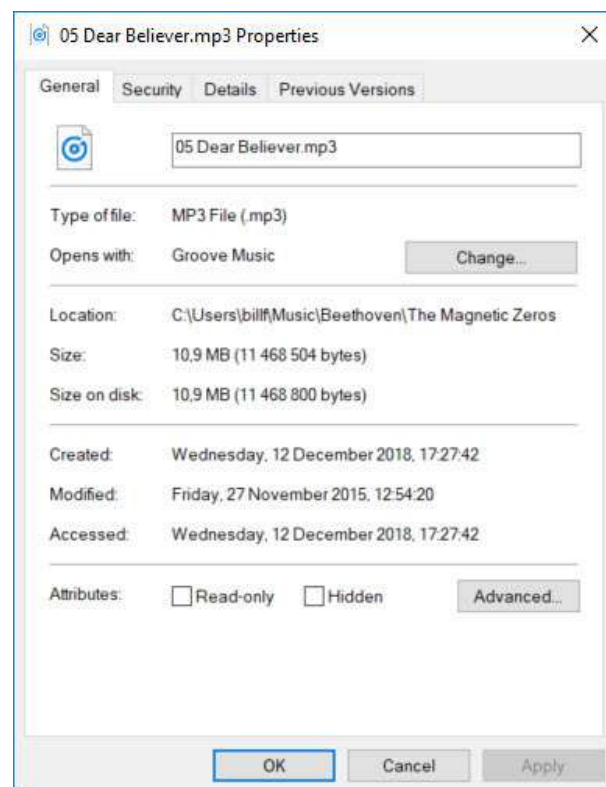


Figure 3.12: Properties dialogue box



Something to know

When giving the new file a name, make sure that you do not accidentally change the file extension. Changing the file extension will prevent Windows from opening the file correctly.

RENAMING FILES

The *Rename* command allows you to change the name of files. As with *Cut* and *Copy*, there are three different ways to rename files.

Table 3.1: *How to rename files*

ACTION MENU	SHORTCUT KEY	LEFT CLICK
1. Select the files you want to rename. 2. Right click on one of the selected files. 3. From the <i>Action</i> menu, select the <i>Rename</i> option. 4. Enter a new name for the file and press the <i>Enter</i> key.	1. Select the file you want to rename. 2. Press the <i>F2</i> key on the keyboard. 3. Enter a new name for the file and press the <i>Enter</i> key.	1. Select the file you want to rename. 2. Using the mouse, click on the name of the selected file. 3. Enter a new name for the file and press the <i>Enter</i> key.

When naming files, it is important that you name the files clearly and in an orderly way. The following are pointers to help you rename files and folders in an organised way:

- Avoid using a structure that is too difficult to follow.
- Since files are organised alphabetically, you need to think about the way in which you name files.
- Make sure that you can identify similar files based on the file names.
- Use the dash symbol to separate different items in a file name (for example, the name and date).

COPYING FILES

When you copy a file, a duplicate of the file is created and stored on the computer's clipboard. This duplicate file can then be copied onto any storage medium connected to the computer, using the *Paste* command.

There are three ways that you can copy and paste files.

Table 3.2: *How to copy and paste files*

ACTION MENU	SHORTCUT KEY	DRAG AND DROP
1. Select the files you want to copy. 2. Right click on one of the selected files. 3. From the drop-down <i>Action</i> menu, select <i>Copy</i>. 4. Navigate to the folder where you want to place the copied file. 5. Right click on the folder and select the <i>Paste</i> option from the <i>Action</i> menu.	1. Select the file you want to copy. 2. Press <i>Ctrl+C</i> on the keyboard. 3. Navigate to the folder where you want to place the copied files. 4. Press <i>Ctrl+V</i> on the keyboard to paste the files.	1. Select the files you want to copy. 2. In a different <i>File explorer</i> window, navigate to the folder where you want to place the copied files. 3. While pressing the <i>Ctrl</i> key on the keyboard, drag the files from the source folder to the destination folder. 4. Drop the files once they are over the destination folder.



Something to know

Applying the **keyboarding** skills you have learned will help you rename files and folders much quicker.



Something to know

Files are copied when you want to share them, or back them up for safety.



COPYING AND MOVING FILES

Use the section of the following video to create an animation about showing what it means to copy a file:



<https://www.youtube.com/watch?v=F7mAoaOf0ts&t=18s>



Something to know

Files are moved when you want to organise them.

MOVING FILES

When moving files, the file or files that you move are copied to a new destination, but then deleted from its original destination. When doing this, you have not created a copy file, but you have moved a file from one folder to another folder.

To move files, you need to *cut* and *paste* them, and NOT copy and paste them.

Table 3.3: *How to move files*

ACTION MENU	SHORTCUT KEY	DRAG AND DROP
1. Select the files you want to copy. 2. Right click on one of the selected files. 3. From the <i>Action</i> menu, select the <i>Cut</i> option. 4. Choose the folder where you want to place the copied files. 5. Right click on the folder and select the <i>Paste</i> option from the <i>Action</i> menu.	1. Select the file you want to copy. 2. Press <i>Ctrl+X</i> on the keyboard. 3. Navigate to the folder where you want to place the copied files. 4. Press <i>Ctrl+V</i> on the keyboard to paste the files.	1. Select the file you want to copy. 2. In a different <i>File explorer</i> window, navigate to the folder where you want to place the copied files. 3. While pressing the <i>Shift</i> key on the keyboard, drag the files from the source folder to the destination folder. 4. Drop the files once they are over the destination folder.

SEARCHING FOR FILES

If you are struggling to find a file on the computer, you can always search for it. The search function allows you to look for any file that is on the computer.

To search for a file in Windows 10, you can do the following:

1. Next to the *Start* button, you can type to search for the file you are looking for.

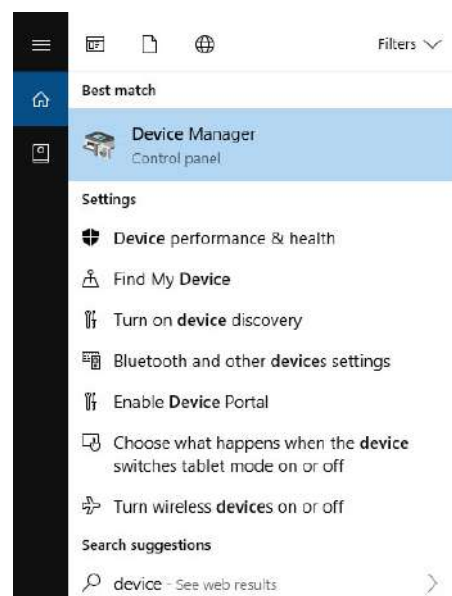


Figure 3.13: *Searching for files*

2. In the example above, the user typed in the word “device”. You can see the search results above the search box.

When searching for files,

- **Use different search terms:** For example, if you are looking for a spreadsheet file, try using a different file name. You could have changed the file name while saving the file.
- **Last application:** If you used a specific application to open and edit the file, you can click on *File > Open > Recent*. The file you are looking for might appear in the list of files that you edited recently.
- **Move and rename the file:** After you have found the file, you can give it a new file name. Then, move it to a folder so that you can find it more easily at a later stage. For example, if it is a new song that you downloaded, move it to the *Music* folder.

DELETING FILES

The *Delete* command enables you to remove files from the storage device. This command can be used to either remove files that you do not need anymore, or to make more space on the storage device.

In Microsoft Windows, deleted files are moved to the *Recycle Bin*. The *Recycle Bin* is a temporary folder where deleted files are stored until you remove them from the bin. Any files you delete will first be moved to the *Recycle Bin* before they are permanently deleted.

The following tables show how you can delete files from the computer and how these files can be deleted from the *Recycle Bin*.

Table 3.4: *How to use the Recycle Bin*

ACTION MENU	SHORTCUT KEY
1. Select the files you want to remove. 2. Right click on one of the selected files. 3. From the <i>Action</i> menu, select the <i>Delete</i> option. 4. Click on the <i>Yes</i> button to remove the files.	1. Select the files you want to remove. 2. Press the <i>Delete</i> key on the keyboard. 3. Click on the <i>Yes</i> button to remove the files.

Table 3.5: *How to remove files from the computer permanently*

EMPTY THE RECYCLING BIN	SHORTCUT KEY
1. Remove the files from the computer, as shown in the previous steps. 2. Go to the computer's desktop. 3. Right click on the <i>Recycle Bin</i> icon and select the <i>Empty Recycle Bin</i> option. 4. Click on the <i>Yes</i> button to remove the files.	1. Select the files you want to remove. 2. Hold <i>Shift</i> and press <i>Delete</i> on the keyboard. 3. Click on the <i>Yes</i> button to permanently remove the files.



Take note

Do not delete any computer software files, as this could cause the computer to malfunction.



Something to know

Files that are deleted from a flash drive *cannot* be restored from the *Recycle Bin*.



Something to know

Previous versions are, at times, referred to as shadow copies.

RESTORING FILES

If you cannot find a file on the computer, or you have deleted it by mistake, you can still recover the file by restoring it from the *Recycle Bin* to its previous version.



Guided Activity 3.6

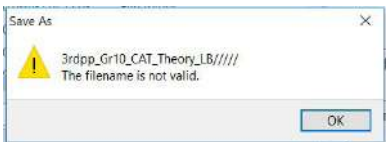
To restore files from the *Recycle Bin*, the following should be done:

1. Open the *Recycle Bin* by double-clicking on the icon that is usually found on the desktop.
2. Browse through the folder and select the file(s) or folder(s) that you need to restore.
3. Right click on the file or folder and from the *Action* menu, choose *Restore*.
4. The *Recycle Bin* will then restore the deleted file(s) or folder(s) to their original location.



Activity 3.2

1. Which solution in Column B can be used to solve the problem in Column A? Note that more than one action can apply to a problem.

COLUMN A	COLUMN B
1. Struggling to find files on the computer	A. Save the file with another extension
2. 	B. Delete unnecessary files
3. File was copied as a shortcut	C. Name the most recent version of the file correctly
4. File was deleted by mistake	D. The wrong action is used; the user needs to use the correct shortcut key
5. User opens file, but Windows cannot open the file	E. Organise files into folders and sub-folders in a logical, hierarchical (tree) structure
6. User is using <i>Ctrl+C</i> on the keyboard to paste files into a new folder	F. Compress files
7. Lots of versions of the same file on the user's computer	G. Recover the file by restoring it from the <i>Recycle Bin</i> to its previous version
8. The wrong file name is used for a document; it needs to be Maths.docx but it appears as Jenny.docx	H. Keep file names short
9. There is a mix of Word files, video files, music files, PDFs, and text documents in the <i>Downloads</i> folder	I. Convert the file to .txt
	J. Avoid file characters
	K. Save the file as another file type
	L. Rename the file
	M. Organise files by their file type

... continued



Activity 3.2

... continued

2. Identify the drive letter, folder, sub-folder, file name and extension in the file paths below. Make a copy of the table below in your workbook and fill in your answers under the correct heading in the table:

- a. c:\Games\Solitaire\Solitaire.exe
- b. e:\Music\Pop\Jenni – Roses.mp3
- c. c:\Program Files (x86)\Calculator.exe

#	DRIVE LETTER	FOLDER	SUB-FOLDER	FILE NAME	EXTENSION
a.					
b.					
c.					

3.2 File types

Each file has a file extension identifying the type of file and how the computer should interpret it. A `.txt` file is a simple text document that contains words with no formatting. A `.doc` file allows for formatted text with different fonts and images.

This section will look at the following file types:

- Text files
- Document files
- Locked document files
- Web pages
- **Archives** and **compressed** files
- Presentation files
- Spreadsheet files
- Database files
- Image files
- Animation files
- Video files
- Audio files
- Adobe Acrobat

TEXT FILES

There are two types of text files:

1. `.txt` (text)
2. `.rtf` (rich-text format)

Text files are files used to store plain text. Plain text refers to text that has no special formatting, such as fonts, font sizes, images and so on. In Microsoft Windows, text files are identified with the `.txt` file extension. Notepad is a popular application in Microsoft Windows that is used for opening and creating text files. Figure 3.14 shows an example of a `.txt` file:

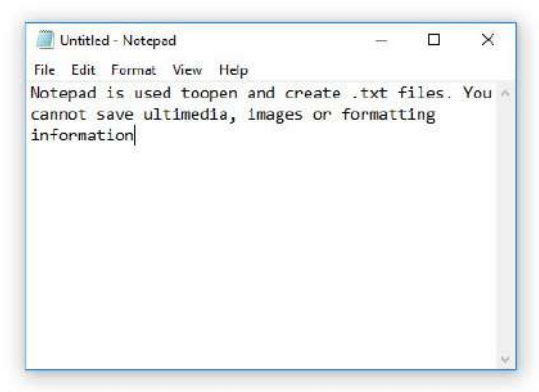


Figure 3.14: An example of a `.txt` file

Rich-text format (RTF) files can save more information than text files. This includes different font types, such as **BOLD**, *italic*, as well as font colour. RTF files cannot be opened in Microsoft Notepad, but can be opened in **Microsoft WordPad**. Both these applications are installed with Microsoft Windows for free.

DOCUMENT FILES

Although document files (.doc, .docx and .odt) are used to store documents. It differs from text files and rich-text files because it can store more complex information such as custom formatting options and styles. Word processors – for example, Microsoft Word, Google Docs and LibreOffice – can be used to create and edit document files.

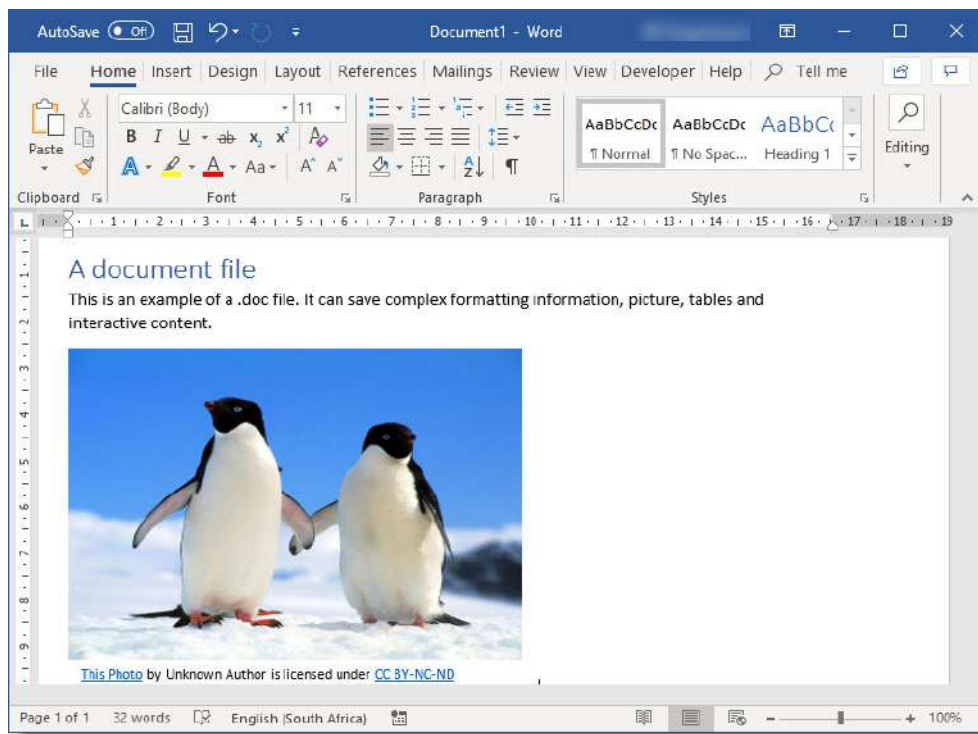


Figure 3.15: Document files can store text with advanced formatting

DIFFERENT TYPES OF TEXT FILES

Jabulani is a Grade 12 learner who works almost every day on his computer on various school assignments. He recently learned that, although two different files may look the same when not formatted, for example a .txt file and a .doc file, the data actually differs. This means that he cannot just change the file type of one file to another and expect the document to open.

However, computer applications have improved quite significantly in that applications, such as Microsoft Word or LibreOffice, can convert one file type to another.

Example 3.1

To save the file in a different file format using the *Save As* function, Jabulani does the following:

1. By clicking on the *Save As* button, a dialogue box opens. This allows him to select the type of file he wants to save his file as, for example PDF, plain text, web page and so on. The application then makes sure that the raw data is saved in the correct way for that file format. An example of this dialogue box is shown in Figure 3.16 on the next page.

... continued

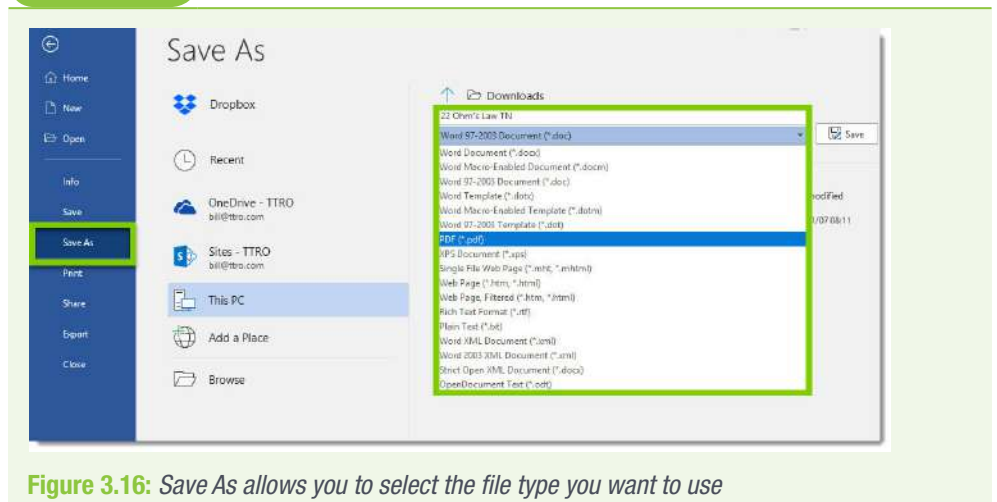


Figure 3.16: *Save As* allows you to select the file type you want to use

LOCKED DOCUMENT FILES

Locked documents are documents with content that is difficult to change, or modify once it has been created. Locked documents are generally used in situations where you want to use the file in the format in which it was saved. Examples of locked document files would be contracts, formal letters and shop catalogues. The two most common locked document formats are the .pdf (portable document format) and .epub (E-PUB) file formats.

WEB PAGES

A **web page** is a computer file that relates to websites and web servers. Web pages include information, such as the colours of text, backgrounds and images. It also includes links to images and other types of multimedia.

The two most common extensions for web pages are:

1. .html
2. .htm

The only difference between the two is that .htm is used as an alternate to .html by a few web servers that do not accept four-character extensions. Both extensions can be used on most web servers.

ARCHIVES AND COMPRESSED FILES

Archives are types of files that can combine and compress multiple different files into one file. For example, if you need to send 50 files in an email to a group member, it will be much easier for you to combine and compress the files into a single .zip file, before sending the .zip file through email. This will make it so much easier for the other person to download the file. The .zip file will also be smaller since all the files are compressed into one file, making it easier to send and download. After receiving the files, it can be **decompressed** and accessed normally. The three main file extensions for archives are:

1. .zip
2. .rar
3. .7z

PRESENTATION FILES

A presentation file stores information using slides that can be shown consecutively. Presentation files are used by teachers, businesses and learners where the speaker discusses the topic using the content of the slides. Slides are short, they can contain pictures and animations, and are generally attractive. The three main file extensions for presentations are:

1. .ppt
2. .pptx
3. .odp

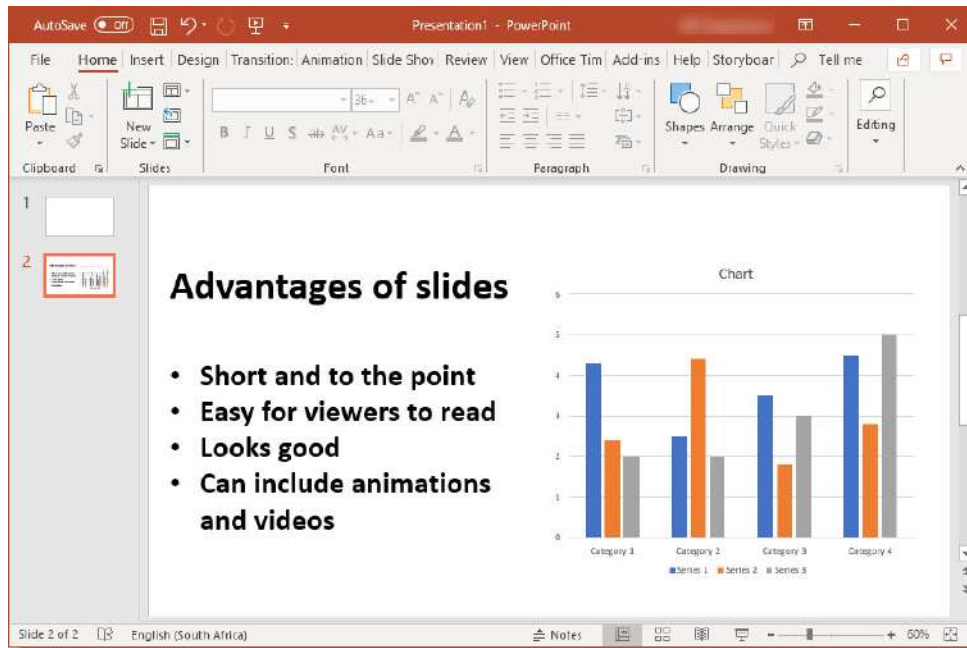


Figure 3.17: A slide in a presentation

SPREADSHEET FILES

A spreadsheet file is a file that stores information in a large table. Companies use spreadsheets when they need to analyse large amounts of data, or do calculations. It is also used to show data, such as checklists and class lists. The three main file extensions for spreadsheets are:

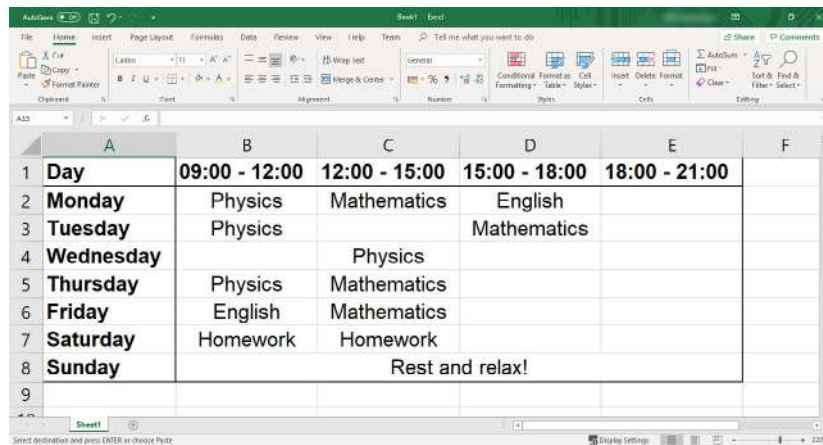
1. .xls
2. .xlsx
3. .ods

The spreadsheet file gets its .xls (and .xlsx) extension from the most popular spreadsheet software – i.e. Microsoft Excel. Figure 3.18 is an example of someone's academic schedule.



Something to know

Excel spreadsheets can have many sheets. Each sheet is a separate page that has its own data.



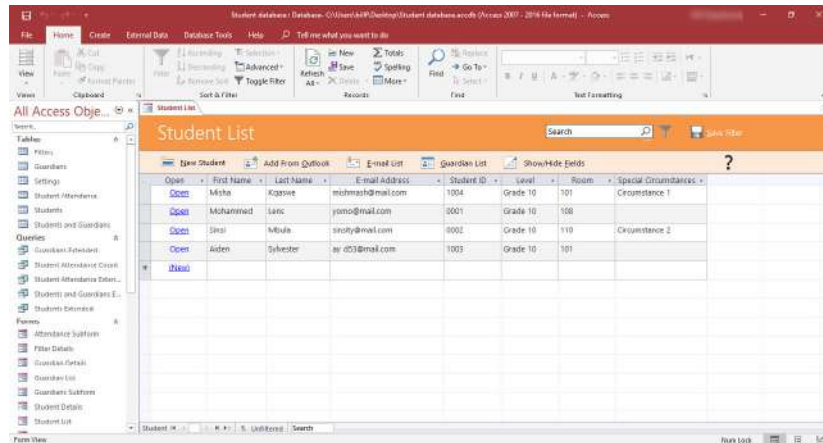
The screenshot shows a Microsoft Excel spreadsheet with a weekly schedule. The columns represent time slots: 09:00 - 12:00, 12:00 - 15:00, 15:00 - 18:00, and 18:00 - 21:00. The rows represent the days of the week from Monday to Sunday. The data is as follows:

	09:00 - 12:00	12:00 - 15:00	15:00 - 18:00	18:00 - 21:00
Monday	Physics	Mathematics	English	
Tuesday	Physics		Mathematics	
Wednesday		Physics		
Thursday	Physics	Mathematics		
Friday	English	Mathematics		
Saturday	Homework	Homework		
Sunday	Rest and relax!			

Figure 3.18: Spreadsheets show data in tables

DATABASE FILES

Databases are similar to spreadsheets, except that the sheets, called *tables*, are all connected to each other. For example, your school might have a database for each grade with the full names of the learners, names of their parents, their contact details, addresses and the amount of school fees paid. Even though these items are listed in different tables, they are still linked together. Because of this, at the end of the month when the school needs to send out fee statements, the database software can create a statement where your name and surname, parents' contact details and fees owed are all displayed on the same page.



The screenshot shows a Microsoft Access database window titled 'Student List'. It displays a table with the following columns: First Name, Last Name, E-mail Address, Student ID, Level, Room, and Special Circumstances. The data is as follows:

First Name	Last Name	E-mail Address	Student ID	Level	Room	Special Circumstances
Misha	Kogee	misham@gmail.com	1004	Grade 10	101	Circumstance 1
Muhammad	Lanc	yom@gmail.com	0001	Grade 10	106	
Sami	Mkulu	snoby@gmail.com	0002	Grade 10	119	Circumstance 2
Aiden	Sylvester	ayr03@gmail.com	1003	Grade 10	101	

Figure 3.19: Databases can store related information in different tables

Most programmers use databases to store important data for their software.

IMAGE FILES

There are five main formats used to store graphics:

1. .bmp (bitmap image)
2. .gif (**Graphics Interchange Format**)
3. .jpg or .jpeg (Joint Photographic Experts Group)
4. .png (portable network graphics)
5. .tiff (tagged image file format)

These image formats are used for cameras, scanning and printing. Most of these file formats need not be compressed and are, therefore, ideal for printing and web graphics. Photos taken by a camera, or pictures created using software, for example, Microsoft Paint or Adobe Photoshop, are often saved in one of these formats.

ANIMATION FILES

Animation files are files that interpret moving images. The two most common animation file formats are:

1. .gif
2. .swf (Small Web Format Flash)

Although both these formats create animations, they do so in two different ways. GIF animations show normal GIF images one after another to create an animation; SWF files use programming to move shapes on the screen, creating an animation.

VIDEO FILES

A **video file** format is used to store digital video data on a computer. Video file formats are usually compressed to reduce the file size. Video files are made up of a “**container**” and a “**codec**”. A codec is used to compress and decompress the video where the video files are too big, causing the video not to play, or be downloaded. Examples of codecs include *FFMpeg*, *DivX*, *XviD* and *x264*.

A container is a set of files that stores information about the digital file. For example, a container in a video file would be a combination of both audio and video in a single file so that the file can play both audio and video at the same time. Some common types of containers would be *AVI*, *FLV*, *WMV* and *MP4*.

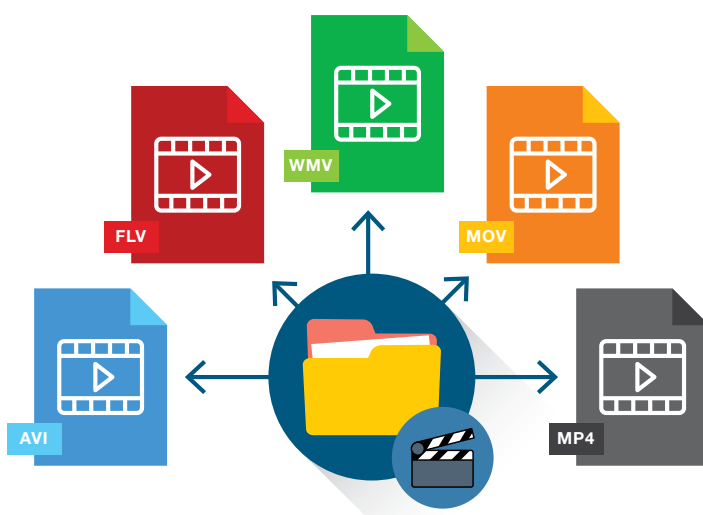


Figure 3.20: Different types of video file formats



Something to know

The opportunity to create videos on PCs and mobile devices are endless. MP4 is one of the most common file formats used. It uses a separate compression for audio (m4a) and video (m4v), and is mainly used for sharing videos over the Web. MP4s are generally small in size; however, they have a high quality even after being compressed. MP4 files are popular, because they are compatible with both online and mobile **browsers**.

SOUND FILES

Sound (audio) files are similar to video files, as they are compressed before they are stored on a computer. They are used to store audio files, such as music. Because of the popularity of using computers to store music and videos, the *.mpeg* and *.mp3* file types have become very popular in recent years. Common audio file formats consist of the following:

- *.aac* (Advanced Audio Coding)
- *.flac* (Free Lossless Audio Codec)
- *.mp3* (Moving Picture Experts Group Layer 3 audio)
- *.wma* (Microsoft Windows Media Audio).

PORTABLE DOCUMENT FORMAT (PDF)

Adobe Acrobat is an application developed by Adobe Systems where you can view, create, print and manage files in a portable document format (PDF). PDFs are a very good file format for saving books, or published documents that can be easily sent to another user through email, etc.

CONVERTING A DOCUMENT TO A PDF

You can convert a document to a PDF by doing the following:

1. Click on the file that you want to be converted into PDF format.
2. Click on the *File* tab and choose *Save As*.
3. Type in the file name.
4. Using the drop-down menu, click on the *PDF* under the *Save As* type.
5. Click *Save* and the document will be saved in a PDF format.



Activity 3.3

1. Complete this activity by following these steps:
 - a. Create a sub-folder in *My Documents*. Name the new sub-folder: *CAT Grade10*.
 - b. Open the Word-processing application. Copy and paste any picture from the computer into the document.
 - c. Save the file with the file name: *Practice* in *My Documents*.
 - d. Move the *Practice.docx* to the folder: *CAT Grade10*
 - e. Create a PDF of the *Practice.docx* file. Save the file in the same location.
 - f. Rename the PDF file to *Practice_new*.
 - g. Delete the file *Practice.docx* in the *CAT Grade10* folder.
2. Answer the following questions about the diagram below.



... continued



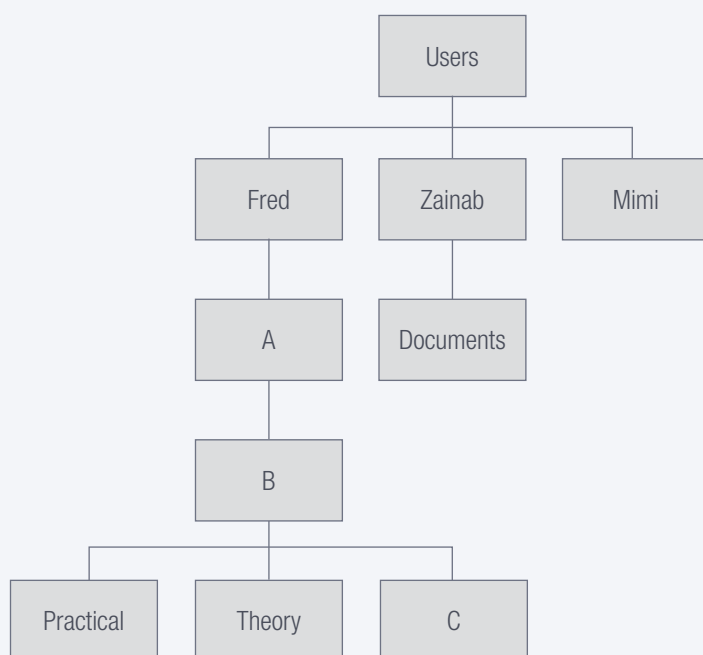
Activity 3.3

... continued

- How many audio files are there?
- What type of file is "08.Internet and the world-wide"? What application can be used to open this file?
- "Image (16).png" can be opened with Adobe Acrobat Reader. Is this statement true? If not, provide the correct answer.
- What type of program can be used to open "4.3 Handover (171129) (Waiting for SME).pptx"?
- What is the name of the file that Excel can open up from the diagram above?

REVISION ACTIVITY

- Each time Sipho saves a file, he saves it to the desktop of the computer. His desktop is very cluttered and he has trouble finding the files that he needs. Advise Sipho on a more effective way to organise his files. (4)
- Which of the following is NOT a file extension used for graphics files? (1)
 - .bmp
 - .gif
 - .toff
 - .jpeg
- Fred has created a folder called "Grade 10 PAT" on a school computer to save all his PAT files. He checks on the file properties and finds that the location of the folder is c:\users\fred\documents\Grade 10\Grade 10 PAT. Use the diagram to answer the questions below:



- Write down the folder names of folders A, B and C. (3)
- How many users have folders on this computer? (1)

... continued

REVISION ACTIVITY

... continued

2. Match the file extension in Column A with the associated file type in column B. Write down only the letter of the file type (Column B) next to the number of the file extension (Column A), e.g. 1C.

(Note: It is possible that the application from Column B may be used more than once.)

COLUMN A	COLUMN B
1. doc	A. Web page
2. xlsx	B. Open Office presentation
3. zip	C. Audio file
4. html	D. Compressed file
	E. Microsoft Excel spreadsheet
	F. Microsoft Word document

(4)

3. Which shortcut key can you use to carry out the following actions?

a. Rename a file

(1)

b. Copy a folder

(1)

c. Paste a document onto the desktop

(1)

TOTAL: [16]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	CAN YOU ...	YES	NO
1.	Explain the importance of file management.		
2.	Find a file path.		
3.	Find a file using a file path.		
4.	Distinguish between a file and a folder.		
5.	Explain the importance of using the correct file-naming conventions.		
6.	Organise, copy, rename, delete, restore, and move files and folders.		
7.	Search, view, and sort files and folders.		
8.	Describe the different file types and their extensions.		

FILE AND PRINT MANAGEMENT

CHAPTER OVERVIEW



Unit 4.1 File management skills

Unit 4.2 File compression

Unit 4.3 Print management



At the end of this chapter, you should be able to:

- Create shortcuts.
- Use the Snipping Tool and Print Screen to take screenshots.
- Compress and decompress files and folders.
- Add new peripherals, such as a printer and mouse.
- Change the default printer.
- Explain what queue management is.
- Print files.

INTRODUCTION

When you open an application using the menu system, it can sometimes be time consuming. An easy way to save time is to create a shortcut for an application or program that you use often. In this section, you will learn how to create **shortcuts**, take screenshots, as well as how to compress and decompress files.

Print management is the managing and monitoring of various print tasks. Before learning how to print, you will learn how to add a new peripheral device, as well as how to do basic printing and apply printer queue management.

4.1 File management skills

In this section, you will learn about basic computer management tasks that you need to do when working on a computer.

CREATING SHORTCUTS

This section will look at how to create a shortcut on the desktop for your favourite program, application, or website, for easy access. There are different ways to create shortcuts and in this section, we will look at some of these ways.



Guided Activity 4.1

Option 1: If you want to create a desktop shortcut for your favourite program or file, do the following:

1. Right click on the `.exe` file.
2. Select *Send To*:
3. Click on *Desktop (create shortcut)* to create the shortcut.

A shortcut of this program will then be created on the Windows desktop.

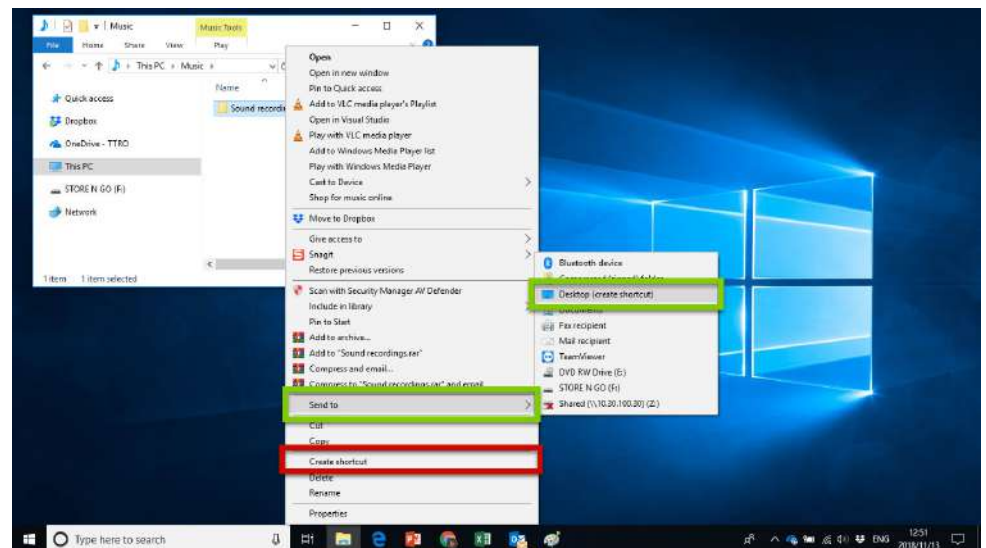


Figure 4.1: Creating a shortcut for a program

If you select *Create Shortcut* (indicated with the red rectangle), the shortcut will be created in the same location. In this example, the shortcut will be created in the *Music* folder.



Guided Activity 4.2

Option 2: Another way of creating a shortcut would be to right click on the desktop, and then select *New* and then *Shortcut*.

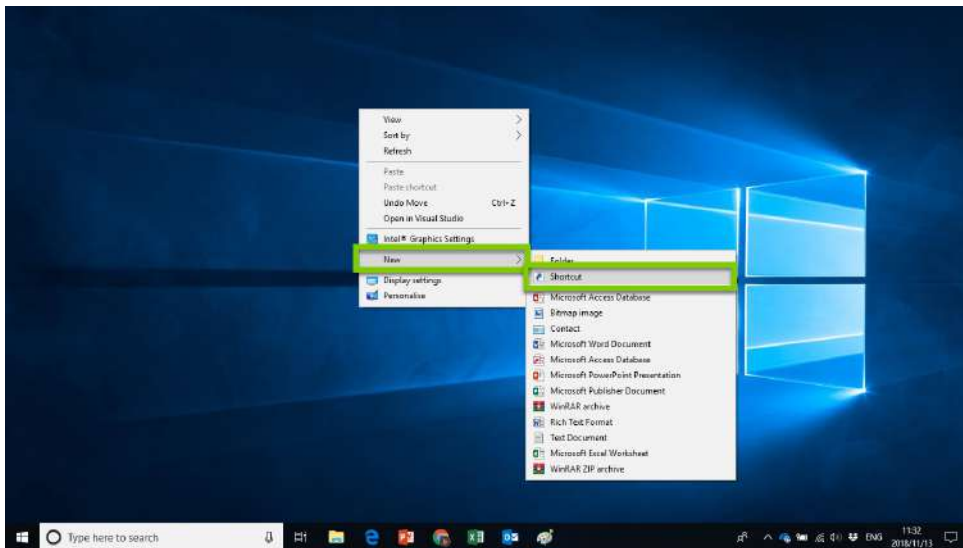


Figure 4.2: Creating a shortcut on the desktop

This will open the *Create Shortcut* wizard dialogue box.

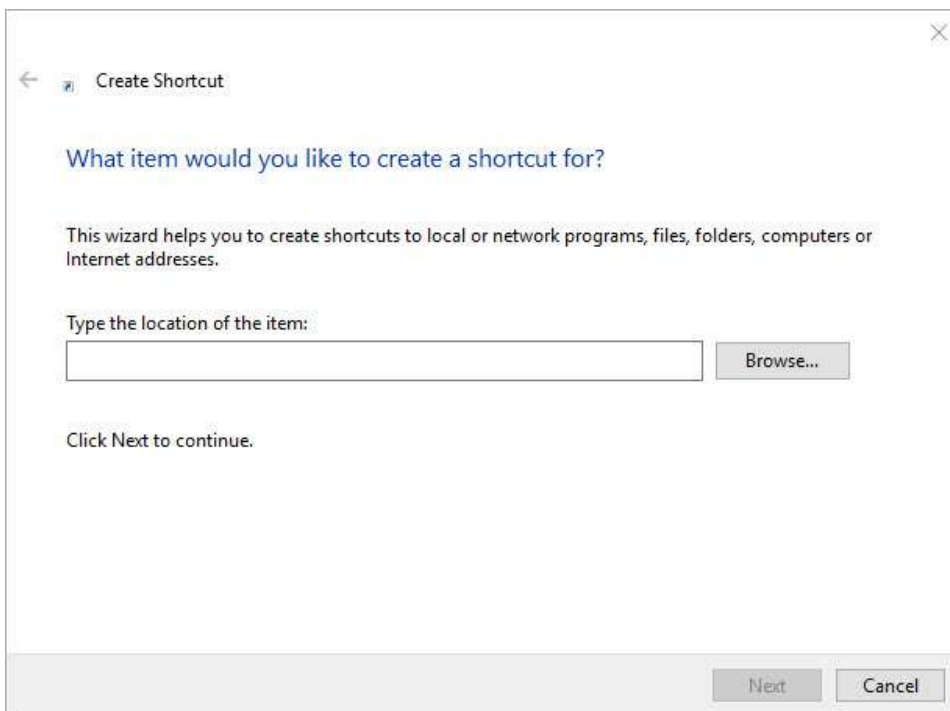


Figure 4.3: Create Shortcut wizard dialogue box

This **wizard** will help you to create a shortcut on the desktop.



Something to know

This only works using the Google Chrome browser; not in Edge or Internet Explorer.



Something to know

Desktop shortcuts can save you time to find a specific file, folder, or even a website.



Guided Activity 4.3

Option 3: Lastly, we will look at creating a shortcut for your favourite website. The method might differ for different operating systems. For this example, we will look at creating a shortcut for a website using the Google Chrome browser.

1. Open your favourite website or web page.
2. Click on the *Options* button on the top right-hand side of the web page, as seen in Figure 4.4:

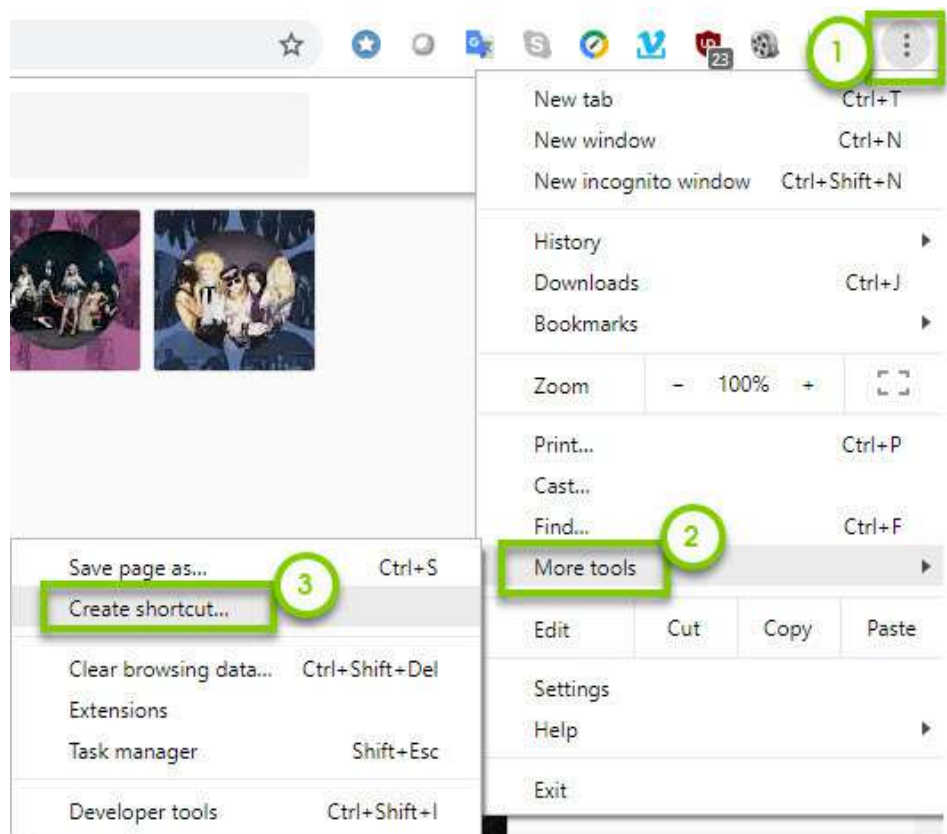


Figure 4.4: Web browser shortcut

3. Go to *More tools*, from the drop-down menu and choose *Create shortcut*. A dialogue box will appear asking you if you want to create the shortcut.
4. Click on *Create*, and the shortcut will be placed on the desktop.

TAKING SCREENSHOTS

A screenshot is an image of how the computer screen looks the moment you capture the screen. For example, it is a snapshot of whatever is displayed on the computer screen at that time.

Different models of computers have different shortcuts on how to take a screenshot. For most cases, you will press the *ALT* key and the *PrtSc* (print screen) key on the keyboard at the same time to take a screenshot of the selected window. *PrtSc* on its own takes a screenshot of the whole desktop.

This image is then saved to the clipboard. The clipboard is a memory space on the computer where data or graphics that are copied or cut are temporarily stored.



Guided Activity 4.4

To save the screenshot permanently, you can paste it into an application. In this example, the application is *Paint*. Do the following:

1. Open *Paint*.
2. Select *Edit* then *Paste* from the *menu* bar of the *Paint* window. An easier way to paste the screenshot will be to use the **hotkeys** (*Ctrl+V*) and the image will be pasted into *Paint*.
3. You can then save the screenshot by clicking *File* and then *Save As* from the *Paint* window's menu bar.
4. You can use the hotkeys (*Ctrl+S*) and save the file with the file name and file location you want, and then select *Save*.



Activity 4.1

1. Create a new folder on the computer. Name it *Practice*.
2. In the folder, create a shortcut to the following:
 - a. Favourite document in *My Documents* folder
 - b. Two of your favourite programs
3. Open the *Practice* folder. Take a screenshot with *Print Screen* (*PrtSc*) of what is displayed on the screen.
4. Paste the screenshot in *Paint*.
5. Save the image as *Practicescreen.jpeg*.
6. Move this image to the *Practice* folder.

4.2 File compression



Something to know

By default, the new *.zip* file will have the same name as the file or folder that you compressed. However, if you compress multiple files and folders, the name of the *.zip* file will be the same as the first file that you clicked at the start of the compression.

Compressed files can be referred to as *archived* or *zipped* files and can contain many separate files in one large archive. This takes up much less space than the original files. Decompressing files is pretty much the opposite of compressing the files, which means to expand the archived or zipped folder back into its original form. In Windows, you can compress (zip) and decompress (unzip) files without installing any extra programs or applications. In this section, you will learn about compressing and decompressing files and folders.

COMPRESSING AND DECOMPRESSING FILES AND FOLDERS



Guided Activity 4.5

To compress files or folders using Microsoft Windows, you can do the following:

1. From the *Start* menu, open *This PC*.
2. Select all the files and folders you want to include in the compressed folder by holding down the *Ctrl* key while clicking on each file.
3. Right click on the selected items and from the drop-down menu, choose *Send To*. Then select the *Compressed (zipped) folder* option. This will create a zipped file, in the same location, in which the files are located. The file extension for compressed folders is *.zip*.
4. Finally, enter a name for the compressed folder.

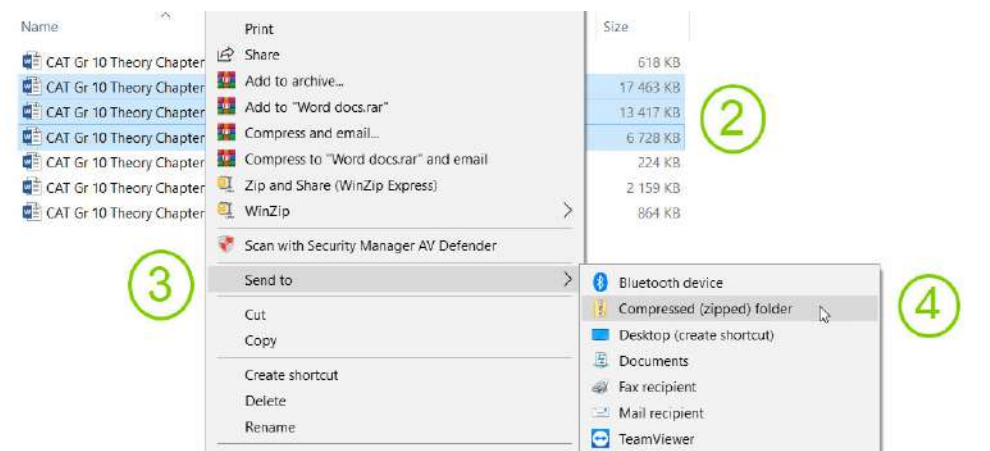


Figure 4.5: Compressing a folder

DECOMPRESSING FILES OR FOLDERS



Guided Activity 4.6

To decompress files or folders using Microsoft Windows, you can do the following:

1. From the *Start* menu, open *This PC*.
2. Look for the *.zip* file that you want to decompress, right click on it and choose *Extract Files...*
A dialogue box will open. You should choose the destination for the decompressed files.
3. Click on *New Folder*.
4. Name the folder.
5. Click *Ok*. The files will be extracted to the folder that you created.

File compression has the following multiple advantages:

- Compressed files require less storage space. This allows you to fit more files onto storage devices.
- Because the files use less storage space, it can be transferred more easily and quickly between storage devices. This is especially true when the files need to be transferred over the internet.
- Because many files can be stored in one archive, it is easier to transfer the data and to ensure that the recipient gets all the correct files.

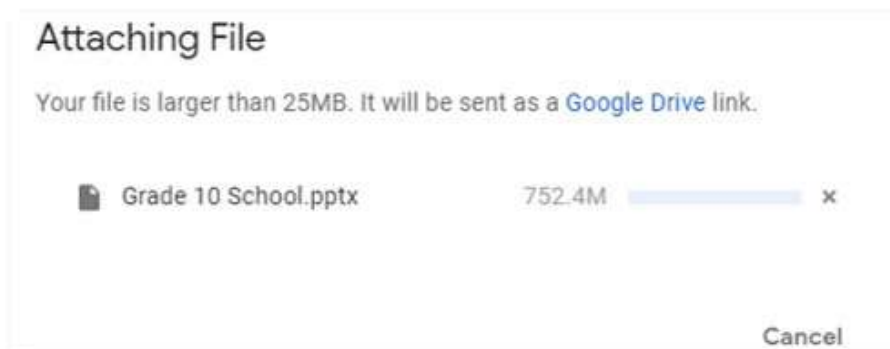
However, compressed files also have some disadvantages, with the most notable being:

- Some compressed files need specific software to decompress them. Fortunately, Windows 10 can decompress *.zip* files without requiring any additional software.
- Compressing and decompressing files can take a bit of time, depending on the level of compression and the size of the files that need to be compressed.



Activity 4.2

1. When sending a file as an email attachment, the email program cannot attach the file to the email. The following message is shown, but you do not want to send it using Google Drive.



- a. Briefly explain how you can use an email to send a file as an attachment, even though it is such a large file.
- b. What is the file extension of the answer in (a)?
- c. After sending the file to the recipient, how does the recipient have access to the file?



Something to know

When zipping or unzipping files or folders, the original files or folders are never deleted. If you need to delete these items, you will have to do it manually.



EXTRACTING ZIP FILES



<https://www.youtube.com/watch?v=nXAYWwiDvc8>

4.3 Print management

Print management allows you to install, view and manage printers. Print management is managing and monitoring various print tasks. Many software companies have proper print management software that coordinates all the print jobs that need to take place.

ADDING A NEW PERIPHERAL

Some printers have software that must be installed before it is connected to the computer. Always read the instructions that come with the printer to see if this is the case. However, in most cases, you can connect printers immediately.



Guided Activity 4.7

To install a new printer, do the following:

1. Connect the printer to the computer, turn it on and turn the computer on.
2. Click *Start* and look for *Control Panel > Devices and Printers > Add Printer*, and then choose the type of printer that you are installing.
3. This opens the *Add Printer* wizard, which will guide you on how to add the printer.

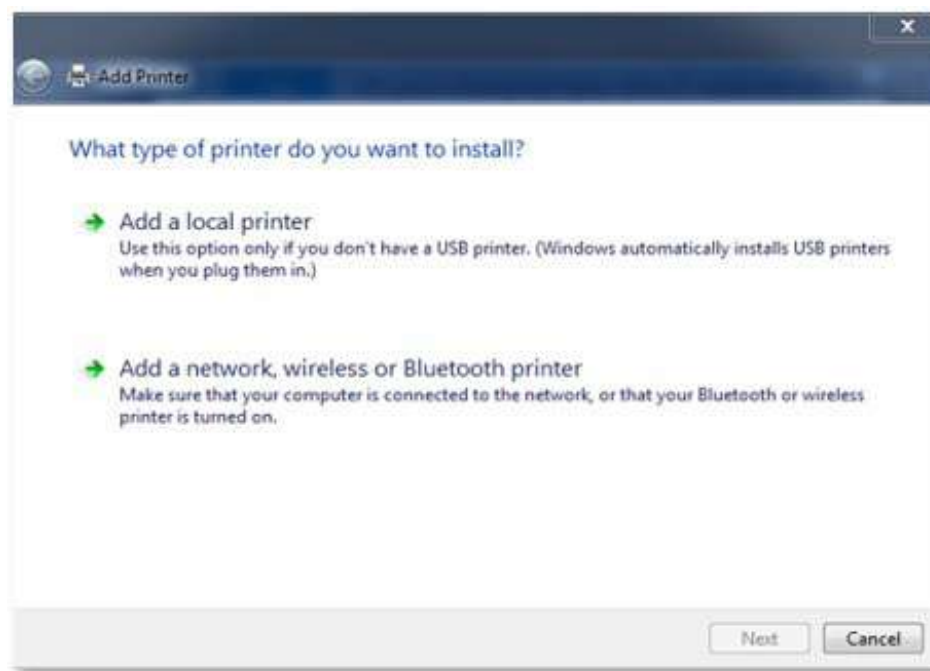


Figure 4.6: Adding a new printer wizard



Guided Activity 4.8

To add a mouse,

1. Click on *Start > Devices and Printers > Add a Device*. The *Add a Device* window will open.
2. The window will show you a list of devices that are available to connect.
3. Click on the device that you want to connect to (in this case, the mouse).
4. Click *Next* and follow the instructions.

CHANGING THE DEFAULT PRINTER

If you already have a printer added to your computer, this printer will always be used as the default printer. However, if you have more than one printer installed on the computer, you can change the default printer.



Guided Activity 4.9

To change a default printer, you can do the following:

1. Click on *Start*.
2. Click on *Control Panel* and choose *Devices and Printers*.
3. Right click on the printer you want and from the drop-down menu, select the *Set as default printer*.

After you have done this, the default printer will have a tick next to it to indicate that this is the current default printer.

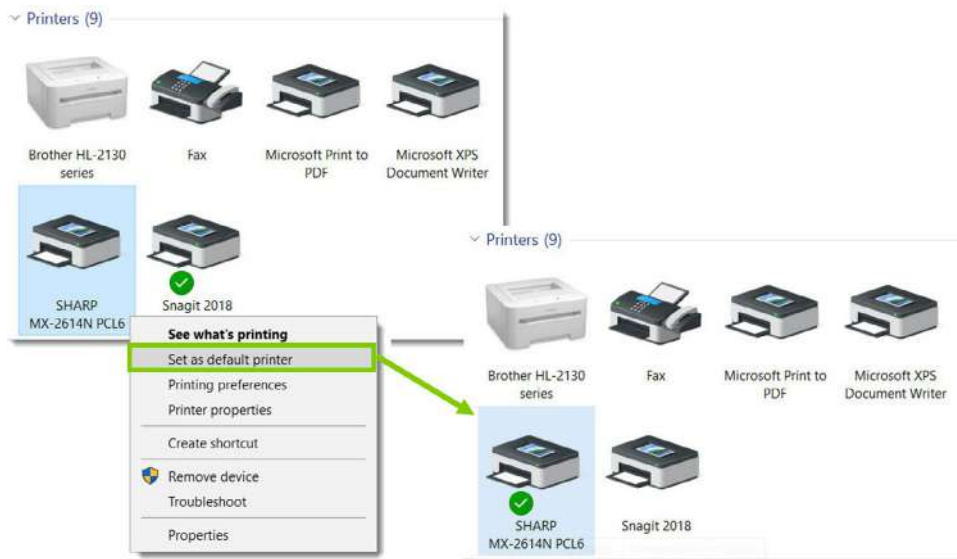


Figure 4.7: Changing the default printer

BASIC PRINTING

If you need to print a document or image, you can do so in the following way.



Guided Activity 4.10

1. Select the document or image that you want to print.
2. Right click on the file and choose *Print*.
3. If a dialogue box appears, click *OK*, or the *Print* button.

The computer will immediately send the pages to the printer, which should start printing the pages automatically.



Something to know

To print a number of documents one after the other, select all the items, then right click on the selected items and choose *Print*. The documents will be printed in the order in which they were selected.

PRINTER QUEUE MANAGEMENT

When a printer needs to print more than one document, it is placed in a queue and printed according to the sequence that it was selected.



Guided Activity 4.11

To check what you are printing:

1. Click on *Start*.
2. Select *Control Panel > Devices and Printers*.
3. Double-click on the *Default Printer*.

The *Printer* window will open and this will show in the order in which the documents will be printed.

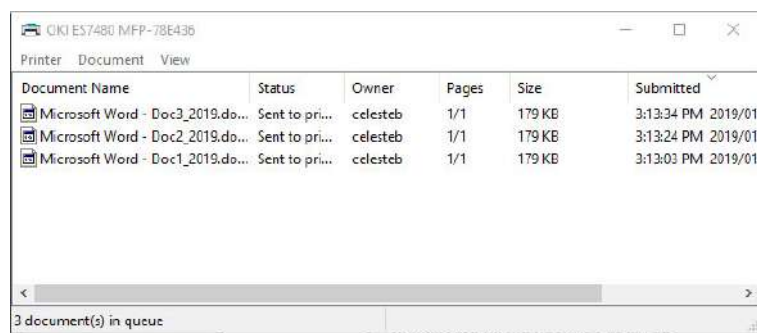


Figure 4.8: Files in printer queue



Activity 4.3

Mimi has her own business and wants to buy a new printer, which she needs to connect to her desktop. She is going to use her printer to print quotations and invoices for her clients. She also needs to connect the printer to the desktop in the office.

1. Should Mimi buy a laser or an ink tank printer? Motivate your answer. Consider the cost of the printer, as well as the printing cost.
2. Mimi's focus is on the quality of the print-outs rather than cost. Which printer should she buy?
3. After she has bought her printer, briefly explain to Mimi how she should connect this printer to the desktop.
4. Explain why nothing is printing after Mimi has connected the printer to the desktop.

REVISION ACTIVITY

1. The following icon appears on the desktop of a computer:



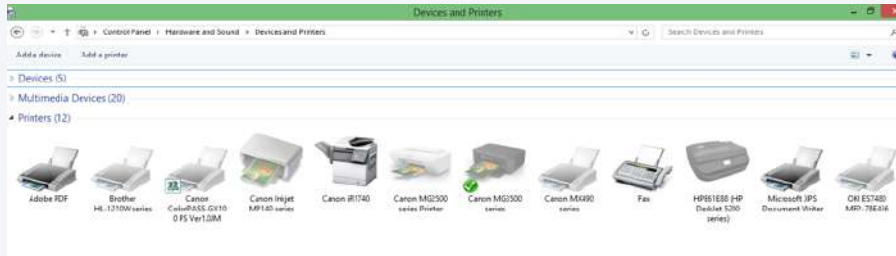
2. a. How can you tell that the icon is a shortcut? (1)
- b. What is the benefit of using a shortcut to an application? (1)
- c. How would you launch the application using the shortcut? (1)
- d. What is the name of the application to which the shortcut refers? (1)

... continued

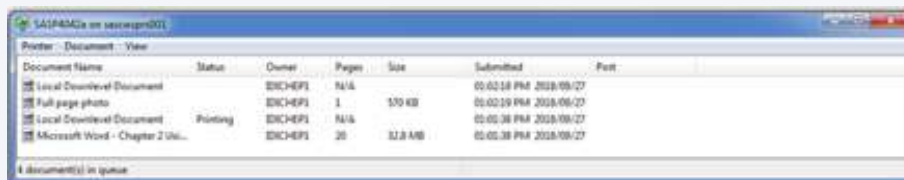
REVISION ACTIVITY

... continued

3. a. What is the name of the Windows 10 tool that can be used to create a screenshot? (1)
- b. What is the difference between pressing *PrtSc* and pressing *Alt* together with *PrtSc* (*Alt + PrtSc*)? (2)
4. The screenshot below shows the printers that are installed on Thalia's computer.



- a. What is the name of the default printer? (1)
- b. Describe briefly how you would make it the default printer. (2)
5. The screenshot below shows a list of documents waiting to be printed.



- a. What is the name given to this list of documents? (1)
- b. You have decided that you no longer want to print the Microsoft Word document. Describe briefly how you would remove this document from the list. (2)
- c. You need special paper to print the full-page photo. You are only able to purchase the special paper the next day. Describe briefly what you would do to temporarily prevent the photo from printing. (2)
6. Samiya, a Grade 10 learner, has a folder called *Grade 9*, which is taking up a lot of space on her computer. How could she reduce the amount of space the folder takes up without deleting any of the files inside the folder? (2)

TOTAL: [17]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	CAN YOU ...	YES	NO
1.	Create different types of shortcuts.		
2.	Take screenshots.		
3.	Compress and decompress files and folders.		
4.	Add a new printer and mouse.		
5.	Change the default printer.		
6.	Describe what queue management is.		
7.	Print a file.		

HARDWARE

CHAPTER OVERVIEW

Unit 5.1	What is hardware?
Unit 5.2	Input devices
Unit 5.3	Output devices
Unit 5.4	Storage devices
Unit 5.5	Green computing

At the end of this chapter, you should be able to:

- Define and describe what hardware is.
- List the various types of ports and connectors.
- Define input.
- Discuss the types of input.
- Discuss the common input devices used.
- Define output.
- Identify the different types of output.
- Discuss the common output devices used.
- Define storage.
- Identify the different storage devices used.
- Use different methods to connect peripherals.
- Understand the importance of green computing and recycling.

INTRODUCTION

Have you ever wondered what we would do without computers in this day and age? Pretty much everything in this world works with computers; from using an ATM, to buying groceries and making new friends. We live in a world where, after you have bought a new computer, taken it home and, before even unpacking it, there is an advertisement for a new computer that makes yours now outdated.

According to Moore's law (1970), computer processor speeds will double every two years. For this to happen, a computer's components must be improved on a continuous basis.

Looking back to where computers started, you will realise that it was not at all as complex or technologically advanced, such as the ones we have today. In the early 1600s to 1800s, people started thinking of building a device that could solve complex problems and calculations easier and faster than what they could do manually. They wanted a device that could add and subtract. As a result, the first digital calculator that could add and subtract was built by the 18-year-old Blaise Pascal in 1642.

As the years went by, newer and better calculators and computers were built. One of the first complex machines that forms an important part of the computer history, was the ENIAC, built for the US army between 1943 and 1946. It was a very big computer that could do thousands of calculations.

After that, computers started changing the world more and more as it became smaller, faster and smarter.

5.1 What is hardware?



HARDWARE BASICS

Have a look at this video to see a 3D virtual tour of hardware:



https://www.youtube.com/watch?v=vG_qmtdBPTU

A computer consists of two major parts – i.e. hardware and software.

Hardware refers to all the physical parts or components of a computer, such as the monitor, keyboard, computer data storage, graphics card, sound card and motherboard. Software refers to the instructions that can be stored and run by hardware.

There are many different kinds of hardware that can be installed inside, or connected to the outside of a computer. To make it easier for you to understand, we will divide it into the following categories:

- Input devices
- Output devices
- Storage devices

PORTS AND CONNECTORS

Ports and **connectors** form an important part of a computer. Ports allow computers to connect with different types of hardware, such as a keyboard, mouse or monitor. Many devices use cables to connect to the computer. Devices can also be connected using Wi-Fi (**wireless**), or **Bluetooth** (short-range wireless).

The connector (plug) which is found at the end of a cable fits into a specific port (socket) of the computer. These ports are usually found at the back of the computer.



Figure 5.1: Port (1) and connector (2)

Table 5.1: Types of ports, common connectors and examples

PORT	IMAGE OF PORT	CONNECTOR	USES
USB port			Used to connect many devices, such as smartphones, flash drives and external hard drives
VGA port			Used to connect monitors and data projectors

... continued

PORT	IMAGE OF PORT	CONNECTOR	USES
DVI port			Used to connect display devices, such as monitors
HDMI port			Used to connect monitors, data projectors and gaming consoles
3.5 mm port			Used to connect audio devices, such as speakers, head phones and microphones
Ethernet			Used to connect to a network device, such as a switch or router
RJ-11			Used to connect an internet router to a telephone line

ADAPTERS AND CONVERTORS

Almost every peripheral device uses an adapter to communicate with a computer. With new technology, the type of computer ports and other computing devices change. However, instead of buying a new computing device, you can use an adaptor convertor to provide additional compatibility and functionality.

Types of common adaptors include the following:

- HDMI to VGA
- HDMI to DVI



Activity 5.1

1. Define the concept *hardware*.
2. Match the port in Column A with the correct device in Column B.

COLUMN A	COLUMN B
1. USB Port	A. Smartphone
2. USB Type C	B. Flash disks, keyboard and mouse
3. VGA Port	C. Network switch
4. HDMI cable	D. Monitor
5. Ethernet cable	E. Laptops and television

3. Briefly explain when adaptors are used.

5.2 Input devices

Input devices are hardware devices that allow you to enter data into the computer, or interact with the computer.

Remember, when entering data, you are actually giving the computer an instruction to do something. In this section, you will learn about the different types of commonly used input devices.

COMMON INPUT DEVICES

KEYBOARD

A keyboard is used for entering information into a computer, such as letters, words, numbers and symbols. Pressing any key on the keyboard is an input that the computer uses as data. There are different types of keyboards, as shown in the table below.

Table 5.2: *Different types of keyboards*

TYPE OF KEYBOARD	IMAGE
Standard keyboard (external)	
Laptop keyboard (built-in)	
Ergonomic keyboard	
Gaming keyboard	

... continued

TYPE OF KEYBOARD	IMAGE
Foldable keyboard	
Laser keyboard	



Something to know

The concept of pressing a key to type text has been around for a very long time. One of the early inventions, which later became the typewriter, was invented by Pellegrino Turri for his blind girlfriend so that she could write letters to him.



Something to know

The *Oxford Dictionary* acknowledges both “computer mice” and “computer mouses” as the correct plural forms of the term. For the purpose of this book, however, we will be using the plural form “mice”.

POINTING DEVICES

Pointing devices are used to control the movement of the **cursor** on the screen. The major pointing device is the mouse for the desktop computer and the touchpad for the laptop.

MOUSE

The mouse is the second most important input device of a computer. Using the mouse, you can move the mouse pointer or cursor on the computer screen, allowing you to interact with the computer.

There are different types of mice, as shown in the following table.

Table 5.3: *Different types of mice*

TYPE OF POINTING DEVICE	DESCRIPTION OF MOUSE	IMAGE
Wireless mouse	A mouse that connects to a computer using Wi-Fi or Bluetooth	
Cable mouse	A cable mouse uses a USB connector to connect to the computer or laptop	

FIRST COMPUTER MOUSE

Douglas Engelbart invented the first computer mouse in 1963. The mouse had two wheels positioned at a 90-degree angle to each other to keep track of the movement. The ball mouse was only invented in 1972 and the optical mouse, around 1980.

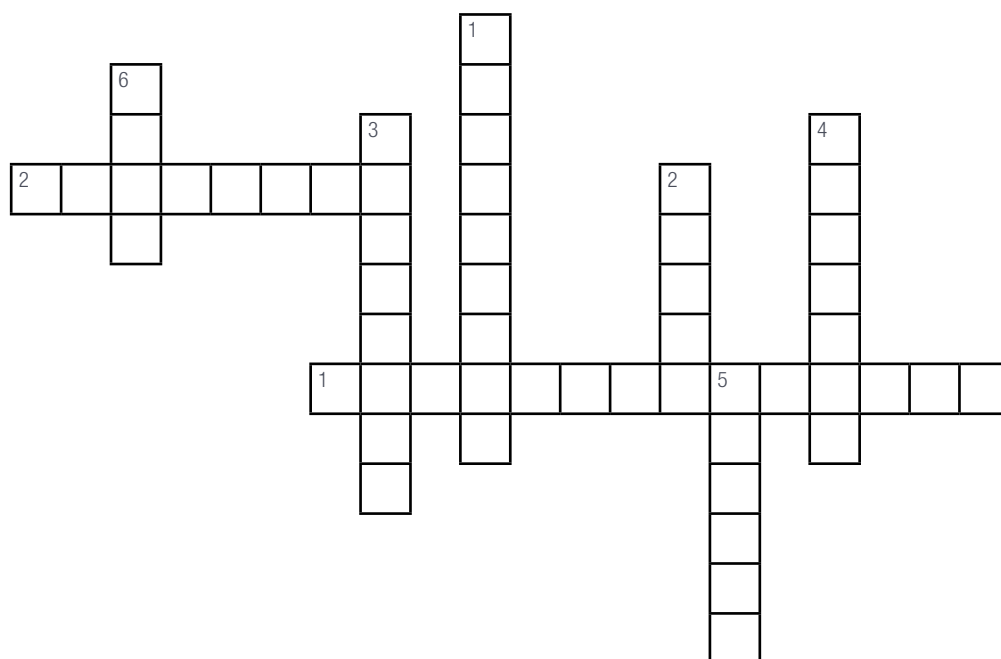


Figure 5.2: Engelbart's mouse



Activity 5.2

1. Fill in the following crossword puzzle using the clues that are provided.



Across

1. A type of wireless technology mice used
2. Physical component of a computer like a mouse

Down

1. Which type of keyboard is used to minimise muscle strain
2. Used to move the pointer on a screen
3. For entering information, such as letters, words, numbers and symbols into the computer
4. Controls movement of the cursor
5. The standard keyboard
6. This allows computers to connect with different types of hardware, such as a keyboard, mouse or monitor

Output devices receive data from a computer. The output device converts this data to a display, projected image, or print-out.

COMMON OUTPUT DEVICES

MONITORS

A monitor, also known as a visual display unit (VDU), is the main output device of a computer, as it represents data in a visual form.



Figure 5.3: *An example of an LCD monitor*

The size of a monitor is measured in inches diagonally from one corner to the other.

Another important aspect of a monitor is its resolution. The clarity of text and images displayed on the monitor depends on the screen resolution. Screen resolution is measured in **pixels** (which is short for *picture element*). A pixel is a tiny dot of colour displayed on a monitor. The more pixels there are, the more detailed the picture will look. The screen resolution shows you how many pixels the screen can show horizontally and vertically. It is written as $1\ 280 \times 800$, which means that the screen can show 1 280 pixels horizontally and 800 pixels vertically.

A third important characteristic is the refresh rate of a monitor. This can be defined as how quickly the image on the monitor can be updated with the newest information.

PRINTERS

Printers allow a computer to convert electronic data (soft copy) into a hard copy (i.e. printed out on a piece of paper). Printers are generally used to print large amounts of text and documents, and can also be used to print graphics, such as photos.

For the purpose of this book, we will only look at the following four printers:

1. **Inkjet printer:** This is a very popular printer that can combine black, white and colour printing, all at the same time. These printers operate by using cartridges that contain the ink. When the print job starts, the ink heats up and flows through a set of tiny holes. The heating process results in a small droplet of ink forming, which is then released as a single dot, forming a part of the image or text. This happens continuously until the whole image or text has been completed. This is why inkjet print-outs are



Something to know

An inch is about 2.5 cm; therefore, a 22-inch monitor has a diagonal length of $22 \times 2.5\text{ cm} = 55\text{ cm}$.



Something to know

The resolution of a monitor is measured in PPI (pixels per inch).

sometimes still a bit wet. This type of printer can be seen in Figure 5.4 and are commonly used in homes.

2. **Ink tank printer:** An ink tank printer uses the same technology as an inkjet printer to print documents. The difference is that an inkjet printer's ink cartridges are replaced with ink tanks, which hold a much larger volume of ink (up to 70 ml). The ink tanks can be filled by the user when they are empty. Each colour is stored in a separate tank and only the colour that is used up, has to be refilled. Ink tank printers can print 1 000 pages or more. Ink tank printers are suitable for printing documents where a large number of copies are required.



Figure 5.4: Inkjet printer

3. **Laser printer:** This is a common printer used in the workplace. It can print a lot of sheets quickly and with high quality. A laser printer works by directing a laser beam onto a drum, which forms a mirror image of what has to be printed. This causes parts of the drum to be magnetised. When the drum rotates, it picks up the ink powder, called *toner*. The toner is then transferred to the paper, printing the letters or images on it with a hot roller bonding the toner to the paper. An example of this printer can be seen in Figure 5.5.
4. **3D printer:** 3D printing is the process of making a three-dimensional (3D) solid model from a digital file. It works by transforming a digital file into a 3D model by adding one layer of material at a time.



Figure 5.5: Laser printer



Figure 5.6: 3D Printer

Print resolution is measured in dots per inch (DPI). This is similar to the way pixels work on a computer. The printer settings can be adjusted to increase or decrease the DPI. The higher the DPI, the more dots there are, resulting in a high-quality print-out.



INKJET PRINTER VERSUS
LASER PRINTER



[https://www.youtube.com/
watch?v=x4lgj5zkB9U](https://www.youtube.com/watch?v=x4lgj5zkB9U)

The following table shows the comparison between various kinds of printers.

Table 5.4: *Advantages and disadvantages of different printers*

TYPE OF PRINTER	ADVANTAGES	DISADVANTAGES
Inkjet	- Portability: These printers are quite small in comparison to other printers and are, therefore, easy to move from one place to another (portable). - Cost: The initial cost of buying the printer is low. - Page cost: Cost per page is lower than the laser type. - Space: Compared to a laser printer, an inkjet printer takes very little space on a desk.	- Cost: Although the initial cost of the printer is affordable, the ink cartridges are quite expensive. - Speed: These printers are slow and are not suitable for high volumes of printing. - Quality: Copies might be slightly wet after the print job and might require time to dry.
Ink tank printer	- Portability: These printers are comparable in size to inkjet printers and can easily be moved from one place to another. - Cost: The cost of refilling ink tanks is cheaper than replacing an ink cartridge. - Page cost: Cost per page will be lower than an inkjet printer.	- Cost: The initial cost is more than that of an inkjet printer, but comparable to that of a colour laser printer. - Speed: Comparable to an inkjet printer, but can be used for high-volume printing.
Laser	- Quality: Has high-quality print-outs. - Speed: It can print more pages at the same time than the inkjet. - Cost: Replacing toner in this type of printer is cheaper than inkjet printers. - Noise: This type of printer hardly makes any noise.	- Cost: A laser printer is more expensive than an inkjet printer. - Space: Laser printers are much bigger and heavier than inkjet printers. It therefore takes more space. - Maintenance: Servicing or fixing can be quite costly.
3D	- Speed: Speed of the parts of a design can be produced much quicker than normal manufacturing methods. - Labour cost: One of the main advantages of 3D printers is the low cost of labour. - Customisation: 3D printers allow complete freedom for customising designs.	- Quality: The quality of a part is not as efficient as manufacturing a part. - Expensive: 3D printing equipment and material costs are expensive. - Material: At the moment, the type of materials that can be used in 3D printers is mainly plastic. - Emissions: 3D printers in homes can produce toxic emissions.



Activity 5.3

Lesedi's dad got an inkjet printer for the house; however, it is not working and no printing is taking place. It seems that there is no communication between the printer and the computer.

Answer the following questions regarding the scenario:

1. Which connector do printers usually have?
2. Briefly explain why a printer is an output device.
3. List two advantages and two disadvantages of the printer that Lesedi's dad bought.
4. Explain one possible reason why the printer and computer are not communicating.

5.4 Storage devices



Something to know

There is a difference between *storage media* and a *storage device*. *Storage media* is anything that holds information. It refers to the digital medium that is used in the storage devices, for example magnetic disks, cards, barcodes, etc. A *storage device* is a device that uses the storage medium and provides a physical interface, such as a CD/DVD player, an external hard drive, a USB flash drive, etc.

Storage devices are used to store data. There are a variety of storage devices with differences in storage capacity, speed, portability, as well as functions. In this section, you will learn about the different types of storage devices and how they are used – i.e. how much data needs to be stored, how quickly the data needs to be transferred and how portable the storage needs to be.

CONCEPTS USED IN STORAGE CAPACITY

Before we discuss storage devices in great detail, let's quickly look at the different capacities storage devices can have.

Data on computers is stored as numbers, which consist of only 0s and 1s. Computer memory is made up of electronic components, which can either be on (1), or off (0). The smallest unit of data storage (0 or 1) is called a **bit**. A number made up of eight bits (11111111) is called a byte.

Table 5.5: *Measuring capacity*

CAPACITY	DESCRIPTION
Kilobytes (Kb)	This is defined as 1 024 bytes. A normal word document is usually about 100 kilobytes.
Megabytes (Mb)	This is defined as a 1 024 Kb. Programs, videos and music are examples of media that are measured in Mb. An average music file would be about 6 Mb.
Gigabyte (Gb)	This is defined as approximately 1 024 Mb. In 1988, a computer had a 30 Mb hard drive. Now, the hard drive of the average computer has about 400 Gb of storage space.
Terabyte (Tb)	One TB is approximately 1 024 Gb. External hard drives and modern computers have at least 1 Tb of storage available.

COMMON STORAGE DEVICES

Nowadays, we are used to hundreds of gigabytes of storage capacity in our computers. With looking at storage devices, there are three main things to remember:

- 1. Storage capacity:** This determines how much data can be stored on the device.
- 2. Storage speed:** This determines how quickly new information can be written to the device.
- 3. Reliability:** This determines the likelihood of the device breaking down.

We will look at the following storage devices:

1. Internal hard drive (**fixed**)
2. External hard drive (portable)
3. SD card
4. **Solid-state drive (SSD)**
5. Flash drive

Table 5.6: Types of common storage devices

STORAGE DEVICE	DESCRIPTION	USES	IMAGE
Internal hard drive	Internal hard drives are found inside computer systems. Hard drives are not really used to transfer data between computers, or even to people. This is because most hard drives are fitted in the computer case and this is usually done with screws, which makes it difficult to move. Modern hard drives can have up to 12 Tb of storage space.	Hard drives are the main storage device in computers because of their high speed and high storage capacity. It is used to store data in the form of images, music, documents and videos.	
External hard drive	External (or portable) hard drives are non-volatile hard drives that are generally used outside the computer case and can be attached to the computer through a USB connection or wirelessly.	It is quick to connect to other computers and is really good in transferring files from computer to computer.	
SD card	SD cards, especially MicroSD cards are small storage devices that can have a storage capacity of up to 512 Gb.	Due to their small size, SD cards are often used in portable devices, such as smartphones and cameras to provide extra storage capacity.	
SSD	Solid-state drives (SSDs) are a type of internal hard drive that does not contain any moving parts, such as internal hard drives. It is quite similar to a CPU that has many small transistors, with each transistor storing one bit of data. Because of this, it has a much lower storage capacity and much higher cost per gigabyte.	Stores data and can be used in some laptops to transfer data.	
Flash drives	Flash disks are small, portable devices that connect to a computer using a USB port.	Flash drives are used when transferring files from computer to computer.	

LIFE SPAN OF A HARD DRIVE

Hard drives contain sensitive moving parts and because of this, they can fail. Data stored on hard drives should always be backed up (stored on another device). Modern hard drives can last for more than ten years.

The following signs show that the hard drive might be failing:

- The computer is slowing down
- There is a clicking or grinding sound
- The computer freezes frequently



Something to know

In January 2018, *SanDisk* (a flash disk manufacturer) unveiled the world's smallest one Tb flash disk at the consumer electronics show in Las Vegas. Although this is still only a **prototype**, it may eventually completely eliminate the need for external hard drives.

CAPACITY AND ROBUSTNESS OF STORAGE DEVICES

The following table summarises the average capacity and robustness for different types of storage devices, provided that they are used correctly.

Table 5.7: *The average capacity and robustness of storage devices*

STORAGE DEVICE	STORAGE CAPACITY	ROBUSTNESS
Internal HDD	Up to 2 Tb	Hard drives contain sensitive moving parts and can be damaged if bumped or dropped, but they can last for many years if they are treated with care.
Portable HDD	Up to 2 Tb	Quite robust, but can easily get damaged if dropped. Some have a protective casing.
SSD	Up to 4 Tb	An SSD has no moving parts and cannot easily be damaged.
Flash drive	Up to 2 Tb	Flash drives are robust as long as they are treated carefully. They do not have moving parts.
SD card	Up to 512 Gb (but 1 Tb is possible)	SD cards are robust and cannot easily be damaged.

CONNECTING PERIPHERALS

A peripheral is an external device that is connected to a computer. A PC on its own cannot function efficiently. It needs peripherals, such as a mouse or keyboard for the PC to operate properly.

Peripherals can be both cabled (wired) or wireless, and uses USB, Bluetooth or Wi-Fi. In this section, we will look at the different methods of connecting peripherals.



Guided Activity 5.1

CABLED CONNECTIONS (USB)

To connect a keyboard with a USB cable, you can do the following:

1. Plug the USB cable of the peripheral device into the computer's USB port.
2. The computer will register the keyboard. You can then start using the keyboard immediately.

WIRELESS CONNECTIONS (BLUETOOTH)

To connect a new peripheral using Bluetooth, you can do the following:

1. Open *Settings*.
2. The *Settings* window will open.
3. Click on *Devices*.
4. From the left panel, click on the *Bluetooth & other devices* tab.

... continued



Take note

Install any drivers if prompted.



Guided Activity 5.1

... continued

5. Turn the *Bluetooth* toggle switch on.
6. Click on the *Add Bluetooth or other device* button.

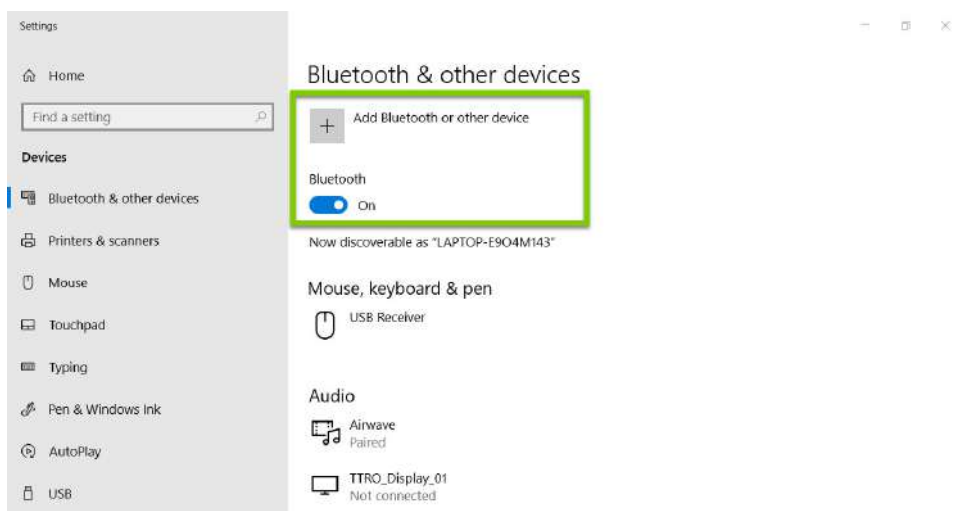


Figure 5.6: Adding a new peripheral using Bluetooth

7. Select the *Bluetooth* option.
8. Switch on the peripheral device.
9. A list of peripheral devices will appear. Choose the device and click it to connect.
10. Click on the connect button to pair.
11. Click *Done*.

Wi-Fi

To connect a new peripheral using Wi-Fi, you can do the following:

1. Click on *Start > Devices and Printers > Add a printer*.
2. The *Add a printer* window will open.
3. Choose the *Add a network, wireless or Bluetooth printer* and follow the instructions.



Activity 5.4

1. What is the smallest unit of storage capacity?
2. Which one of the following options describes the type of devices used to keep an electronic record of work done on a computer?
 - A. Input
 - B. Output
 - C. Processing
 - D. Storage
3. Fill in the missing word: There are 1 024 kilobytes in a _____.
4. Generally speaking, a hard drive has an average life span of three to five years, depending on the brand, size and type of environment.
 - a. Do you agree with the statement? Motivate your answer.
 - b. List two ways that indicate a hard disk is starting to fail.
 - c. Briefly explain why an internal hard disk is more prone to damage compared to a portable hard disk.

5.5 Green computing

Green computing is about using computers in an environmentally responsible and eco-friendly manner. Green computing aims to reduce the use and impact of hazardous materials, maximise energy efficiency and promote recycling of computers that are no longer being used.

Green computing tries to reduce the environmental effect of computers, by making computers more efficient and making sure the computers are built from biodegradable materials.

Some examples of green computing include:

- Computers going to “sleep” after they have been inactive for a certain amount of time.
- Data stored on SSDs use considerably less electricity than data on disk drives.
- Improved CPU architecture allows less electricity to create more power.
- More efficient software requires less processing and storage space, which requires less electricity.
- Better resource allocation means unnecessary computers in data centres can be turned off, saving a lot of electricity.

These improvements have already made significant improvements to the energy use of computers.

GOING GREEN WHEN PRINTING

With printing, you need to understand the negative effect that printing has on the environment. Currently, too much paper is being wasted for printing; most printed copies are generally ignored or thrown away after they have been used once. We should print documents when necessary, and rather file and send documents electronically (for example, through email). You can also save paper by printing on both sides of the paper (double-sided printing), as well as by recycling paper.

Disposing printer cartridges in the wrong way (throwing them out in the environment) can cause serious damage to the environment. Rather, find a disposal outlet that has special bins for this purpose, for example, this type of outlet can be found at a local supermarket, such as PicknPay.



Figure 5.7: Example printer cartridge recycling outlets

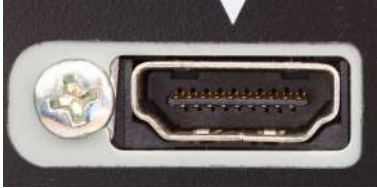


Activity 5.5

1. Explain the aim of green computing.
2. Describe three ways in which green computing can be achieved. Use ONLY examples related to hardware and software.

REVISION ACTIVITY

1. Match the ports in Column A to the connectors in Column B.

COLUMN A	COLUMN B
1. 	A. 
2. 	B. 
3. 	C. 
4. 	D. 

(4)

2. Write down the missing numbers:
 - a. 1 Mb = _____ Kb (1)
 - b. 1 Tb = _____ Mb (1)
3. In some laptops, internal hard disk drives are being replaced by SSDs.
 - a. What do the letters SSD stand for? (1)
 - b. Mention one advantage of using an SSD. (1)
 - c. Mention one disadvantage of using an SSD. (1)

... continued

REVISION ACTIVITY

... continued

4. Mr Chotia has a small work-from-home business. He has a computer that he uses to do basic accounting and keeps a record of appointments with clients. He also needs to print out quotations and receipts.
 - a. Mr Chotia has purchased a desktop computer. Other than a printer, list three hardware devices that are essential for Mr Chotia to be able to use the computer. (3)
 - b. Which of the devices mentioned in (a) above can be used as both an input and as an output device? (1)
 - c. Mr Chotia has been advised to purchase a laser printer. Comment on this advice. (2)
 - d. Mr Chotia loads a file with a customer receipt into a word processor. He then prints out the receipt. Which of the above (file or print-out) would be referred to as a hard copy? (1)
 - e. Mention two ways in which Mr Chotia could connect the printer to the computer. Which method would you recommend? (4)
 - f. Mr Chotia is an environmentally aware member of the community. Mention two ways in which he can save paper in his business. (2)

TOTAL: [22]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Learn what hardware is.		
2.	List the different ports and connectors.		
3.	Learn what input is.		
4.	Learn about the different input devices.		
5.	Learn what output is.		
6.	Learn about the different output devices.		
7.	Learn what storage is.		
8.	Learn about the different storage devices and their capacities.		
9.	Understand the importance of green computing.		

EXTENDED HARDWARE CONCEPTS

CHAPTER OVERVIEW



Unit 6.1	More about input devices
Unit 6.2	More about output devices
Unit 6.3	Storage media and devices
Unit 6.4	Processing devices
Unit 6.5	Caring for your PC and storage devices

At the end of this chapter, you should be able to:



- Describe the different input devices.
- Describe the different output devices.
- Explain the different storage media and devices.
- Explain what a processing device is.
- Describe the different processing devices.

INTRODUCTION

Hardware devices can also come in the form of pointing devices, scanning and reading devices, microphones and many more. In this chapter, you will learn more about the different types of hardware devices, their features and uses.

6.1 More about input devices

As you learned earlier, an input device is any hardware device that allows you to enter data into or interact with the computer. This section will focus on the following types of input devices:

- Pointing devices
- Digital cameras
- Scanning and reading devices
- Video input devices
- Audio input devices
- Biometric input devices

POINTING DEVICES

Pointing devices are used to control the movement of the cursor on the screen. Two examples of a pointing device are the mouse for the desktop computer and the touch pad for the laptop.

Table 6.1: *Different types of pointing devices*

TYPE OF POINTING DEVICE	DESCRIPTION	IMAGE
Touch pad	These are small rectangular pads that are mostly found on notebooks and laptops. You can move the mouse pointer on the screen by moving your fingers on the pad. Although notebooks and laptops do not need additional devices to move the mouse pointer, it is not the most efficient way to do it.	
Trackball	This is a stationary pointing device; an “upside down mouse” as some people call it. It has the same functions as a mouse, although the user makes use of the ball to move the cursor.	
Touch screen	Touch screens enable you to use your fingers to press the keys directly on the screen. Most smartphones and tablets are designed with touch screens, and it seems that notebooks are also following this design.	

... continued

TYPE OF POINTING DEVICE	DESCRIPTION	IMAGE
Stylus	A stylus is a pen-like device that is used draw, write, or select an object on a flat surface, such as the computer screen, mobile device, or tablet.	
Joystick	A joystick or game controller is a type of input device mainly used for games. It is made up of one or more buttons for special functions, such as moving an on-screen character and other action buttons linked to specific actions inside a game.	

DIGITAL CAMERA

A digital camera is designed to capture and store images on a [memory card](#). Digital cameras are available in different price ranges and require no film.



Figure 6.1: An example of a digital camera

SCANNING AND READING DEVICES

SCANNING DEVICES

Scanning devices are commonly used in offices, schools and even homes to scan documents, such as photographs and pages of text, and then convert it into a digital format. It reads typewritten, computer-printed or handwritten characters from ordinary documents and translates them into a form that the computer can understand.

Table 6.2: *Types of integrated scanning devices*

SCANNING DEVICE	DESCRIPTION	IMAGE
Flatbed	After the document or picture has been scanned, an electronic file is created by the scanner in the computer's memory.	
Multi-functional	Some printers come with four-in-one functions, which means that it can scan, email, copy and print.	
Handheld	A barcode scanner is an example of a handheld scanner. It is generally found in supermarkets or warehouses and is used to read a product's barcode. It uses a laser beam that uses light patterns that are able to decode vertical barcode lines.	
Mouse	This is a small and portable scanner that can easily be carried around. It can be used for text scanning, but because of its limitation in terms of resolution, it is not recommended for scanning photographs.	
Smartphone	A smartphone has the ability to act as a barcode and QR scanner. It can also scan documents, provided that you download the application for it. Smartphones scan by using the built-in camera of the phone.	

READING DEVICES

There are several fields where reading or scanning technology is applied. In this section, we will explain these in some more detail.

RADIO FREQUENCY IDENTIFICATION (RFID) READERS

RFID is a form of wireless communication that uses radiowaves to automatically identify and track tags attached to objects. These tags, called RFID tags, have the advantage in that they do not have to be positioned precisely relative to the reader, but can work within a few metres away (more or less five metres) from the reader.

Although this technology has been available for many years, mostly to track cattle, it has become a common technology for tracking consumer products worldwide. These tiny tags can be added to any product and can be scanned by an RFID reader. For example, manufacturers use these tags to help track the location of each product they produce until each is sold.



Figure 6.2: Type of RFID scanner

RFID tags come in various shapes and sizes, and may be covered in a variety of materials, such as:

- Rice-sized tags inserted underneath the skin of animals to identify them
- Screw shaped tags to identify trees or wooden items
- Credit-card shaped for use in access applications
- Anti-theft hard plastic tags attached to merchandise in stores
- Heavy-duty millimetre-rectangular transponders used to track shipping containers, heavy machinery, trucks and railroad cars

Table 6.3: Types of readers

READER	DESCRIPTION	IMAGE
Magnetic strip	A magnetic strip is often found on bank cards. A magnetic strip reader is able to scan the information contained in the strip, such as a bank account number, card number, name, expiry date of the card and pin number.	
Barcode	This is an electronic device that is used to decode and capture information in a barcode. This type of reader is used in places, such as shops and warehouses. A barcode reader is one of the devices that is used in a POS system.	
Quick response (QR) code	A quick response (QR) code is a type of barcode that contains information about the item to which it is attached. The user can download a QR reader application on his or her phone. You can use a QR reader to pay a bill at a restaurant, for example.	



UNDERSTANDING URLS

Have a look at this video to understand how to use Zapper.



https://www.youtube.com/watch?time_continue=10&v=9SW1dnqctGA



Something to know

Before OCR technology, the only option to **digitise** paper documents was to retype them. This was a very time-consuming process and had the disadvantages of inaccuracy and typing errors.

Example 6.1

Nowadays, you can use your phone to pay the bill, or even split the bill with a friend. This is done by scanning the QR code on the bill with Zapper. First, you have to sign up to the application and when the bill comes, use the application to pay the bill.

OPTICAL CHARACTER RECOGNITION (OCR) SCANNERS

OCR scanners, also called *optical character readers*, can recognise almost any kind of typed, handwritten, or printed text inside images, for example in scanned documents or photos, and convert it into machine-readable text data.



Figure 6.3: An OCR scanner

The most popular application of OCR scanners is converting printed paper documents into machine-readable text documents that can be edited with a word-processing program. However, it can also be used to sort magazines and letters, and therefore, speed up mail delivery, for example.

VIDEO INPUT DEVICES

Video input devices allow you to capture media, such as pictures, videos and sound, which can be stored on a computer, or on a storage device and accessed whenever you need them. The following are examples of common types of video inputs.

Table 6.4: Types of video input devices

DEVICE	DESCRIPTION	IMAGE
Video camera	Video cameras are designed to capture and store videos on a memory card. A video camera functions as an input device, as it can be connected to a computer or television to view the videos directly through these devices. The videos can also be transferred to a computer for storage purposes and viewing later.	
Web cam	A web cam is short for 'web camera' and is used to capture still pictures, or video images of a scene in front of it. It can be either built into a computer, or connected using a USB cable. It is used when making video calls, recording interviews on a computer, or taking part in a video Skype meeting.	

AUDIO INPUT DEVICES

Audio input devices allow you to communicate with a computer using your voice. Because of the world in which we live, you can now talk to your friend in Spain via Skype, or video calling with the proper audio input. Some examples are discussed in the table below.

Table 6.5: *Types of audio input devices*

DEVICE	DESCRIPTION	USES	IMAGE
Microphone	A microphone is an input device used to input sound into a computer that is then stored in a digital form to be played back later. It can be used, for example, to add sound to a multi-media presentation, or to mix music.	Recording voice for podcasts: A podcast is an audio broadcast that is often listened to on a computer.	
Voice recorder	A voice recorder refers to any device capable of recording a voice message.	VoIP: VoIP stands for “voice over internet protocol ”, which makes it possible for users to make calls over the internet. Computer gaming: Used to talk to your team mates during a game.	

VOICE RECOGNITION

Voice recognition, or speech recognition, is a type of computer program that is able to interpret spoken instructions and respond to them. In most cases, a microphone is necessary to be able to use the software. This microphone can be a separate device (desktop computer), or built-in (laptop and smartphone).

Computers using the Windows operating system include a speech-recognition program included called *Windows Speech Recognition*. Most smartphones have a built-in speech recognition software.

BIOMETRIC INPUT DEVICES

Biometric inputs measure a person's unique physical characteristic, with the most commonly used biometric being fingerprints. Other biometrics are hand or palm geometry and retina, iris, or facial characteristics. Behavioural characters include signature, voice, keystroke pattern and a person's manner of walking.

Biometrics can be integrated into any application that requires security, access control and the identification and verification of people.



Something to know

These days, smartphones have built-in voice recognition, which is designed to decode your voice and carry out verbal commands. The idea behind voice recognition is for the user to save time by designing software that can decode the human voice, eliminating the need for a keyboard. So, next time you are driving, and you need to find your friend's location and text him something, remember the following:

With Google, tap the microphone symbol and state your command, for example: “Search for no. 84, Charles Street”, or “text Ronny”. The phone will do everything else for you!

Table 6.6: *Different biometric devices*

DEVICE	DESCRIPTION AND USES	IMAGE
Fingerprint scanner	Fingerprint scanners are one of the most developed and commonly used biometric devices. It serves as a type of authentication device that verifies the user's fingerprint by comparing it with the saved fingerprint data. It can, therefore, not be stolen, forgotten, borrowed, or forged. Nowadays, it is also a feature of most smartphones.	
Retina scanner	Uses the unique patterns on a user's retina blood vessels to identify the user. It is commonly used in high-security areas, such as research labs, banks, as well as government organisations.	
Iris scanner	Uses the patterns found in the coloured ring of tissue that is found around the pupil to identify the user. It is highly accurate and is used all over the world. It is used as a security at some banks and on some smartphones. In some countries, such as the United Arab Emirates (UAE), it is used at border posts to identify travellers.	
Facial recognition scanner	Facial recognition identifies the user by matching the nodal points of the user's face to the print that is on the system. Nodal points are different characteristics on a person's face that measure the length and width of the nose, cheekbones and eye socket width, for example. It is used in security systems. Social sites, such as Facebook, use facial recognition to tag users in pictures.	



Something to know

Biometric inventions, such as fingerprint sensors and iris scanners, are commonly used in big companies and airlines to identify a person. However, experts predict that soon these biometric inputs will fall away, and facial recognition will rule the biometric security market.



Activity 6.1

1. Touch screens enable users to use their fingers to press keys directly on the screen.
 - a. Give two examples of devices that use touch-screen technology.
 - b. What device can be used on a touch screen to draw or write on it? Also, give a brief description of this device.
2. Briefly explain the function of scanning devices.
3. Barcode readers are used in a variety of environments.
 - a. List any two instances where a barcode reader can be used.
 - b. Explain briefly how a barcode reader works.
4. Video inputs measure a person's unique physical characteristic, with the most commonly used biometric being fingerprints.
 - a. Which type of reader uses the patterns found in the coloured ring of tissue that is found around the pupil to identify the user?
 - b. List two places where this device is used.
 - c. What other scanner uses the eye as a form of identification?
 - d. Biometric devices are excellent in verifying a person's identity. List two advantages and two disadvantages.

6.2 More about output devices



Something to know

Smart speakers are a new technology that is fast growing. It is currently available as single-unit wireless speakers with **artificial intelligence (AI)** from Amazon, Apple, Google, Microsoft, or Samsung built into them. This means that you can ask your smart speaker, for example, to play a song or playlist, to control the lights in a room, or ask for a weather report, and it will give you that.

Output devices are used to send information from a computer to another device, or to you as the user. In this section, you will learn about the following output devices:

- Headsets and speakers
- Multi-function devices
- Data projectors

AUDIO OUTPUT

Audio output devices convert data on a computer into sound. This allows you to listen to music, chat to friends, or watch a movie or episode of a series on a computer.

Table 6.7: *Types of audio output devices*

DEVICE	DESCRIPTION	IMAGE
Headsets	Headsets are also referred to as earphones and are a hardware output device that is plugged into a computer, smartphone, or MP3 player so that you can listen to audio privately, without anyone else hearing what you are listening to.	
Speakers	Speakers convert data on a computer into sound. You can listen to music, or watch a show using your speakers. Determining the quality of speakers is a difficult task; however, having more speakers can increase the surround sound , but not necessarily the quality. Therefore, it is always advised to have two quality speakers to produce quality sound rather than many low-quality speakers.	

OTHER OUTPUT

There are a variety of other output devices used with computers and this section will look at some of these devices.

MULTI-FUNCTIONAL DEVICES

Multi-functional devices are able to perform various functions of several devices, but only takes the space of one, such as a four-in-one printer, where you can print, copy, fax and scan using just the one hardware device.

DATA OR DLP PROJECTOR

Projectors use a bright light to project the images that would be displayed on a monitor on any flat surface, such as a wall or a screen. This then replaces the image on the monitor and an enlarged magnified image is shown on the screen, which can be viewed by a group of people. Projectors are used in schools and offices, with bigger projectors used in cinemas.

There are two types of data projectors:

1. Liquid crystal display (LCD) projectors
2. Digital light processing (DLP) projectors

Most projectors use DLP technology.



Figure 6.4: Example of a projector



Activity 6.2

Match the terms in Column A with the descriptions in Column B.

COLUMN A	COLUMN B
1. Headsets are also referred to as _____	A. Projector
2. Converts data into sound	B. Watts
3. Multi-functional device	C. Speaker
4. Replaces the image on the monitor and magnifies the image on the screen	D. Earphones
5. Loudness of a speaker	E. Printer

6.3 Storage media and devices

In the previous chapter on hardware, you learned that storage devices all serve the same general purpose – i.e. to store data. In this section, you will learn more about:

- CDs, DVDs and Blu-Ray
- Memory cards
- Card readers

Table 6.8: *Types of storage devices*

DEVICE	DESCRIPTION	USES	IMAGE
CDs, DVD and Blu-Ray	Compact discs (CDs), Digital Versatile Discs (DVDs) and Blu-Ray discs are still three of the most commonly used forms of portable storage. The sizes of the different media are: CD: Up to 0.9 Gb DVD: Up to 9.7 Gb Blu-Ray: Up to 50 Gb	This is mainly because music, series and movies are stored and sold on these discs. However, data files from a computer can also be copied on the discs, using the correct CD, DVD or Blu-Ray writer. Even though DVD drives can read CDs, CD drives cannot read DVDs and a special Blu-Ray drive is needed to view and use Blu-Rays.	
Memory card	A memory card is a storage device. You will find it in many portable electronic devices, such as phones, digital cameras and media players.	It is mainly used for storing digital information. Although it can be used to transfer files between different devices, it is usually too small and too expensive to use for this purpose.	
Memory card reader	This device reads memory cards. It is usually built-in in some printers and smartphones, as well as most personal and tablet computers.	Used to read memory and SD cards.	



Something to know

CD/DVD drives in laptops are starting to become obsolete. Nowadays, manufacturers have started to leave optical drives out, which enables the laptops to be lighter and less expensive.



Activity 6.3

There are many storage devices that are available currently.

1. Which type of storage device is best used in a smartphone?
2. Briefly describe the storage device you mentioned in (a).

6.4 Processing devices

When a computer receives data from an input device, for example the mouse or keyboard, the data must be processed before it can go to an output device, such as the printer or monitor. A processing device is responsible for converting that data into useful information.

In this section, we will look at the following processing devices:

- Motherboard
- **Central Processing Unit (CPU)**
- **Random Access Memory (RAM)** and **Read-only memory (ROM)**

MOTHERBOARD

The motherboard can be compared to the skeleton of the human body; it connects all the parts of a computer together. You can also think of it as a mother that holds all her children (pieces) together.

The motherboard is a large printed circuit board connecting all the different components in a computer, such as the CPU and RAM.



Figure 6.5: An example of a motherboard

There are also connectors on a motherboard allowing you to connect a power supply to the computer and finally, it has a number of ports in the back that allow you to easily plug in devices, such as keyboards, speakers and monitors.

CENTRAL PROCESSING UNIT (CPU)

Think of a computer as a human body with the CPU being the brain, controlling everything the computer does.

The CPU is the part of a computer responsible for receiving and carrying out computer instructions. It does this by making use of millions (or even billions) of transistors, which can each be switched on or off individually.



Something to know

Phones, tablets and other small computerised devices also have motherboards, often called *logic boards*.



EXPLORING A MOTHERBOARD



<https://www.youtube.com/watch?v=Zjrayb7E9tl>



Something to know

Many people call their computers the CPU – this is incorrect as the CPU is only a chip found inside a computer.

Each CPU can be made up of multiple **cores**, which are independent processing units that can complete tasks on their own. By adding multiple cores to a CPU (which is then called a *multi-core processor*), a computer can dramatically increase the processing power available to it.

The speed of a CPU is measured in **gigahertz (GHz)**, which shows how many instructions can be performed in one second. One GHz is equal to a 100 million hertz.

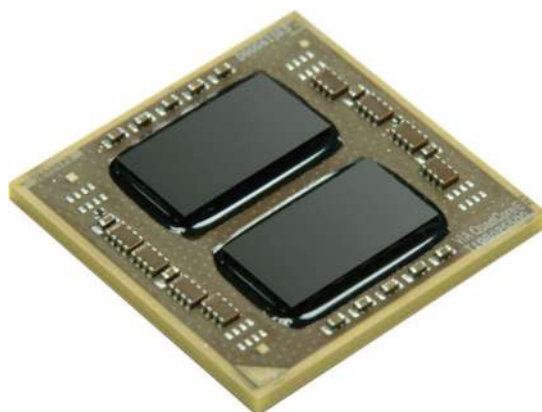


Figure 6.6: An example of a CPU quad core

The CPU was first developed during the 1970s at Intel, with the first processor, the 4004 processor, released.

The type of processor determines the speed and its capabilities. In the past, computer processors used numbers to identify and name the processor. For example, the Intel 80486 (386) processor is faster than the 80386 (286) processor. However, after the Intel Pentium processor was developed, processors started getting names, such as Pentium, Celeron and Duron.

MEMORY

You learned about hard drives earlier on. Other than the hard drive, there is the random access memory (RAM) and read-only memory (ROM).

Table 6.9: RAM and ROM

DEVICE	DESCRIPTION	FEATURES	IMAGE
RAM	RAM is used to store the programs and data used by the CPU in real time. Data on RAM can be read, rewritten, or erased when needed to.	The data that is stored on the RAM is erased the minute you switch off the computer. It requires a constant supply of power to keep the data safe. This is called volatile memory and is why we cannot use RAM as a permanent storage device, even though it is way faster than normal hard drives.	

... continued

DEVICE	DESCRIPTION	FEATURES	IMAGE
ROM	The ROM is used when a computer starts up; it uses the information that is stored on the ROM to start up.	ROM is a non-volatile memory; it will not forget any data that is stored on it, even after the computer is switched off, or if the power supply is turned off.	



Something to know

The first writeable RAM was magnetic core memory invented in 1951.



Activity 6.4

This will be a group activity. The teacher will arrange you in groups. Each group will be asked to research a specific processing device, such as:

- Motherboard
- CPU
- RAM and ROM

Each group should give a description of the device. Their role (use) in a computer and importance thereof. In order to do so, you need to do the following:

1. Each group must create a PowerPoint presentation with a maximum of eight slides (containing a cover, content and reference slide) on research of the assigned topic.
2. Each group will present the research done on the topic.
3. Your teacher will give you an assessment task to complete, based on the information found by all the groups.

6.5 Caring for your PC and storage devices



Something to know

Always remove a flash disk or external hard drive safely from the computer. This can be done by right clicking the *Removable storage* icon in the *Computer* window and select *Eject*. Yes, it might seem tedious, but in the long run, it will help you by preventing any data corruption.

You must always take care of your PC system and storage devices, as some of the components and the information loss can be quite costly. To take care of your PC, keep food and drinks away from it and keep back-ups on a storage device, such as an external hard drive. Also install an **antivirus program** and keep your PC clean. These are just some examples of taking care of your PC.

For storage devices, examples would be that CDs and DVDs can get scratched, making them unreadable. Flash disks are small and can easily land on the floor. When stepped on, they can become damaged. They can also be damaged by water and high temperatures, or other environmental factors. Even normal wear can cause them to become unusable.

Although external hard drives are normally resistant to damage, care should be taken not to drop them as they contain small parts that can easily be damaged.

Always take care of your computer, shut it down in the correct way when you are done using it, so that the computer can last for a long period of time.



Guided Activity 6.1

Do the following activity with the rest of your class mates. Your teacher will help guide you in this activity.

To check the size of a hard disk and the space available on a plugged-in storage device, the following should be done:

1. Double click on the *My Computer* icon in the *Start* menu.
2. From the *My Computer* window, right click on the icon that shows the drive that you want to look at.
3. In this case, we will look at the *Windows (C:)* icon.
4. Right click on the icon.
5. Click on *Properties* from the drop-down menu, this will display the properties of *Windows (C:)* in the dialogue box.
6. The dialogue box will show the disk space, available free space and the capacity of the disk, as seen in Figure 6.7.

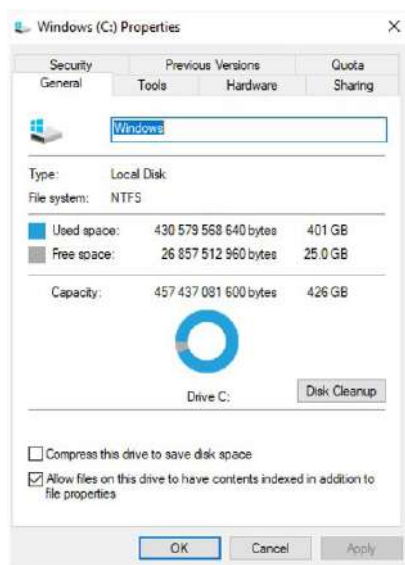


Figure 6.7: Properties window of Windows (C:).

REVISION ACTIVITY

For questions 1 and 2, write down the number of the question and the letter that matches the correct answer, e.g. 9 C.

1. Which of the following devices is MOST likely to have a touch pad?
 A. Desktop computer
 B. Foldable keyboard
 C. Laptop computer
 D. Touch-screen monitor (1)
2. Which of the following is NOT a pointing device?
 A. Trackball
 B. Stylus
 C. Digital camera
 D. Joystick (1)
3. Petrus works in the office of a large company. One of his responsibilities is to scan many documents and store them in digital format on the company server. Some of the documents can be up to 20 pages long. Should the company use a stand-alone flatbed scanner, or a scanner included with a multi-functional printer? Give a reason for your choice. (3)
4. Many people have a small RFID chip implanted in their pets. The chip stores a unique identity number. If the pet goes astray and is found by someone else, it can be taken to a veterinary practice where the vet can use an RFID scanner to check who owns the pet.
 a. What do the letters RFID stand for? (1)
 b. Mention one other use for RFID chips. (1)
5. Study the image below showing the components inside a desktop computer's tower.



- a. What is the function of the motherboard? (2)
- b. Without referring to the human brain, explain what the function of the CPU is. (2)
- c. Computers have two types of memory, RAM and ROM. Compare RAM with ROM by listing two differences between them. Write your answer in a table format. (5)
6. List two devices that use memory cards as additional storage. (2)

... continued

REVISION ACTIVITY

... continued

7. CDs, DVDs and Blu-Ray discs can all be used as storage media.
 - a. Which of these storage media has the lowest capacity? (1)
 - b. Which of these storage media would be most suitable to distribute a 3D movie? (1)
8. In a number of places in this textbook, you will find images like the one below that take you to a website or resource when scanned.



- a. This image looks like a barcode in two dimensions. What is it called? (1)
 - b. What software do you need to scan the image? (1)
 - c. Use a smartphone to scan the image. Write down the URL of the website that it opens. If you do not have a smartphone, ask a friend or your teacher to help you. (1)
9. Chris Hani Secondary School has installed a fingerprint scanner. Staff have to use the scanner when they arrive at school and again when they leave. The scanner software is used to keep records on staff attendance.
 - a. The fingerprint scanner is a biometric input device. Explain briefly what is meant by biometric input. (2)
 - b. Name one other type of biometric scanner. (1)
10. Faizal has bought a portable speaker from an online shopping site.



The following specifications were found on the website:

- Speaker Size: 3 inch
- Speaker output: 9W, 40hm
- Battery Capacity: 1000mAh
- Frequency response: 100-20kHz
- Bluetooth version: V2.1
- Supports up to 32 GB on a USB and SD card
- AUX-in port. Rechargeable lithium battery

- a. Faizal wants to use the speaker to play music that is stored on his smartphone. How would he connect his smartphone to the speaker? (1)
- b. Faizal wants to play music without connecting to his smartphone. Which two storage media could he use? (2)

TOTAL: [29]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Learn about pointing devices.		
2.	Learn about scanning and reading devices.		
3.	Understand what video and audio output devices are used for.		
4.	Understand the different types of biometric devices.		
5.	Learn about audio and video output devices.		
6.	Learn about the different types of projectors.		
7.	Understand the functions of the different processing devices.		
8.	Learn about storage media, such as CDs, memory cards and DVDs.		
9.	Understand the importance of taking care of your PC.		

CHAPTER 7

COMPUTER SOFTWARE

CHAPTER OVERVIEW

Unit 7.1 Basic software concepts

Unit 7.2 Operating system

Unit 7.3 Being more secure

At the end of this chapter, you should be able to:

- Describe what software is.
- Describe the types of software.
- Describe the differences, advantages and disadvantages of each type of software.
- Describe what an operating system is and give examples thereof.
- Describe what GUI is and give examples of some of the elements.
- Explain basic PC security.

INTRODUCTION

In Chapter 1, you learned that software is a set of programs or instructions that tells the computer and its hardware how to perform specific tasks. Without software, most computers would be useless, for example without Microsoft Word software it will be difficult to type out assignments and use specialised fonts and WordArt.

In this chapter, we will look at the basic software concepts, different software components and different types of software.

Something to know

The term “software” was not used until the late 1950s. In 1948, computer scientist, Tom Kilburn, was responsible for writing the world’s first piece of software. Kilburn and one of his colleagues created one of the earliest computers, which was called the Manchester Small-Scale Experimental Machine (SSEM), also known as the “Baby”. The SSEM used coded instructions programmed into it to do mathematical calculations. This piece of software took 52 minutes to correctly calculate 2^{18} , which is 262.144.

7.1 Basic software concepts

In this unit, we will look at the different concepts used in software. To begin, however, we will look at the different types of software.

Software consists of the following two categories, which will be explained in detail later on:

1. **Application software:** Application software allows you to perform tasks, such as playing games, creating slide show presentations, creating spreadsheets, and so on. There are many different types of application software programs.
2. **System software:** This includes the operating system and other software that controls and manages the operation of the computer.

APPLICATION SOFTWARE

Whether it is creating a report for work, typing out an assignment for school, messaging a friend, or just trying to find out where you can buy the latest PlayStation, there is probably an **application (app)** that can help you find what you are looking for. Unfortunately, there is no application that explains what it is, or what it does. So, in this section we will look at this term and how it has developed – and you do not even need an application to read it!

As you have learned, computer software is a set of instructions that tells a computer what to do and how to do it. Although the computer cannot run without system software, application software is the most useful type for computer users, as it helps us perform many of the tasks that make us use computers in the first place, such as web browsing, word processing and spreadsheet processing.

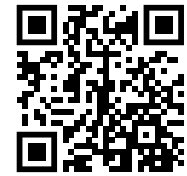


Figure 7.1: Different software applications

Over recent years, classifying applications according to computing platforms has become more difficult. A platform is the combination of hardware and software required to make an application run. In the past, the type of software running on a computer depended on the



WHAT IS SOFTWARE?



<https://www.youtube.com/watch?v=grrYbJqnSzY>

operating system, for example, some software works with a Windows operating system; other software only works with an iOS or Unix system. That has changed, however; new development in software has resulted in many applications no longer relying on a specific operating system, it runs on the **web** and can be accessed through a browser.

TYPES OF APPLICATION SOFTWARE

The table below shows some of the different types of application software installed on a computer. You should note that although software is identified as a program, it can be anything that runs on a computer.

Table 7.1: Examples and types of application software

APPLICATION SOFTWARE	EXAMPLES
Office suites	Microsoft Office, Libre Office, Google G-Suite
Internet browser	Chrome, Firefox, Internet Explorer, Edge
Movie player	VLC, Windows Media Player
Presentations	PowerPoint
Word processor	Microsoft Word
Portable Document Format (PDF)	Adobe Acrobat Reader, Foxit Reader
Operating system	Microsoft Windows, Linux, Android, iOS
Antivirus	Norton, AVG, McAfee, Symantec, Windows Defender
Spreadsheets	Microsoft Excel
Accounting	Pastel, QuickBooks
Gaming	Minesweeper, Solitaire, Counter Strike
Designing and graphics	Adobe Photoshop, AutoCAD

Some application software, such as *Wordpad*, *Notepad*, *Calculator*, *Paint* and the *Snipping Tool*, all come packaged with the system software of the Microsoft Windows operating system. We will briefly look at these basic programs and how they can be useful to you.

ACCESSORIES

Microsoft Windows includes a few programs, such as the *Calculator*, *Paint* and *Snipping Tool*. You can start any **accessory** from the *Start* menu. An accessory is any device or program that is not necessarily required, but it can be added to a computer to perform an additional function.

WINDOWS CALCULATOR

To use the Windows Calculator accessory you can do the following activities:



Guided Activity 7.1

1. Click the *Start* button.
2. Type in *calc*.
3. Click *Calculator* from the search results.

... continued

Something to know

In a few years' time, application software will develop even more. According to software developers, we are likely to see less emphasis on desktop PC software and an increased emphasis on **cloud** and mobile applications. Although this may benefit users in terms of user-friendliness, it can lead to potential privacy and security risks.

Something to know

You can access these programs by clicking on the *Start* menu and searching for the program.



Guided Activity 7.1

... continued

The calculator will appear on your desktop.

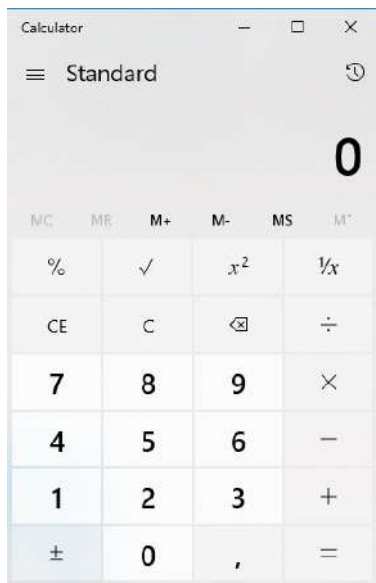


Figure 7.2: The Calculator application



Guided Activity 7.2

To perform a simple calculation, you can do the following:

1. Use the keyboard to type in the required data.
2. Or, you can type in the data by using the mouse and clicking on the buttons of the calculator.
3. Press the *Enter* key to get your answer.

SNIPPING TOOL

This tool captures all, or part of the computer screen as an image. You can save this image, attach it to an email, or even paste it into another application, such as Microsoft Word or Microsoft Paint.



Guided Activity 7.3

To access this program, you can do the following:

1. Click on the *Start* menu.
2. Type in *snip*.
3. Click on the *Snipping Tool* from the search results.

To capture a snip, you can do the following:

4. In the *Snipping Tool*, select *Mode*. You can choose any of the following types of snips:
 - Free-form snip
 - Rectangular snip
 - Window snip
 - Rectangular snip

... continued



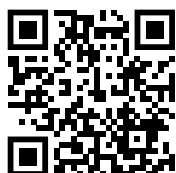
Something to know

Did you know that you can change the type of calculator from standard to scientific, or to a programming calculator by clicking on the icon () on the left-hand side of the calculator. So, next time you have some homework that requires a calculator, maybe this program could be of some use.



USING THE SNIPPING TOOL

You can learn how to snip the computer screen by opening the QR code shown below.



https://www.youtube.com/watch?v=J6S09zf_QL0



Guided Activity 7.3

... continued

5. Select the area that you want to snip.

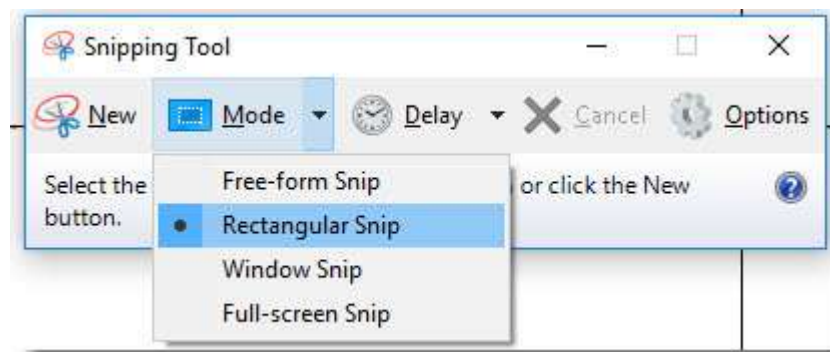


Figure 7.3: The Snipping Tool

6. After you have captured your snip, save the snip.
7. This is done by clicking on *File > Save As*.
8. Type a file name and then click on *Save*.

PAINT

Microsoft Paint is an accessory that provides virtual pens and brushes to create art or even text.



Guided Activity 7.4

To use this program, you can do the following:

1. Click on the *Start* button.
2. Type in *paint*.
3. Click *Paint* in the search results.

The *Paint* window will then open and you can create your own canvas by clicking or dragging the mouse, or by using the brushes.

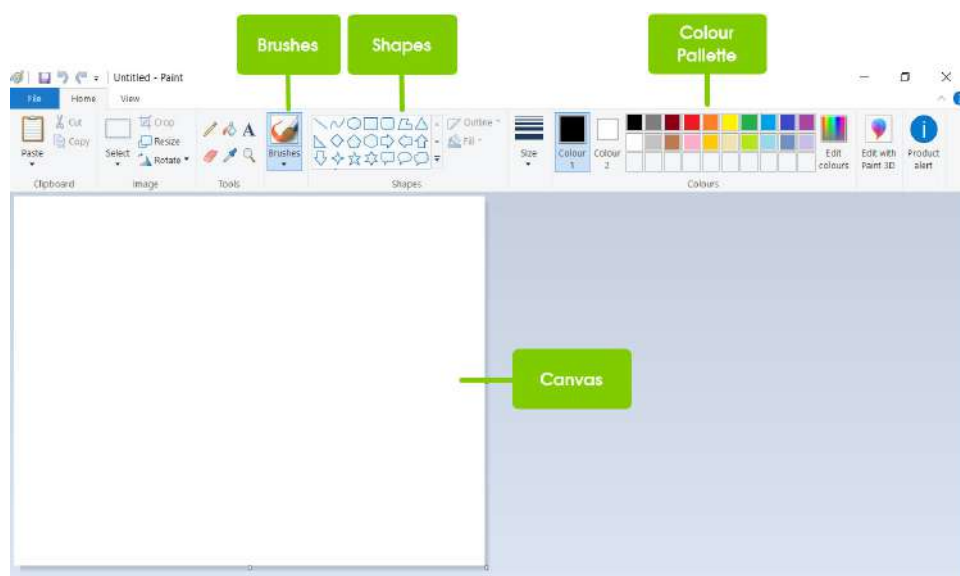


Figure 7.4: The Paint tool

SYSTEM SOFTWARE

System software is the specific type of software that manages the computer. Much like a traffic officer at a busy intersection, the system software manages the flow of data and information in the computer. This is done in the following ways:

- Translating the instructions from the software to a language that the hardware can understand
- Sending these instructions to the correct hardware
- Receiving information from the hardware (such as, the button press on a keyboard) and sending it to the software
- Allowing different applications to speak to each other
- Making sure all the hardware and software on the computer are in a good condition

The operating system (OS) is the most important type of system software as it manages all the other programs in the computer.

TYPES OF SYSTEM SOFTWARE

The table below shows some of the different types of system software installed on a computer.

Table 7.2: Examples and types of system software

SYSTEM SOFTWARE	EXAMPLES
Operating systems	Microsoft Windows, Linux, Android, iOS
Device drivers	Drivers for the mouse, keyboard, sound card, display card, network card, printer, etc.
Firmware	Firmware is the operational software embedded within a flash drive, ROM, or EPROM memory chip for the operating system to identify it
Utilities: This kind of software is designed to maintain the computer	Antivirus and security software Disk partition services Disk defragmentation File compression Data back-up for security reasons Hardware diagnostic services Data recovery Firewall Screensaver



Activity 7.1

1. What is a platform?
2. Match the following software in Column A with the correct example in Column B:

COLUMN A	COLUMN B
1. Internet browser	A. AVG
2. Presentations	B. Chrome
3. Antivirus	C. VLC
4. Movie player	D. PowerPoint

7.2 Operating system



Take note

Before everything else, the operating system is the first thing loaded onto the computer. Without the operating system, a computer is pretty useless.



UNDERSTANDING THE OPERATING SYSTEM



<https://www.youtube.com/watch?v=pTdSs8kQqSA&list=PL4316FC411AD077AA&index=8&t=0s>

When you turn on a computer, it is nice to be able to move the mouse, play music, or even browse the web with just a click. However, as easy as it looks, there is a lot going on inside the computer and the real mastermind behind handling all the necessary tasks, is the operating system. Most desktops or PCs come preloaded with Microsoft Windows while **Macs** are preloaded with Mac OS X. Other PCs and desktops may use the Linux or UNIX operating systems.

Recently, operating systems are found in smaller computers and devices that we use every day; from smartphones to **wireless access points**. The purpose of an operating system is to organise and control both the hardware and software to perform specific tasks.

Any device that has an operating system allows you to change how the device works. This means that you can add a new security update, a new application, or even install a new operating system (depending on compatibility) whenever you want to. So, no need to throw your computer away each time you need to add a new application or update! As long as you understand how an operating system works, you can change some of the ways in which it operates; the same applies for your phone.

What exactly can an operating system do? In the following section, we will take a brief look at the role that the operating system plays.

THE ROLE OF THE OPERATING SYSTEM

The operating system of a computer manages all the software and hardware of the computer. In other words, the operating system has many different functions and jobs. In most cases, people generally have a number of programs running at the same time on their computers. These programs all need to access the computer's CPU, storage and memory. The operating system manages and coordinates these processes so that the program gets what it needs to run.

THE OPERATING SYSTEM AND GUI

Operating systems are usually preloaded onto any computer you buy. Most users use the operating system that comes with their computers; however, this can be updated or even changed when required. The most popular operating systems for PCs are Microsoft Windows, Mac OS X and Linux.

Modern operating systems use a graphical user interface (GUI). This interface allows you to use the mouse to click on buttons and the keyboard to type text; while everything is shown clearly on the display screen as graphics and text.

Each operating system has a different look; so, if you decide to change operating systems at some point, it might initially feel weird and unfamiliar. However, modern operating systems are designed to be easy to use and the basic operating principles are the same.

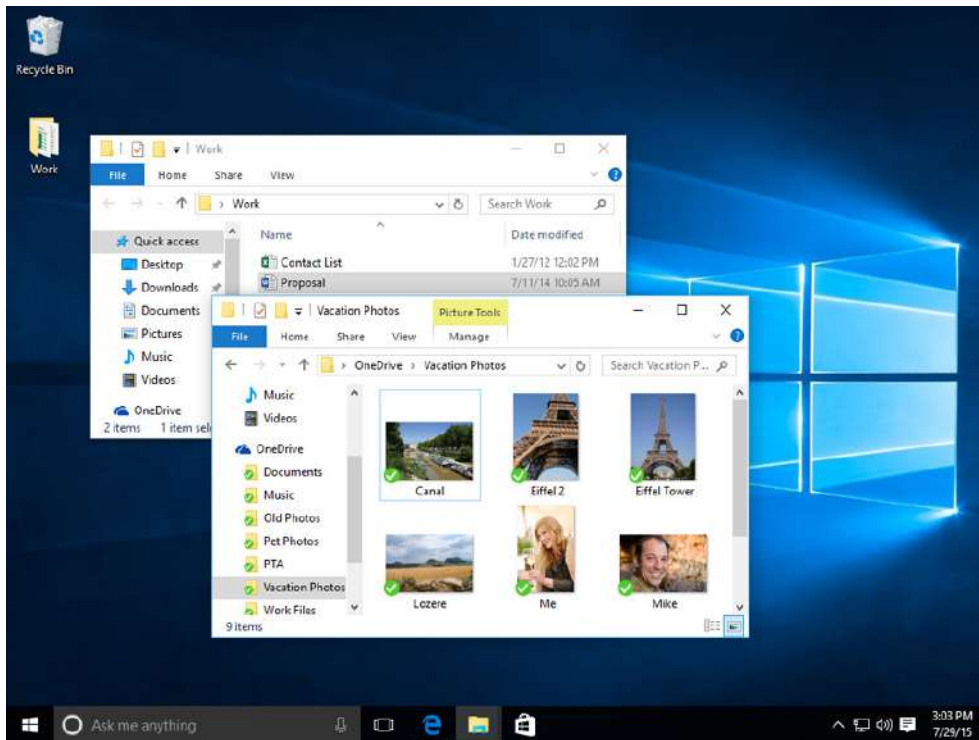


Figure 7.5: GUI of a Microsoft Windows operating system

TYPICAL FEATURES OF A GUI

A GUI uses windows, icons and menus to perform commands, such as opening, moving, closing and deleting files. Although most GUI systems use the mouse to navigate, it can also use hotkeys and arrow keys.

GUI elements are included, but not limited to the following:

- **Input controls:** Checkboxes, option buttons, list boxes and dialogue boxes
- **Navigational components:** Icons, menu usage and navigation
- **Informational components:** Toolbars and message boxes

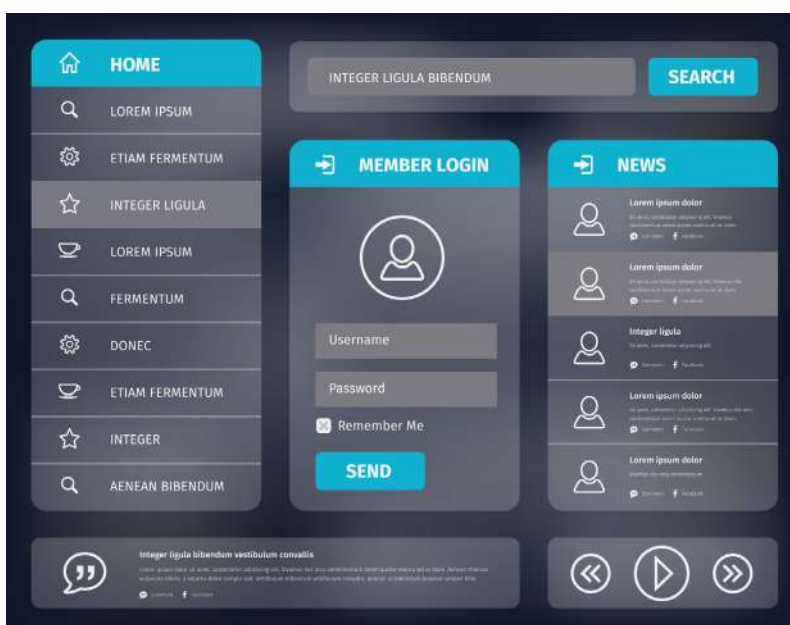


Figure 7.6: Examples of user interface elements

These operating system elements are grouped in tables and are briefly discussed below.

Table 7.3: Input controls

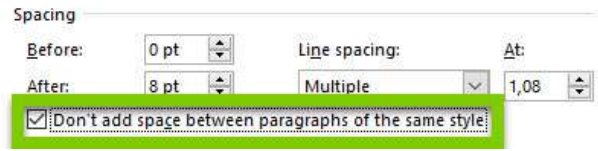
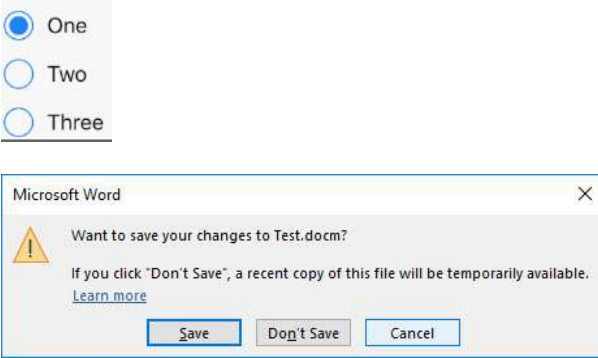
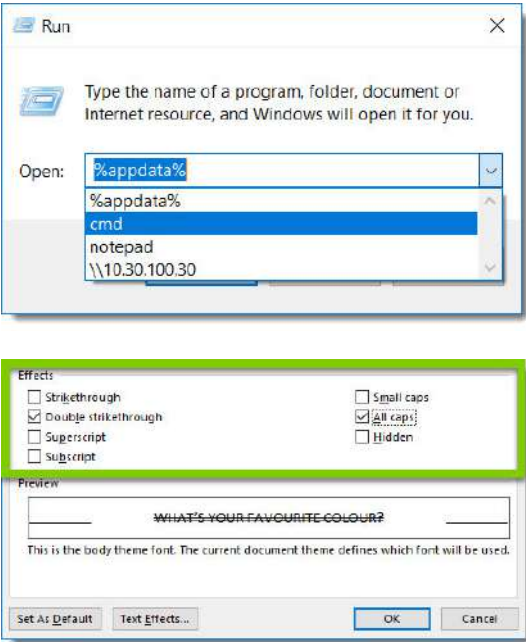
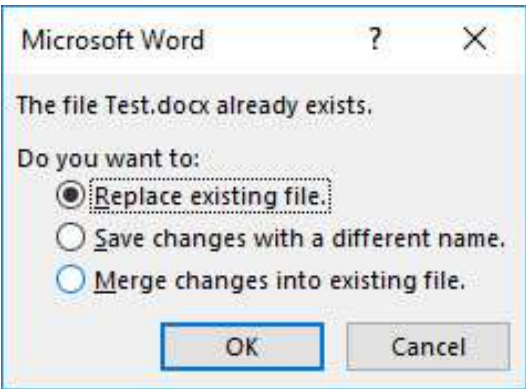
ELEMENT	DESCRIPTION	IMAGE
Checkboxes	Checkboxes allow you to select one or more options from the list.	
Option buttons	These buttons let you choose or select one item at a time.	
List and combo boxes	List box: This is a box that has a variety or list of options; however, only one option can be selected. Combo box: This is a dialogue box that has a variety of controls, for example text boxes, sliders and drop-down lists.	
Dialogue boxes	A dialogue box is a temporary window that an application creates to receive input from the user. It prompts the user for additional information.	

Table 7.4: Navigational components

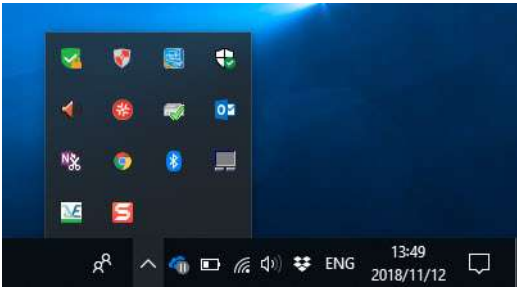
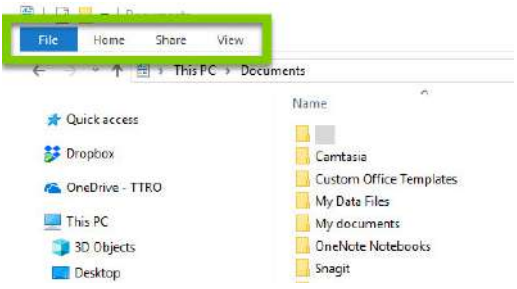
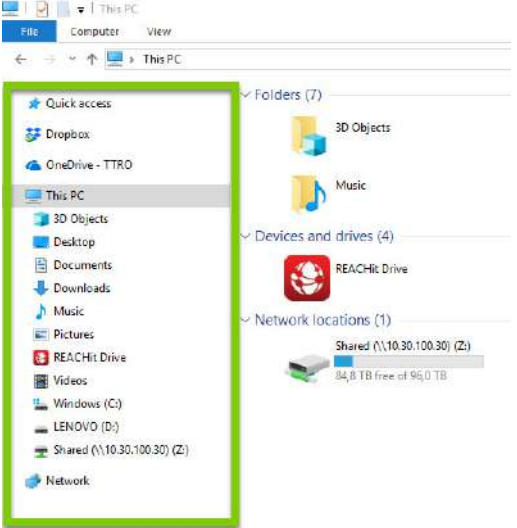
ELEMENT	DESCRIPTION	IMAGE
Icons	An icon is a picture that has a symbol, which can help you navigate to a specific program. It can be found on the taskbar , or PC Home screen.	
Menu	A menu bar is a thin, horizontal bar that has drop-down menus. It allows you to find a program's functions, such as opening or closing files.	
Navigation bar or pane	This is a set of buttons or graphics in a row that allow you to click on an option that will give you more information.	

Table 7.5: Informational components

ELEMENT	DESCRIPTION	IMAGE
Toolbars	A toolbar gives you quick access to frequently used commands.	
Message boxes	A message box is similar to a dialogue box; the computer gives you a message based on the data that you entered.	

DIFFERENT OPERATING SYSTEMS

The three popular operating systems for PCs are:

1. Microsoft Windows
2. Mac OS X
3. Linux

MICROSOFT WINDOWS

Microsoft Windows is one of the most popular operating system. Microsoft is calling Windows 10 “the last version of Windows”. This is not because it is killing off Windows, but because it decided to rather keep improving and updating Windows 10.



Figure 7.7: Windows user interface

MAC OS X

This is a range of computer systems created by Apple. All Macintosh computers, better known as Macs, come preloaded with this operating system. All their recent versions are known as OS X with, the latest version that was released in 2018, called *Mojave*.



Figure 7.8: Mac “El Capitan” user interface



Something to know

According to *StatCounter Global Stats*, there are only about 10% Mac OS X users of global operating systems, which is really low compared to Windows users (about 80%). A reason for this is that Apple computers tend to be much more expensive; although most people prefer the look and feel of Mac OS X over Windows.

The iPhone and iPad both have the iOS 9 user interface.

LINUX

Linux is a group of open-source operating systems. This means that it can be modified and distributed across the world by anyone. The advantages of Linux are that it is free and there are many different versions from which you can choose, for example, Linux Mint, Ubuntu, CentOS 7, etc.



Figure 7.9: Linux user interface

MOBILE OPERATING SYSTEMS

All of the previously mentioned information on operating systems is pretty much designed to only run on desktops or laptops. Mobile devices, such as phones, tablets and iPads differ from PCs or laptops and, therefore, have different operating systems designed just for them. Examples of these systems are Google Android, Apple iOS, Nokia Symbian, BlackBerry OS, etc.

Most mobile operating systems are tied to specific hardware, with little flexibility. It also does not have as many features as the operating systems of PCs and laptops, and the different devices cannot all run on the same software.

Mobile devices with communication abilities, for example smartphones, have the following two operating systems: the main user-facing software **platform** and a second lower-level **proprietary** real-time operating system, which operates the radio and other hardware.

A mobile operating system starts when the device is powered on, presenting a screen with various icons or **tiles** that provide specific information and application access. It also manages cellular and wireless network connectivity, as well as phone access.



Something to know

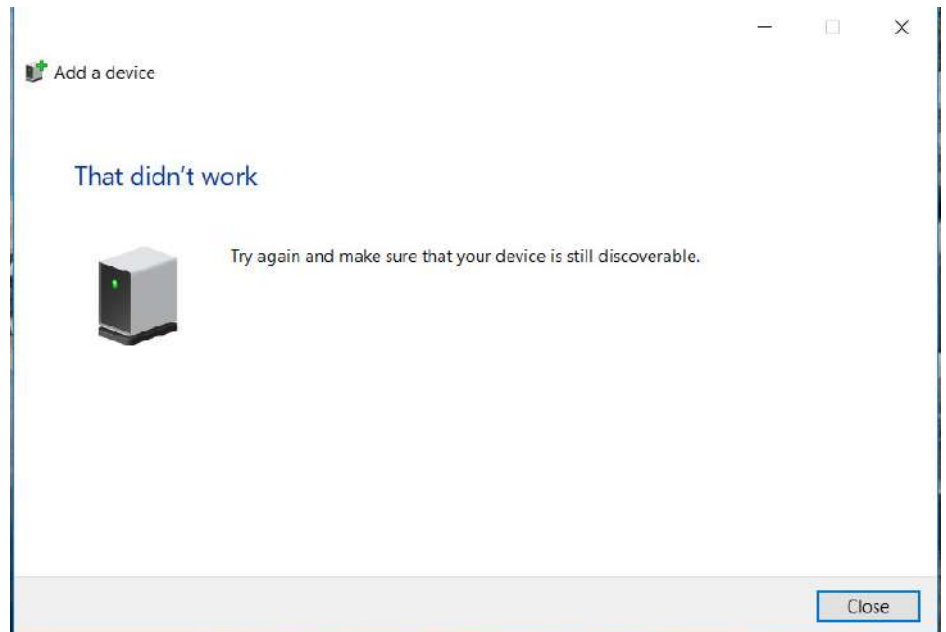
According to *StatCounter Global Stats*, Linux users are less than 2% of global operating systems. Most servers run on Linux, which makes it easier to customise.



Activity 7.2

Are the following three statements true or false? If false, provide the correct answer.

1. The operating system is a type of hardware.
2. Modern operating systems use a graphical user interface (GUI).
3. A menu bar gives you quick access to frequently used commands.
4. Which one of the following systems is an open-source operating system?
 - a. Linux
 - b. Windows
 - c. Mac OS
5. Look at the diagram below and answer the questions that follow.



- a. What type of GUI element is shown in this diagram?
- b. Explain the type of element shown. Give a brief description of this element.
- c. The diagram shows that the user was trying to connect to a peripheral device. Is this statement true? Motivate your answer.
- d. According to the diagram, what method of connection is being used (this will be based on your answer in question c above)?

7.3 Being more secure

Computer security is something every computer user should take very seriously. A breach in security can cause major problems. This means that you must do certain things to prevent and detect unauthorised access to your computer. *Unauthorised use* refers to anyone that gets access to your computer without your permission. In the worst-case scenario, this can refer to hackers who break into computer systems to steal, change or destroy information.

One of the steps you can take to be more secure, is to create a password that allows access to your computer.

CREATE A PASSWORD

Computer passwords are used to help identify the correct user and also prevent unauthorised use. A password consists of a string of secret characters, numbers, or words that are used to verify the identity of a user during the **authentication process**. It is used together with a username, and is designed to be known only to the user and allow that user to gain access to a device, application or website.

You can create a password in Windows 10 by doing the following:

1. Click on the *Start* button.
2. Type in *Control Panel*.
3. Open the *Control Panel*, and click on *User Accounts*.
4. Open *User Accounts* and choose *Make changes to accounts*.
5. Click on *Add a new user in PC settings*.
6. Choose *Make changes to my account in PC settings*.
7. Click or tap *Sign-in options* from the left.
8. In the *Password* area, click on the *Add* button.
9. Enter the new password correctly in both text fields, where required.
10. In the *Password Hint* field, enter something that will help you remember the password. Click *Next*.
11. Click on *Finish* to complete the password setup.

To access your computer, you need to key in the correct login name and password. You must ensure to log off when you are done with your computer so that no one else has access to your information.

TIPS FOR CREATING A PASSWORD

- The longer the better – a minimum of 12-characters is recommended.
- Mix it up; use variations on capitalisation, spelling, numbers and punctuation.
- Avoid names, places and dictionary words.
- Use a sentence that you will remember, for example, your name and date of birth.
- Use the first letters of each word as your password, for example, “My name is Julia Mabula and I was born on 13 March 2001!” The password would then be “MniJMalwbo13M21!” – it consists of 16 characters: letters – upper- and lowercase, numbers and symbols.



CAN'T REMEMBER MY PASSWORD

Should you forget your password, click on this QR code to find out how you can reset it:



<https://www.techrepublic.com/article/how-to-reset-your-windows-10-password-when-you-forget-it/>

OTHER AUTHENTICATION METHODS

The world of computers is changing every year. Nowadays, with the touch of your finger, you can unlock your computer and phone. Businesses that are really serious about their security use fingerprint scanning, facial recognition, or an iris scanner to protect the information on their computers.

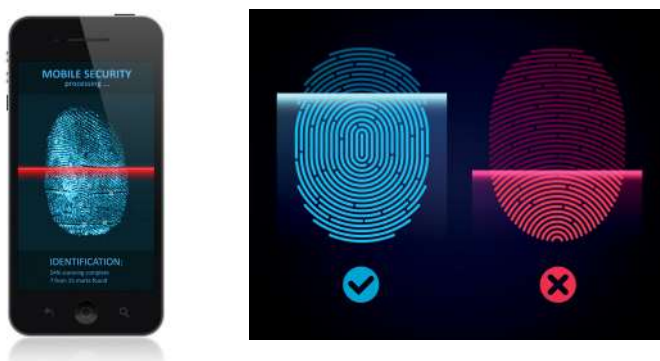


Figure 7.10: *Authentication safety measures*

SCREEN LOCK PATTERN

Users can secure their smartphones or tablets by using a screen lock. So, every time the user turns the phone on, or wakes it up from standby mode, it requires a screen lock pattern. This can be in the form of a pin, password, fingerprint scan and nowadays, users can unlock their devices by scanning their retina.

More about the pattern:

- It is a simple pattern that is drawn with the user's finger.
- It makes the device more secure and less accessible by others.
- If you take privacy very seriously, using a pattern to lock the device can actually be handy.



Activity 7.3

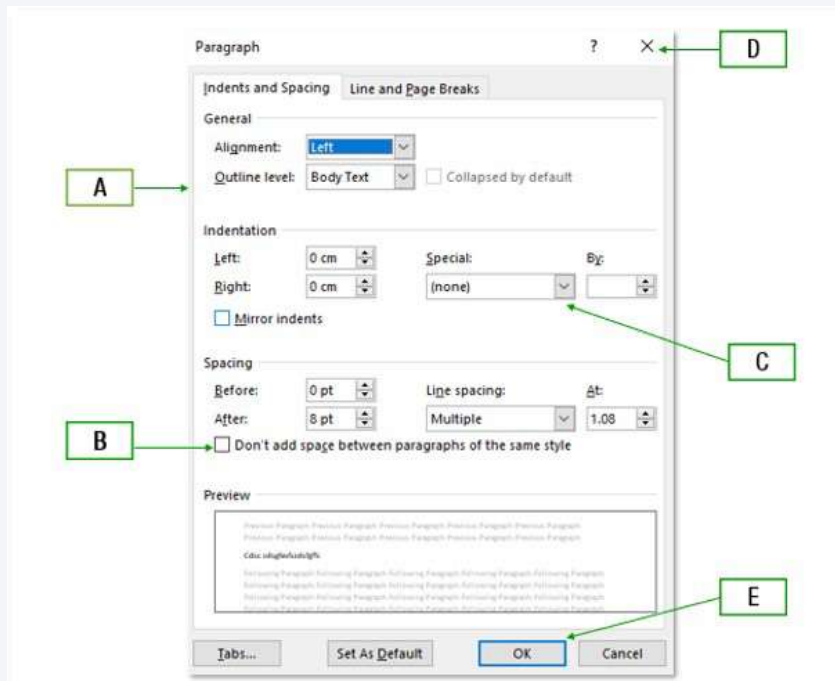
Read the following scenario and answer the questions that follow:

The Mo-ho family used a desktop computer connected to a printer, keyboard and wireless mouse at home for some time. Mrs Mo-ho has now started her own catering business, which she runs from home and requires that she interact with foreign clients.

1. **List** one essential output device, other than a printer, that she will need to perform her duties.
2. **List** one essential input device, other than a keyboard and mouse, that she will need to perform her duties.
3. Mrs Mo-ho stores sensitive sales information on her computer. She uses a login ID, with her numeric birth date (2907) as the password to access her computer. However, she is scared that someone might crack her password and gain access to her computer.
 - a. State two ways in which she could strengthen her password.
 - b. List one example of an authentication device, which could be used as part of her computer system to gain access to her computer, as well as give her an extra layer of security.

REVISION ACTIVITY

1. Explain briefly what is meant by the term “software”. (2)
2. The image below shows a box that you will often use when formatting paragraphs in the Microsoft Word processor.



Write down the letter of the label that matches each of the following components:

- a. Dialogue box (5)
 - b. Close button (1)
 - c. Option button (2)
 - d. Check box
 - e. Combo box
3. What do the letters GUI stand for? (1)
 4. What is the function of a GUI? (2)
 5. Johan often uses Microsoft Paint, the *Calculator* application and the *Snipping Tool* to assist him with homework tasks. Which of the applications would he most likely use to do the following: (3)
 - a. Resize a photograph taken with his smartphone camera
 - b. Take a screen capture of an image found on a website for a project
 - c. Convert a temperature in degrees Fahrenheit to degrees Celsius.
 6. Explain the difference between application software and system software. (4)
 7. Match the number of the application software type in Column A with the letter of the example in Column B. (4)

COLUMN A	COLUMN B
1. Movie player	A. Counter strike
2. Accounting software	B. Adobe Acrobat Reader
3. Internet browser	C. VLC
4. Gaming	D. Microsoft Excel
	E. Edge
	F. QuickBooks

... continued

REVISION ACTIVITY

... continued

8. Mention two functions of an operating system. (2)
9. Give the name of an operating system that is used for smartphones. (1)
10. Rate each of the following passwords as “weak”, “strong” or “very strong”:
 - a. N@t1onal2o19
 - b. Long2Montagu#Electric)Lego
 - c. SamHolly (your dogs’ names)
 (3)
11. Mention two types of access control systems that are found on smartphones other than a password or pin. (2)

TOTAL: [29]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it

NO.	DID YOU ...	YES	NO
1.	Understand what software is.		
2.	Identify the different types of software.		
3.	Learn about the different features, advantages and disadvantages of the different types of software.		
4.	Learn about the different operating systems.		
5.	Understand what the GUI is and its different elements.		
6.	Understand the importance of PC security.		

EXTENDED SOFTWARE CONCEPTS

CHAPTER OVERVIEW

Unit 8.1	Types of software
Unit 8.2	System software
Unit 8.3	Licensing and licensing agreements

At the end of this chapter, you should be able to:

- Describe system software in more detail.
- Identify the different types of system software.
- Explain the advantages and disadvantages of the different types of system software.
- Describe drivers and utilities.
- Understand the importance of computer security and software piracy.

INTRODUCTION

We have already looked at the basics of software and the different types of software you can get. Remember, without the correct system software, the application software and programs will not function correctly on your computer.

In this section, we will look at software in a bit more depth.

8.1 Types of software

System software is designed to control and coordinate the tasks and functions of computer hardware. System software also allows for interaction between hardware, software and the user. You can say that system software is like the “middle man”, it allows the software and hardware to communicate, making the computer easy to work on.

Let's look at the different types of software.

FREWARE, SHAREWARE AND PROPRIETARY SOFTWARE

We will now take a look at the differences between freeware, **shareware** and proprietary software in the following table:

Table 8.1: *Freeware, shareware and proprietary software*

TYPE OF SOFTWARE	FREWARE	SHAREWARE	PROPRIETARY
Description	Freeware can also be called <i>free software</i> , which is copyright software that is available to you, free of charge. However, the developers or company that distributes the software keeps the rights to the software.	Shareware is also copyright software that is distributed for free, but only for a trial period, which notifies users that they might need to pay for it later if they want to keep using the program.	Proprietary software , also known as <i>closed-source software</i> , is where the source code is not given to the public or to the user. In other words, it still belongs to the distributor or developer, even if you have bought the program.
Examples	Gmail, Firefox, Skype, Adobe Acrobat Reader, etc.	Deposit photos, Google Music, BitTorrent, etc.	Adobe Flash Player, Google Earth, WinRAR, Microsoft Office, etc.
Advantages	The main advantage of freeware is that it is free as well as easily available.	The advantage about shareware is that it gives the distributor exposure, while at the same time, gives users the time to decide whether they want to spend money on the program or not.	Proprietary software, on the other hand, is more user friendly and stable, and the technical support is easier to access.
Disadvantages	It can contain viruses and Trojan horses that can affect the way in which the computer performs.	The main disadvantages are that it cannot be modified and may be a cut-down or temporary version.	The only downside is that this software costs money and if there are any technical issues, you will have to wait for a developer to address the problem.



Something to know

Keep in mind that there are safety precautions you should know about when **downloading** software for free or even if you are paying for it. Nothing stops the developer from bundling the software with **malware** (virus, worms or spyware) and still make you pay for it. There is also a lot of freeware that is completely useless or ineffective. Always be aware of the importance of practicing good software safety so that you can choose the best software for your needs.

OPEN-SOURCE SOFTWARE

When the program is developed, the developers must decide whether they want to make its code proprietary or **open-source software**. The following table explains more about open source software.

Table 8.2: *Open-source software*

OPEN-SOURCE SOFTWARE	
Definition	Open source software has a source code that is available to the public or anyone that wants to use it. It also allows the user to change or modify the code if need be. You can also distribute this type of software to anyone and for any purpose.
Advantages	• It is free and you do not need to pay for using it, which saves you money. • Since the source is available to anyone, the public can collaborate and fix bugs. • Collaborators can improve the performance, or even add features. • It is high-quality software that is well designed.
Disadvantages	• Open software might not be as user friendly especially to unskilled users. • Difficult to find proper technical support. • Difficult to find drivers for some devices. • You can be prone to viruses and other cyber attacks.
Examples	Linux Kernel, Firefox, Gimp, etc.



OPEN-SOURCE OR
PROPRIETARY
SOFTWARE



[https://www.youtube.com/
watch?time_continue=
5&v=2q91vTvc7YE](https://www.youtube.com/watch?time_continue=5&v=2q91vTvc7YE)

8.2 System software



Something to know

If the device drivers need an update and the manufacturer is still around, then the drivers are usually listed on their web page.

Sometimes, the manufacturer might be out of business and you can find out-of-date, or even obsolete drivers on third-party websites that specialise in this type of business.

You can update the Windows system with the latest software. Other than a few drivers and Microsoft built-in programs, there is not much to a Windows update, and this is where driver and software update tools come into play. These utilities keep the Windows, PC, laptop or tablet up to date on a regular basis.

We will now learn a bit more about drivers and utility programs.

DRIVERS

Device drivers, more often known as drivers, are a group of programs that allow the hardware and the operating system to communicate with each other. Each hardware device has a specific driver that must be installed on the computer to which it is connected.

If the wrong driver is installed, if the driver is outdated, or if there is no driver installed at all, the hardware may not work properly, or it may not work at all. If you are using Microsoft Windows and there are compatibility issues, open the *Device Manager* to see what is happening.

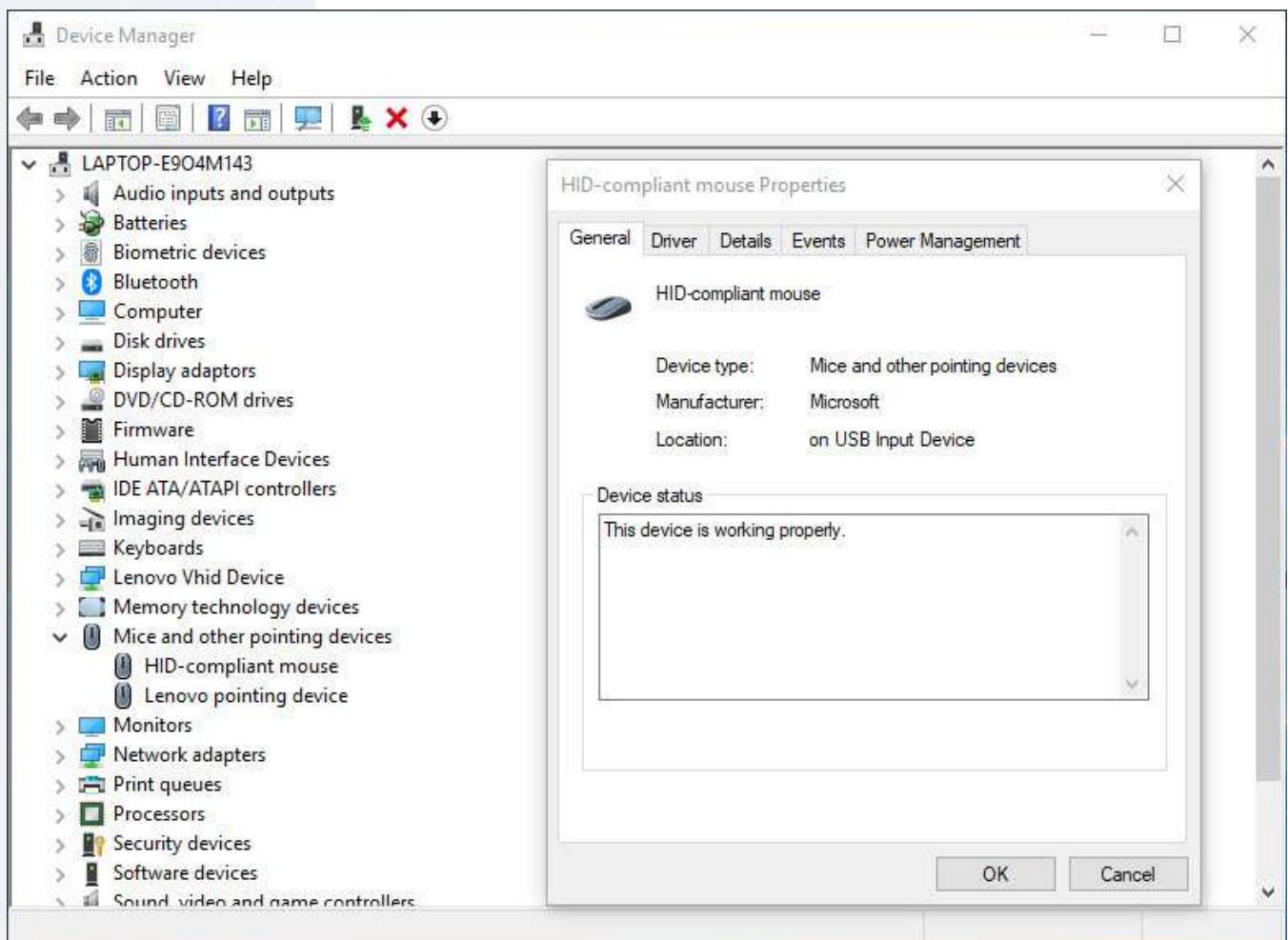


Figure 8.1: Device Manager

AUTOCONFIGURATION OF DEVICES

Autoconfiguration means to add devices to a computing device, without the need of user-intervention or software-configuration programs. In other words, you can connect the device to the computer, and the device will automatically install the drivers and then configure it in the system. The two types of autoconfiguration have been explained below.

PLUG AND PLAY

Plug and play, also known as PnP, is a term used when an operating system recognises a device and it installs the device driver automatically. These devices allow the user to use the device immediately after it has been connected to the computer. Hardware examples would include keyboards, mice, monitors, etc.

HOT SWAPPABLE

This is the ability to plug a device into a computer while it is switched on. It also allows you to use the device without having to switch off the computer. An example of this type of device would be a mouse. A mouse is a type of **plug-and-play device**, after plugging it in. It will work immediately.

UTILITY PROGRAMS

Utility programs or utilities allow you to perform computer management and maintenance tasks. These programs are much smaller in size compared to other software programs and can come packaged with the operating system, or installed at a later stage. Examples of utility programs are back-up software, disk clean-up, compression programs and so on.

Backup software is a program that copies data to a different location for in case you lose or delete the original data. You can then restore this data from the back-up copy. Nowadays, people actually back up their data on flash disks, hard drives and even the cloud.

Compression programs

Compression programs are used to compress or decompress files. In Windows operating systems, examples of programs that decompress and compress files would be 7Zip, WinZip and WinRAR.

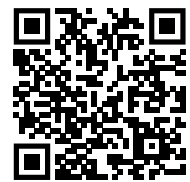
Disk Clean-up is a program that removes unnecessary files and temporary files. This can help speed up the performance of your computer. In order to clean up your computer space, you can do the following:

1. Click on the *Start* button.
2. Type in *Disk Clean-up*.
3. Click on *Disk Clean-up*. It will then show you the suggested files to delete (refer to Figure 8.2).
4. If you agree with the evaluation, click on *Clean up system files*. (Refer to Figure 8.2.)



CLOUD STORAGE

Have a look at this article to see how cloud storage works:



<https://computer.howstuffworks.com/cloud-computing/cloud-storage.htm>

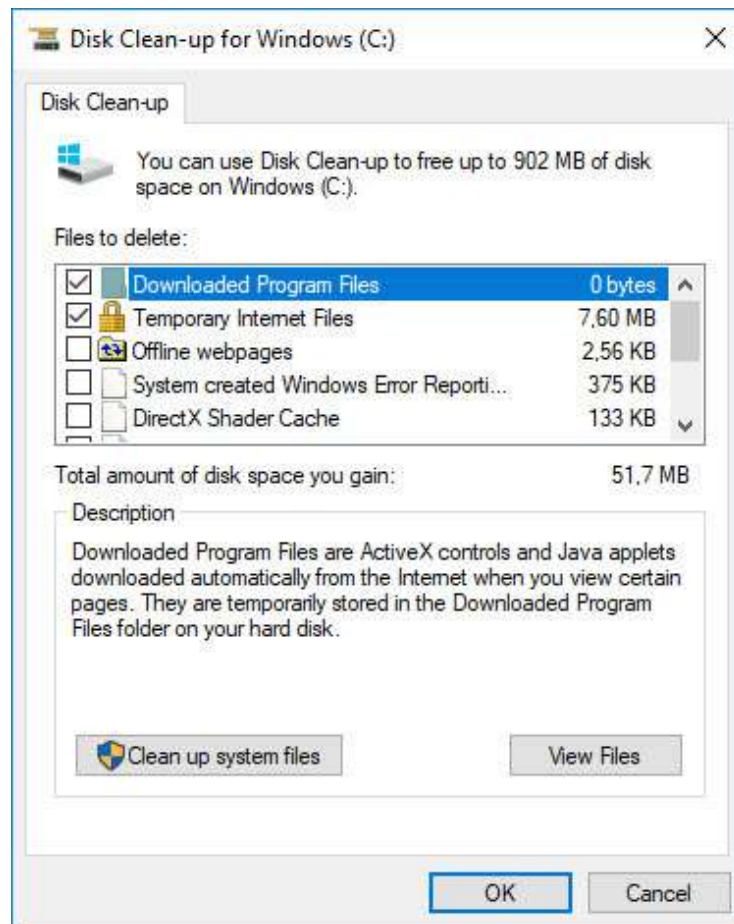


Figure 8.2: *Disk Clean-up*

Using *Disk Clean-up* at least once a month can help improve the performance of your computer!

8.3 Licensing and licensing agreements

SOFTWARE LICENSING AND LICENCE AGREEMENTS

When users pay for proprietary software, they pay for the rights to use it while the software still belongs to the developer. When you install proprietary software, you have to agree to terms and conditions in the licence agreement. This is an actual agreement and can contain information, such as:

- The software cannot be modified or changed
- Copies of the software cannot be made and distributed
- The software cannot be installed on more than one device

The licence agreement is better known as the end-user licence agreement (EULA) and has three types of agreements:

1. **Single-user licence:** This means that you can install the software on only one computer or device.
2. **Multi-user licence:** This means you can install the software on several devices or computers, which will be indicated in the licence agreement.
3. **Site licence:** This licence allows you to install software on a number of computers at the same site. This licence may be unlimited, but may also be limited to a certain number of users at the same time. This type of licence is often used by businesses and schools.

CREATIVE COMMONS

This is a set of copyright agreements that allows the public to freely distribute and use copyright work. It is used when a developer wants to give people the right to share, use and build upon a program that they have created. Creative commons can be abbreviated to CC and there are several CC licences that allow the user to do the following:

- Copy the work – i.e. you can download, copy and photocopy the content that is required.
- Distribute the content, for example to teachers, schools and other educational purposes.
- Communicate the content, for example make it available on the school network.

Examples of organisations that use the CC licence are Wikipedia, some YouTube channels, Boundless, etc.

SOFTWARE PIRACY

Piracy is the illegal copying, distribution or use of software. Illegal copies are sold at a cheap price. When illegal copies are made and distributed in the form of CDs or DVDs or another storage medium, it is called software piracy. There are different types of software piracy:

- **Softlifting:** When a software program with a legal licence is installed or copied with the main purpose of providing it to multiple users, rather than to sell copies for profit.
- **Client-server overuse:** When more users, as stipulated in the licensing agreement, are using a central copy of a program at the same time.
- **Hard-disk loading:** This happens when illegal copies of software are loaded onto the hard disks of new computers to make the purchase more attractive.



Something to know

According to the *Business Software Alliance* (BSA), about 36% of software that is used is pirated. Software piracy causes a lot of monetary loss for developers, making the legal copies more expensive.

- **Counterfeiting:** The illegal duplication, distribution and/or sale of copyright material with the intent of imitating the copyright product.
- **Online piracy:** When proprietary software is downloaded from the internet.



Activity 8.1

1. Define the following types of software:
 - a. Freeware
 - b. Shareware
 - c. Open source
 - d. Proprietary software
2. Describe how one would go about using proprietary software, without having to spend much money. Motivate your answer.

One of Mr Itek's children knows his friend has software for Microsoft Word. He wants to copy the software, then install it on their home computer.

3. Is Mr Itek doing the right thing? Motivate your answer.
4. How would you categorise the way in which Mr Itek's friend intends on using the software?

REVISION ACTIVITY

1. The table below is comparing freeware, shareware and proprietary software. Make a copy of the table in your workbook and fill in the headings, as well as the missing information. (12)

TYPE OF SOFTWARE			
Description	Known as "try before you buy" software		
Example		Gmail	
Advantage			Technical support is freely available
Disadvantage			

2. Fairiza purchases a laptop that she can use to work on assignments for her course at university. The laptop was supplied with the Windows 10 Home operating system.
 - a. She obtained a copy of Microsoft Office 2016 from a friend and installed it on her laptop. Explain why she should NOT do this. (2)
 - b. If she wishes to continue using Office 2016, what type of licence does she need? (1)
 - c. When she updates Office 2016 with a valid licence, she is asked to acknowledge that she has read the EULA. What does EULA stand for? (1)

If she cannot afford to purchase a licence from Microsoft, she could install an Open Source office suite.

 - d. What is open-source software? (2)
 - e. Give an example of an Open Source office suite that she could use. (1)
3. You have purchased a new printer. When you connect the printer to your computer for the first time, you have to install a printer driver that is on a CD provided with the printer.
 - a. What is a driver? (1)

... continued

REVISION ACTIVITY

... continued

- b. Write down the missing words:
When you plugged in a new wireless mouse for the first time, the driver was installed automatically by the operating system. This is known as plug and play, or _____. (2)
 - c. What is a hot swappable device? (2)
 - 4. Mention two utility programs that are part of the Windows 10 operating system. (2)
- TOTAL: [26]**

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Learn about the different types of system software.		
2.	Understand the advantages and disadvantages of open-source software.		
3.	Learn about drivers and utility programs.		
4.	Understand what auto-configuration of devices is and how it works.		
5.	Understand the importance of computer security.		
6.	Learn about the different ways in which software can be pirated.		

INTRODUCTION TO NETWORKS AND NETWORKING CONCEPTS

CHAPTER OVERVIEW

Unit 9.1	Concepts used in networks
Unit 9.2	Uses of networks
Unit 9.3	Social implications: Networks

At the end of this chapter, you should be able to:

- Explain the different concepts used in networks.
- Understand how networks are used.
- Discuss the advantages and disadvantages of networks.
- Identify the social implications that are associated with networks.

INTRODUCTION

A computer network consists of two or more computers and other hardware devices that are linked together through **communication channels** to share electronic communications and resources, such as printers with various users.

Something to know

The Semi-Automatic Ground Environment (SAGE) computer system was invented by two MIT professors during the 1950s and became operational in 1961. Because of America's size, the speed of the military attack planes and the possibility of nuclear attacks by the Russians, the US military decided that they needed an air-defence system that would identify and intercept enemy planes before they could do any damage.

SAGE was, therefore developed, using techniques learned from World War II radar development. Weighing over 250 tonnes, SAGE was one of the largest computers ever built and consisted of 23 bunkers all over the US and one in Canada; each having a SAGE computer that could track 400 planes. As it consisted of a network of long-distance communications technology through landlines and ground-to-air radio links and **interactive** display terminals, it could differentiate between enemy and friendly planes by keeping track of flight plans. What made SAGE so extraordinary, is the fact that it was the first network-based computer system, laying the foundation for the entire internet age.

In this chapter, you will learn about networks, the different concepts used in networks, and the advantages and disadvantages of networks. You will also learn how to use networks in an ethical and correct way.

9.1 Concepts used in networks

Looking at the definition of a computer network in the “Introduction”, you will remember that a computer network consists of two or more computers and hardware devices linked together through communication channels to share electronic communications and resources with various users.

In computer networks, computing devices are called **nodes** and share data with each other through **data links**. These links are sent over cable media such as wires, optic cables, or wireless media such as Wi-Fi. The connected computers also share resources, such as access to printers, with the most common resource being the internet.

There are two common types of networks:

1. **Home Area Network (HAN):** A HAN is a very small network that usually covers a single home or home office.
2. **Personal Area Network (PAN):** A PAN is similar to a HAN, but even smaller! It refers to a network that is built for a single person and contains all the devices connected to this network.

Most organisations, such as schools and businesses, have computers that are connected to a central computer called the **network server**. This server is the main computer that manages the communication between all the other computers on the network. A **network client** refers to the hardware or software that can access services made available by a server.

Network administrators are the people that are part of a technical support team, and who can modify and change the network according to the organisation’s needs. The administrator can, for example, remove computers from the network and restrict access to users, preventing them from having access to confidential information.

There are various devices helping in the communication between nodes, for example:

- **Switches:** The main purpose of switches is to share information between specific computers without involving the other computers in the network. This means that these computers can continue doing what they are doing. Looking at the following diagram, you will see that by using a switch, Computer A sends a message to Computer B, without the other computers seeing the message. Those computers can send other messages at the same time and without interference.

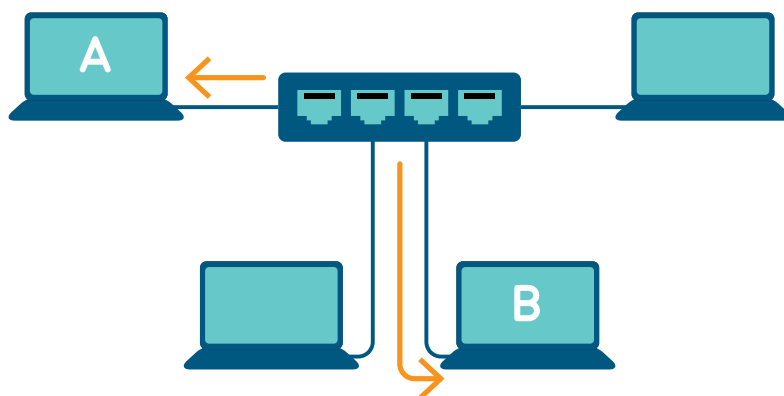


Figure 9.1: Messages sent using switches

- **Routers:** Routers do most of the hard work in a network as they make the decisions about all the messages that pass through the network. This means that they analyse each piece of data and protect information from security threats through a firewall. Routers also choose the route for the data to travel and decide which computer gets priority.
- **Wireless access points:** These points allow devices to connect to a wireless network without cables. It can also give useful data about the devices on the network.



Activity 9.1

1. Write the correct answer to the questions below in your book:
 - a. In computer networks, computing devices are called _____.
 - A. Computers
 - B. Nodes
 - C. Modems
 - D. Switches
 - b. A PAN is a _____.
 - A. Network that operates in a large area, such as across a country
 - B. Network built for a single person connecting all the devices
 - C. Very small network that usually covers a single home
 - D. Network that operates in a small area, such as a house, school or office
 - c. A switch is a device that _____.
 - A. Shares information between specific computers without involving the other computers in the network
 - B. Makes the decisions about all the messages that pass through the network
 - C. Connects a computer or network to the internet
 - D. Allows computing devices to connect to a wireless network without cables
 - d. Ethernet _____.
 - A. Is a set of rules that define the way in which two networking nodes can communicate
 - B. Is a type of data link protocol that defines the types of cables and connections used to wire computers, switches and routers together
 - C. Connects a computer or network to the internet
 - D. Can remove computers from the network and restrict access to users
 - e. The people who can modify and change the network settings are called the _____.
 - A. Network technicians
 - B. Computer nerds
 - C. Network administrators
 - D. Gamers

9.2 Uses of networks

So far, you have learned that networks can be used in a variety of environments; from schools to businesses to homes. These uses (aims and objectives) can be grouped into the following four main categories:

1. Sharing resources
2. Connecting people
3. Organising information
4. Access to entertainment

In this section, we will look at the different uses of networks, as well as their advantages and disadvantages.

AIMS AND OBJECTIVES OF NETWORKS

Networks have become something that we can barely live without. In this section, we will take a look at some of their main uses.

SHARING

One of the primary objectives of a network is to *share information*.

Although networks are designed to share information, they do much more than that; they also *share the resources* linked to the computers on the networks. This includes access to the internet, files, hardware, people and money.



Figure 9.2: *People connected to networks*

The most common reason for creating a home, school, or office network is to provide all the computers connected to that network, access to the internet. In South Africa, the internet is an expensive resource. By sharing the internet between many people, it becomes slightly more affordable and people who would previously not have had access to the internet, can now gain access.



Something to know

According to the United Nations (UN) agency that oversees international communications, more than three billion people all over the world are using the internet. This number increased from 738 million in 2000, to 3.2 billion in 2015. This is fantastic, right? So now, people from South Africa can send a file, text or image from their computers to someone anywhere in the world with just the click of a button.